

TSG15

PAL PULSE & BAR TEST SIGNAL GENERATOR MODULE

Please check for
CHANGE INFORMATION
at the Rear of this Manual

First Printing FEB 1978
Revised SEP 1987

Tektronix
COMMITTED TO EXCELLENCE

Copyright © 1978 Tektronix, Inc. All rights reserved. Printed in U.S.A.
Tektronix products are covered by U.S. and foreign patents, issued and
pending. Information in this publication supersedes that in all previously
published material. Specification and price change privileges reserved.

TEKTRONIX, TEK, PLOT 10, TEKTEST, SCOPE MOBILE, and
are registered trademarks.

For further information, contact: Tektronix, Inc., Corporate Offices,
P.O. Box 500, Beaverton, OR 97077. Phone: (503) 627-7111, TLX 151754,
TWX: (910) 467-8708, Cable: TEKWSGT.

Subsidiaries and distributors worldwide.

TABLE OF CONTENTS

PART I OPERATING INSTRUCTIONS

SAFETY SUMMARY

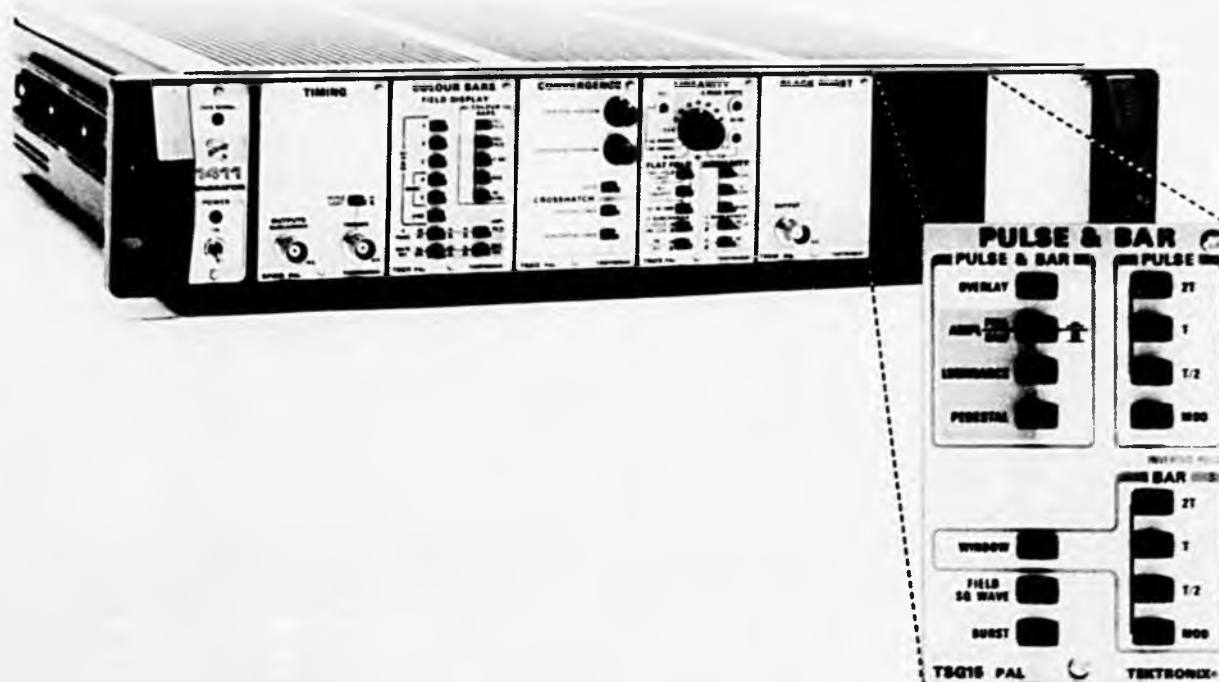
	Page		Page
SECTION 1 OPERATING INSTRUCTIONS		SECTION 2 SPECIFICATION	
Description	1-1	Electrical Characteristics	2-1
Front-Panel Controls	1-1	Performance Check Procedure	2-6
Glossary of Terms	1-4	Test Equipment	2-6
Applications	1-5	Calibration Fixture	2-6
		Procedure	2-6

PART II SERVICE INFORMATION

WARNING

THE FOLLOWING SERVICING INSTRUCTIONS ARE FOR USE BY QUALIFIED PERSONNEL ONLY. TO AVOID PERSONAL INJURY, DO NOT PERFORM ANY SERVICING OTHER THAN THAT CONTAINED IN OPERATING INSTRUCTIONS UNLESS YOU ARE QUALIFIED TO DO SO.

	Page		Page
SECTION 3 INSTALLATION		SECTION 6 MAINTENANCE (cont)	
Installing in the Mainframe	3-1	Troubleshooting	6-1
Installation Adjustment	3-2	Diagrams	6-1
Procedure	3-3	Circuit Boards	6-2
Operating Mode Selection	3-4	Multi-pin Connectors	6-2
Additional Program Considerations	3-4	Troubleshooting Hints	6-2
Remote Disable Options	3-5	Corrective Maintenance	6-3
SECTION 4 CALIBRATION		Parts Location	6-3
Test Equipment	4-1	Selected Components	6-3
Calibration Fixture	4-1	Circuit Board Replacement	6-3
Measurements	4-2	Board Removal	6-3
Short Form Procedure	4-2	Extracting Integrated Circuits	6-3
Calibration Procedure	4-3	Pushbutton Switch Replacement	6-3
SECTION 5 THEORY OF OPERATION		LUMINANCE Switch Replacement	6-4
The Block Diagram	5-1	SECTION 7 REPLACEABLE ELECTRICAL PARTS	
Pulse & Bar Timing (Diagram 1)	5-1	SECTION 8 SERVICE ILLUSTRATIONS	
Pulse & Bar Drivers (Diagram 2)	5-4	SECTION 9 DIAGRAMS AND CIRCUIT BOARD ILLUSTRATIONS	
Pulse & Bar Output (Diagram 3)	5-6	SECTION 10 REPLACEABLE MECHANICAL PARTS	
SECTION 6 MAINTENANCE		CHANGE INFORMATION	
Preventive Maintenance	6-1		
Cleaning	6-1		
Visual Inspection	6-1		
Recalibration	6-1		



2479-00

Fig. 1-1. The TSG15 PAL Pulse and Bar Test Signal Generator Module.

TSG15

PART I

OPERATING INFORMATION

OPERATING INSTRUCTIONS

SAFETY SUMMARY

This manual contains safety information to ensure safe operation and maintenance of the 1411 Generator, including the TSG15.

WARNING information is intended to protect personnel.

CAUTION information is intended to protect the instrument.

WARNING

Electrical shock hazards are present inside the instrument. Only qualified service personnel should remove instrument covers.

DESCRIPTION

The TSG15 is a Sine-Squared Pulse and Bar television Test Signal Generator (TSG) module for use in the 1411 PAL Generator. The TSG15 provides luminance pulse, luminance bar, inverted luminance pulse, modulated pulse, modulated bar, window, and field square wave test signals. Front-panel controls provide most test signal options, but internally selectable options provide even greater versatility.

Some of the features of the TSG15 are:

Full or half amplitude pulse and bar test signals provided with or without pedestal.

Independently-selectable sine-squared shaping of the 2T, T or T/2 signal for luminance pulse and bar.

A modulated pulse and modulated bar may be selected with or without luminance. 10T or internally-selectable 20T pulse is sine-square shaped.

Pulse and bar may be overlaid for convenient amplitude comparison.

A window signal may be selected with a 2T, T, T/2 signal, or modulated bar.

A field square wave is also provided for field time measurements.

FRONT-PANEL CONTROLS

Controls discussed in this paragraph are illustrated in Fig. 1-2. Refer to that figure while reading about control functions. All switches are on when pushed in, except where other conditions are specifically stated.

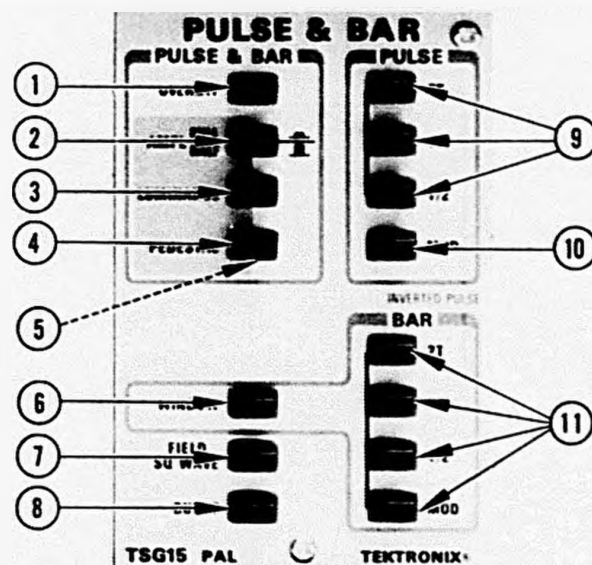


Fig. 1-2. TSG15 front panel controls.

1. OVERLAY

The pulse and bar test signal consists of a modulated pulse, a luminance pulse, and a modulated or unmodulated bar. In the overlay mode the normal pulse and bar, shown in Fig. 1-3, alternates with the altered timing pulse and bar, shown in Fig. 1-4, to permit convenient comparison of pulse and bar amplitudes. Internal jumper selection provides a selection of H/2, H/4, or V/2 overlay rate. Overlay rate is factory-set to H/4 for use with a 1480-Series Waveform Monitor, which triggers at H/2. In the bar calibration mode, any two bars may be overlaid for amplitude comparison by selecting both bars with the front-panel BAR switch.

The inverted and non-inverted 2T pulses may be overlaid for convenient comparison of shape and half-amplitude duration (HAD).

2. AMPLITUDE

Full amplitude is 700 mV. Half amplitude is 350 mV. See Fig. 1-5.

3. LUMINANCE

The luminance pulse, luminance bar, and the luminance components of the modulated pulse and bar may be switched off to provide a chrominance bar and/or chrominance pulse. See Fig. 1-6. The chrominance pulse and bar can be placed on a 350 mV pedestal to prevent chrominance from extending below the blanking level. See Fig. 1-7.

4. PEDESTAL

A pedestal (normally 350 mV) can be added to the pulse and bar signal. A pedestal does not reduce the pulse and bar amplitude. See Fig. 1-7. If full amplitude luminance is on, the pedestal is reduced. The reduced pedestal, normally 0 V, is determined by an internal adjustment.

5. OFF

The pedestal is removed, referencing all levels to the blanking level.

6. WINDOW

A modulated or unmodulated bar may be gated to provide a window signal. The window does not alter pulses.

7. FIELD SQ WAVE

A standard field square wave with full amplitude can be selected. Field square wave takes precedence over pulse and bar if both are selected. See Fig. 1-8.

8. BURST

The burst can be switched off without affecting the chrominance on a modulated pulse or bar. This is useful for forcing equipment under test into monochrome operation, which may alter its response characteristics.

9. PULSE (2T, T, T/2)

Three self-canceling switches permit the selection of a 200 ns, 100 ns, or 50 ns half-amplitude duration (HAD) luminance pulse, independent of bar risetimes.

An internal jumper option permits the addition of an inverted 2T pulse to the T bar whenever the 2T pulse and the T bar are selected together. See Fig. 1-9. The inverted and non-inverted pulses can also be overlaid for convenient comparison.

10. MOD (PULSE)

A modulated pulse, with or without luminance can be selected. An internal jumper option permits selection of either 10T or 20T HAD.

11. BAR (2T, T, T/2, MOD)

Four self-canceling switches that permit selection of a 193 ns, 96 ns, or 48 ns risetime luminance or modulated bar. The modulated bar (see Fig. 1-10) risetime is 964 ns with the 10T pulse, or 1929 ns with the 20T pulse.

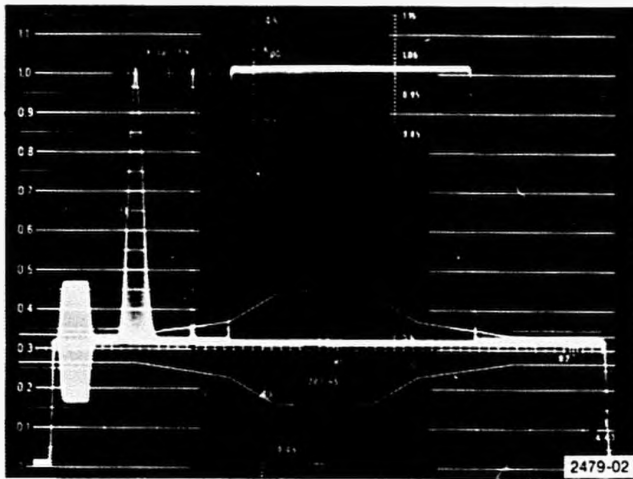


Fig. 1-3. Normal TSG15 Pulse and Bar signal.

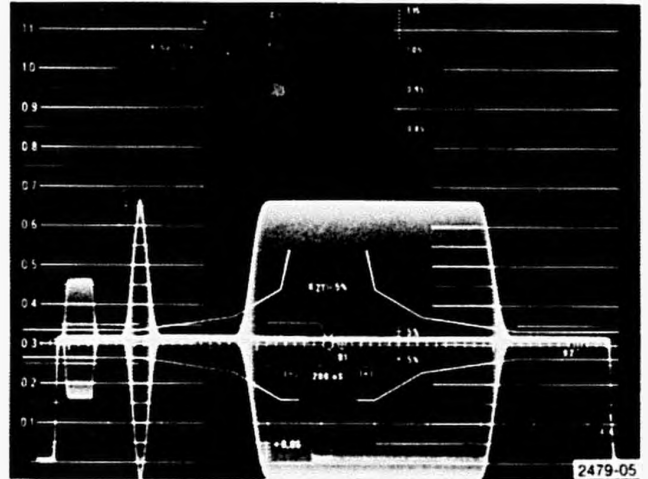


Fig. 1-6. TSG15 chrominance pulse and chrominance bar.

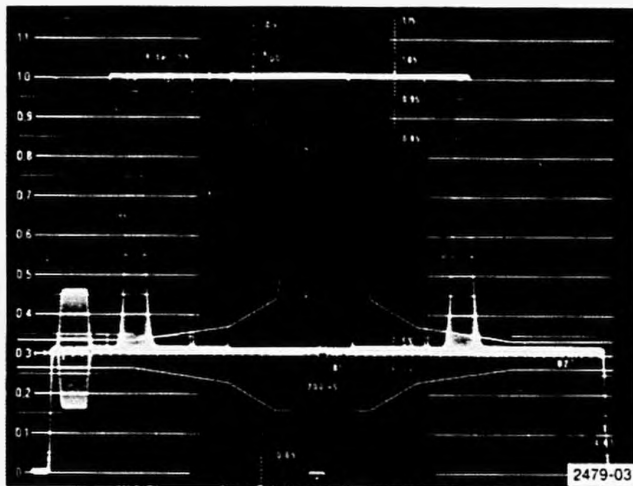


Fig. 1-4. Altered timing Pulse and Bar signal overlayed with the normal Pulse and Bar signal.

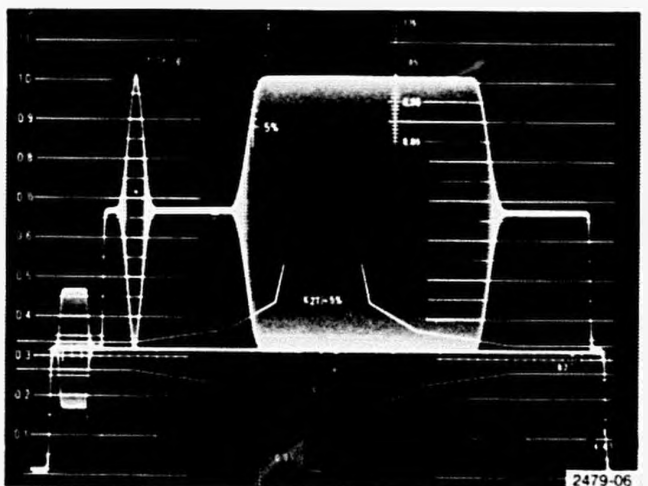


Fig. 1-7. Chrominance pulse and chrominance bar on a 50% pedestal.

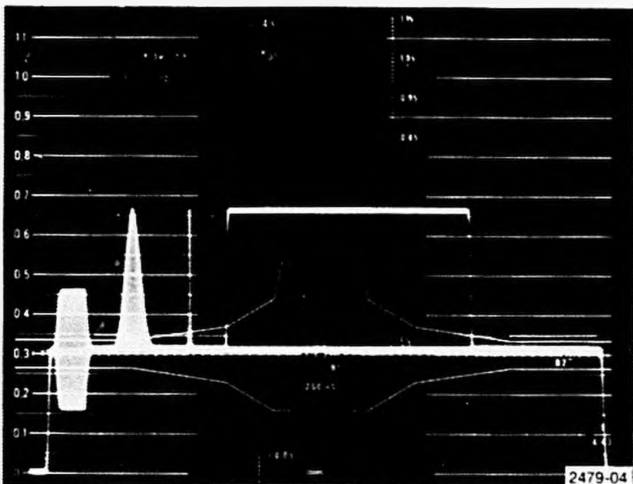


Fig. 1-5. TSG15 half-amplitude Pulse and Bar signal.

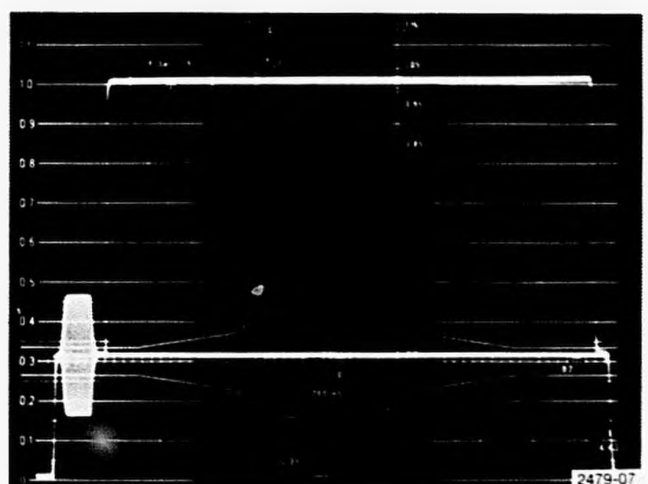


Fig. 1-8. Field squarewave signal.

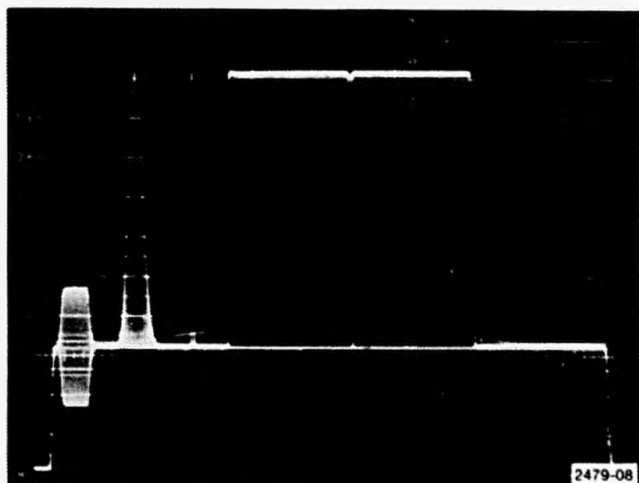


Fig. 1-9. TSG15 Pulse and Bar signal with Inverted 2T pulse.

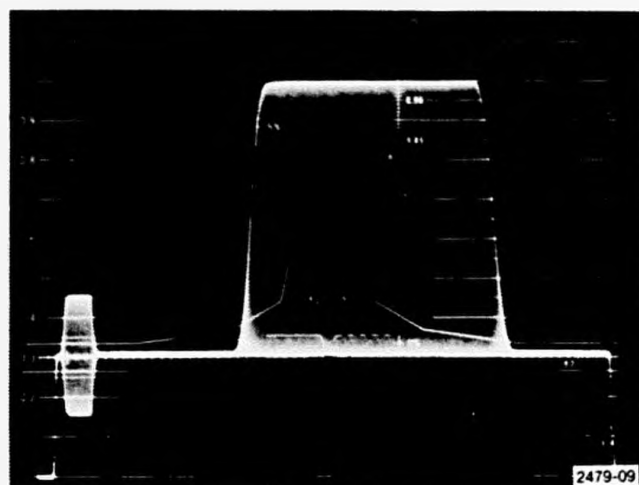


Fig. 1-10. TSG15 20T modulated bar.

GLOSSARY OF TERMS

Chrominance Bar—A chrominance signal whose envelope is a sine-squared bar. Normally in PAL its rise and fall times are 964.4 ns (10T) or 1928.8 ns (20T).

Chrominance/Luminance Delay—The difference in time delay of subcarrier-frequency information compared to low-frequency information.

Chrominance/Luminance Gain—The ratio of gain at sub-carrier frequency compared to gain at low frequencies.

Chrominance Pulse—A chrominance signal whose envelope is a sine-squared pulse. Normally in PAL its half-amplitude duration is 1000 ns (10T) or 2000 ns (20T).

Field Square Wave—A field rate signal composed of lines at 100% amplitude for part of a field and at blanking level for the remainder of the field.

Field Time Distortion—Time domain distortions with 64 μ s to 16 ms duration.

Line Time Distortion—Time domain distortions with 1 μ s to 64 μ s duration.

Modulated Bar—The sum of a sine-squared luminance bar and a sine-squared chrominance bar of equal rise and fall times. Typically 10T or 20T for PAL.

Modulated Pulse—The sum of a sine-squared luminance pulse and a sine-squared chrominance pulse of equal half-amplitude duration. Typically 10T or 20T for PAL.

Short-Time Distortion—Time domain distortions with 100 ns to 1 μ s duration.

T Bar—A sine-squared step with 96.4 ns rise and fall times. The rising and falling edges are the integral of T pulses.

T/2 Bar—A sine-squared step with 48.2 ns rise and fall time.

2T Bar—A sine-squared step with a 192.9 ns rise and fall time.

T Pulse—A sine-squared luminance pulse with a 100 ns half-amplitude duration (HAD). Energy distribution is from 0 to 10 MHz.

T/2 Pulse—A sine-squared luminance pulse with a 50 ns half-amplitude duration (HAD). Energy distribution is from 0 to 20 MHz.

2T Pulse—A sine-squared luminance pulse with a 200 ns half-amplitude duration (HAD). Energy distribution is from 0 to 5 MHz.

Window—A pulse and bar signal formed by gating the bar off for approximately the first and last 1/4 of every field.

APPLICATIONS

Linear distortions of the television signal may be measured in terms of frequency- and phase-distortion, or in terms of time-domain distortion.

The measurement of linear distortions, such as overshoot, undershoot, ringing, gain, delay, and tilt, are especially suited to time-domain measurement. These time-domain measurements relate directly to the perceived quality of the final picture:

Short-time distortion is seen as horizontal blurring or echoes.

Line-time distortion shows up as a brightness variation from left to right.

Field-time distortion is seen as a brightness variation from top to bottom.

The TSG15 provides many test signals that are useful in making time-domain measurements of these linear distortions, as well as numerous nonlinear distortions of the television signal.

Chrominance/Luminance Gain

The gain near subcarrier frequency may be considerably different from the gain at low frequencies. The resulting chrominance/luminance gain error may alter the saturation of a color picture.

The modulated pulse and modulated bar test signals provide a convenient method of measuring these errors. The test signals consist of luminance and chrominance components proportioned so that correct chrominance/luminance gain results in a flat baseline beneath either signal. See Fig. 1-11.

Incorrect proportion of chrominance and luminance results in a baseline error similar to the one shown in Fig. 1-12.

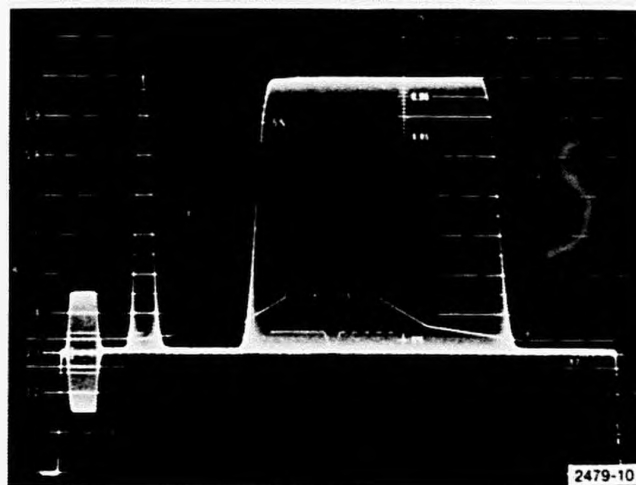


Fig. 1-11. Correct chrominance/luminance gain.

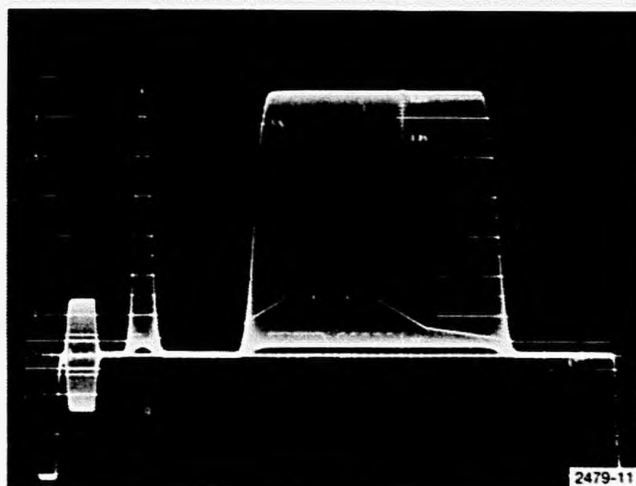


Fig. 1-12. Chrominance/luminance gain error, easily observed as reduced chrominance at the baseline, beneath a modulated pulse or modulated bar.

Operating Instructions--TSG15

The spectrum of luminance and chrominance components making up a modulated 10T and 20T pulse is shown in Fig. 1-13.

For a 20T modulated pulse, the luminance component occupies the 500 kHz region where the typical video signal luminance information is concentrated. The chrominance component occupies a region extending 500 kHz either side of the subcarrier frequency, representing a typical color picture's chrominance distribution.

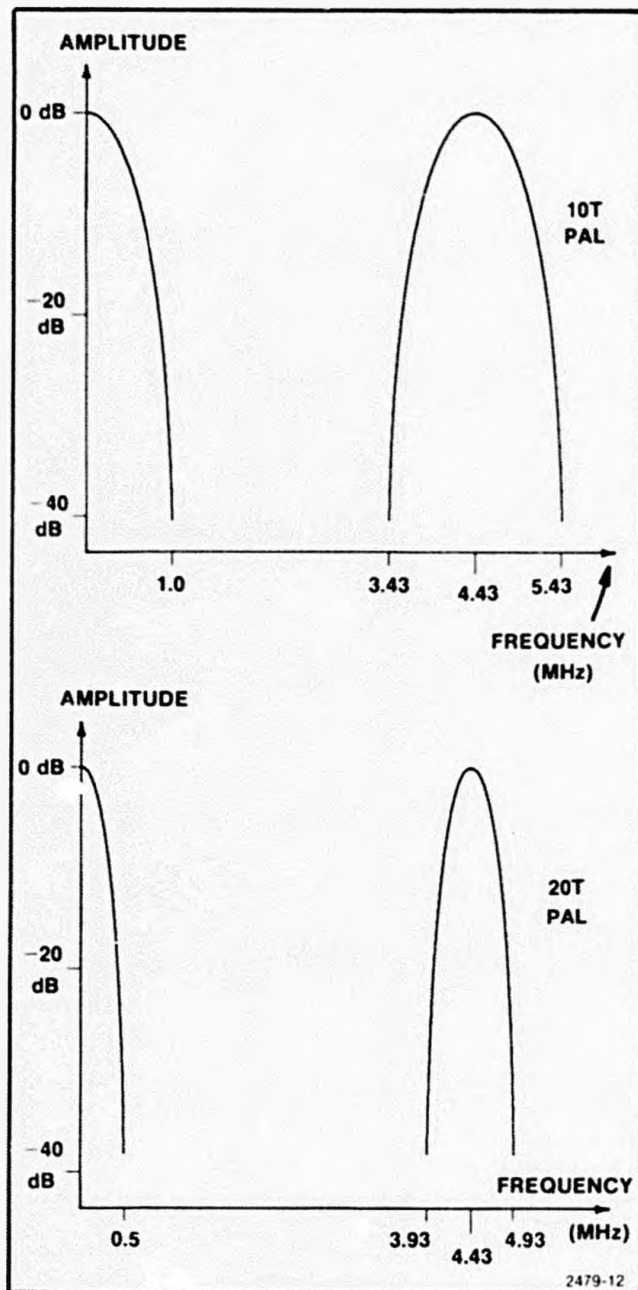


Fig. 1-13. Energy Spectrum for modulated pulses, a. 10T b. 20T.

The modulated pulse allows short-time measurement of chrominance/luminance gain errors. The modulated bar, or window, allows line-time, or even field-time, measurement of chrominance/luminance gain errors.

Modulated pulse and modulated bar measurements of chrominance/luminance gain errors may not always yield the same results in cases where the chrominance signal has gone through bandpass filtering. Examples of this are some processing amplifiers and video tape recorders.

Chrominance/Luminance Delay Distortions

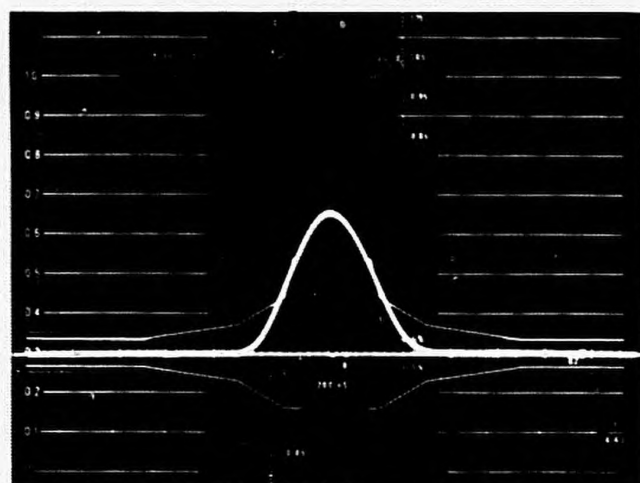
The delay near subcarrier frequency may be different from the delay at low (luminance) frequencies. The resulting chrominance/luminance delay error may adversely affect the registration of color with respect to luminance in the color picture.

The modulated pulse test-signal is useful for measuring chrominance/luminance delay errors. If the luminance, Fig. 1-14a and chrominance, Fig. 1-14b, components experience equal delays, their sum will result in a symmetrical modulated pulse. Gain errors will alter baseline flatness but will not change pulse symmetry, Fig. 1-14c. Delay errors will result in an asymmetrical pulse shape as shown in Fig. 1-14d and e.

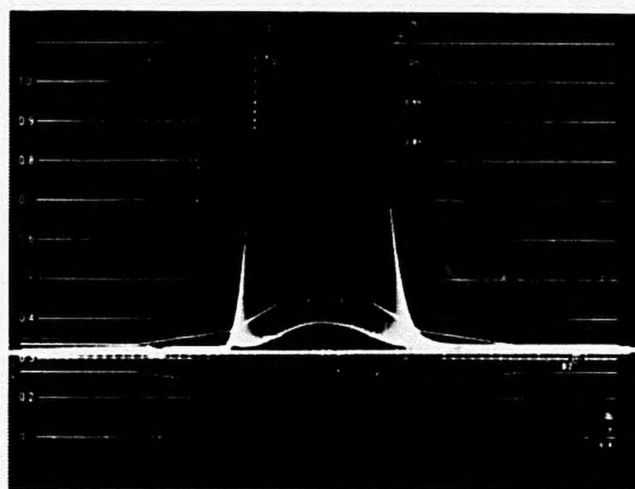
Chrominance/luminance delay can be determined by measuring the baseline errors on the modulated pulse. If the luminance gain is correct and the chrominance/luminance gain error has been normalized, the left peak Y1 and the right peak Y2 will be 10 mV P-P for every 18 ns delay on a 20T pulse. (With a 10T pulse, every 10 mV P-P equals 9 ns delay.)

If there is chrominance/luminance gain error in addition to the delay error, the nomograph in Fig. 1-15 should be used to determine chrominance/luminance delay. The nomograph is convenient when the chrominance to luminance gain error is small. When large gain errors are present a chrominance gain normalizer, such as the TEKTRONIX 1478, should be used.

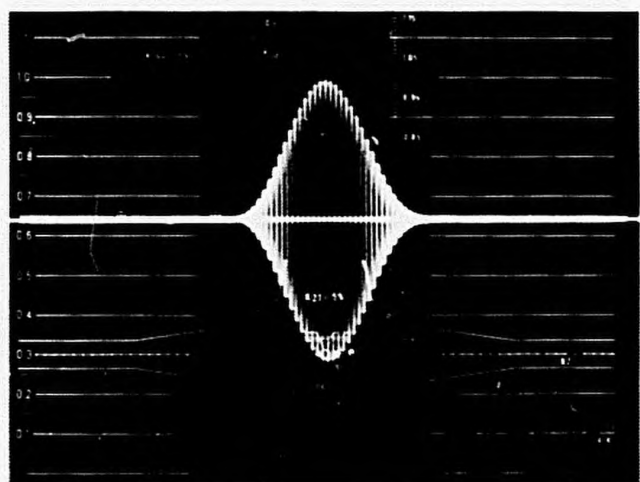
The 10T modulated pulse provides increased sensitivity to chrominance/luminance delay errors, but has significant energy outside 5.0 MHz. The 20T modulated-pulse subcarrier information does not exceed the 5.0 MHz bandwidth but is not as sensitive to chrominance/luminance delay errors. The TSG15 is factory-programmed to deliver the 20T signal. The 10T signal is recommended for studio applications, while the 20T is more useful for transmitter applications.



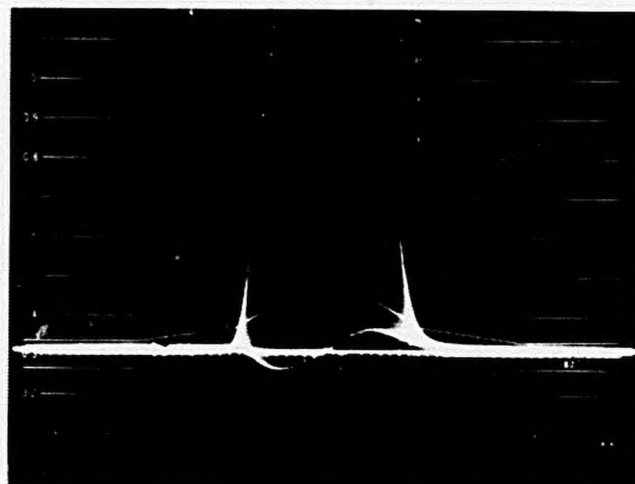
(a)



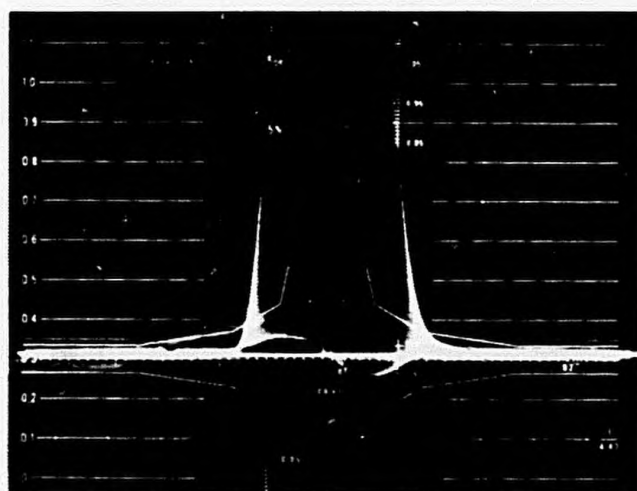
(c)



(b)



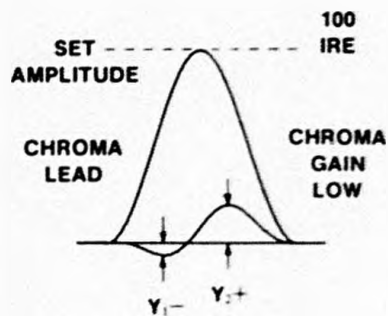
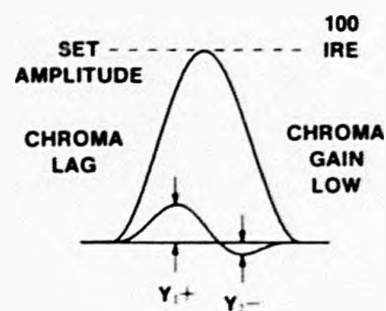
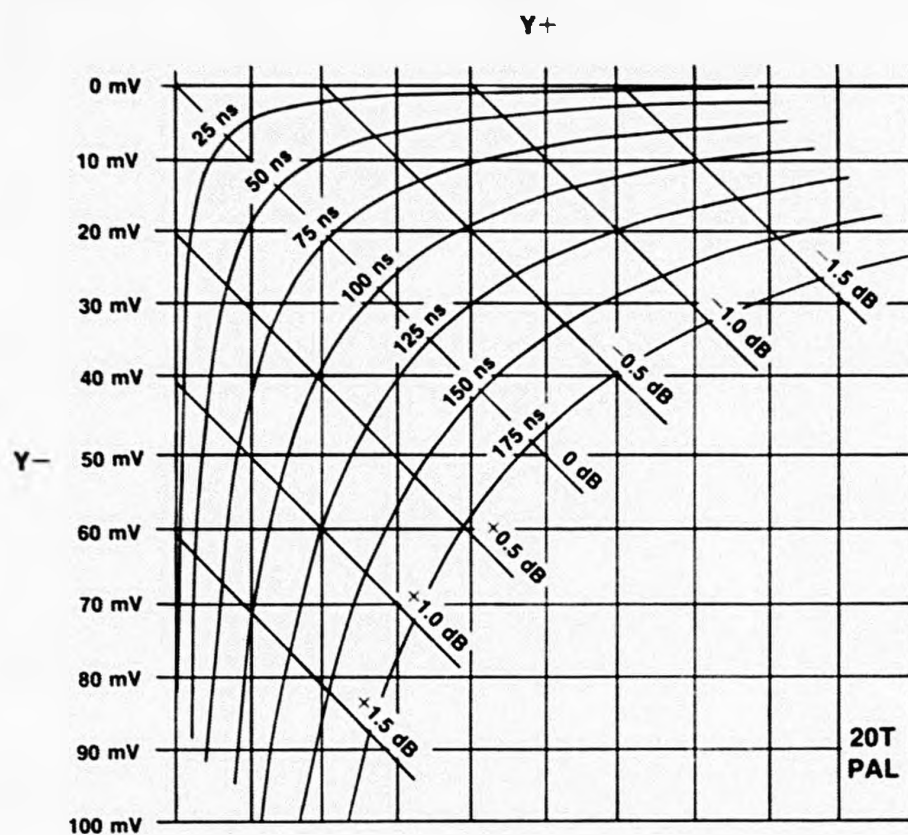
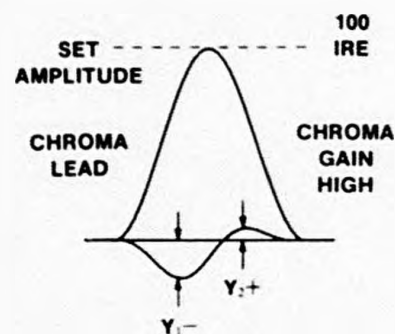
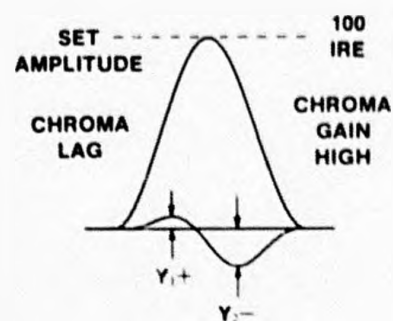
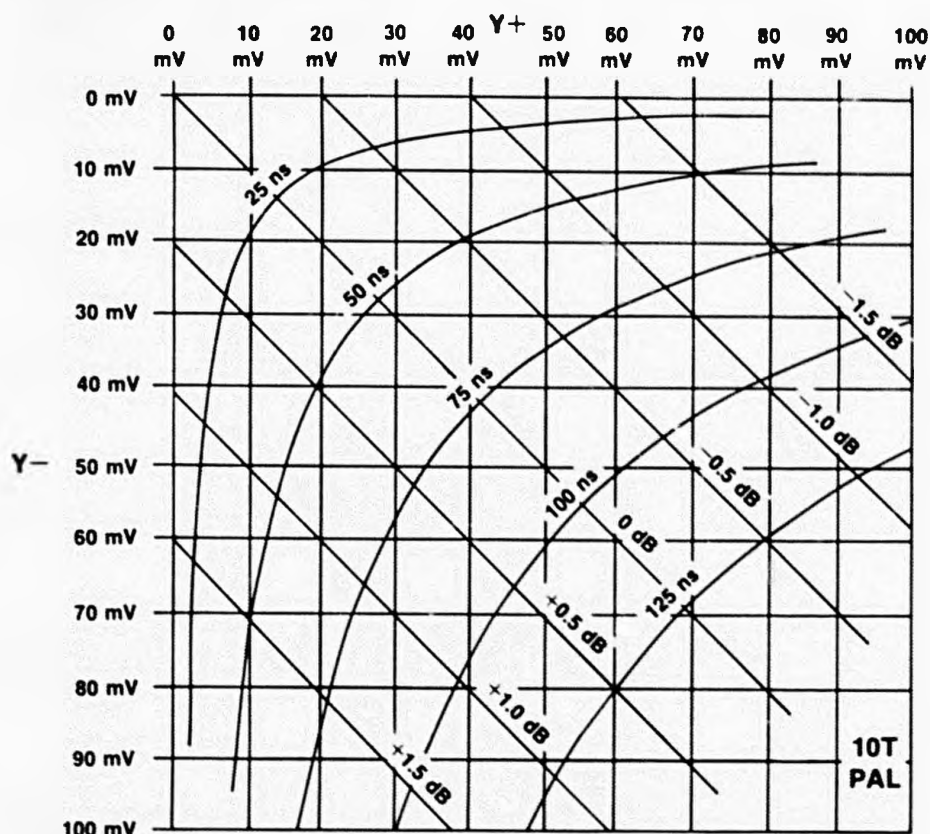
(d)



(e)

2479-13

Fig. 1-14. Chrominance/luminance error measurements. a. Shows the luminance component. b. Shows the chrominance component. c. Gain error. d. Chrominance/luminance delay error with chrominance leading. e. Chrominance/luminance delay error with chrominance lagging.



2479-14

Fig. 1-15. Nomograph for determining chrominance/luminance delay errors.

In some equipment, the chrominance channel gain and/or transient response may be a function of the subcarrier phase. Under those conditions, it is useful to have more than one subcarrier phase present in the modulated pulse and modulated bar. The TSG15 has an internal jumper option which permits the phase of the modulated pulse and modulated bar subcarrier to be phase modulated from 0 up to 360° at a field rate.

It is important to note that many chrominance- and/or luminance-channel problems may interfere with the precise measurement of chrominance/luminance delay. For example, poor transient response in the chrominance channel may result in trailing-edge ringing on the chrominance portion of the modulated pulse. The pulse may be asymmetrical so that its peak and median do not coincide.

The effects of nonlinear distortion can be observed by comparing half-amplitude response at high APL (pedestal on) or low APL (pedestal off) to full-amplitude response.

Nonlinear distortions may also interfere with the measurement of linear chrominance/luminance gain and delay. Differential gain, luminance nonlinearity, chrominance-amplitude nonlinearity, and quadrature distortion will cause different responses to the luminance and chrominance components of the modulated pulse.

Short-Time Distortion

High-frequency response deficiencies and fast thermal time-constant problems in video equipment can cause short-time distortions such as overshoot, undershoot, or ringing. See Fig. 1-16. The evaluation of combined frequency- and phase-response errors can be done with the pulse and/or bar.

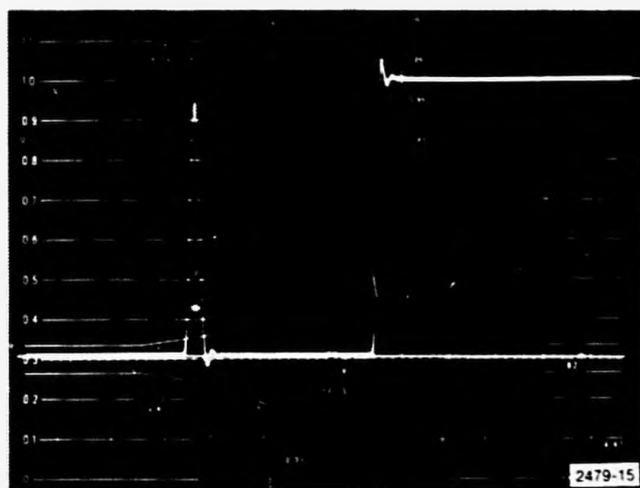


Fig. 1-16. Short-time distortion viewed on a 1480-Series Waveform Monitor.

Short-time distortions are normally taken to be from 100 ns to 1 μ s in duration. The CCIR K Factor Graticule, for the 1480-Series Waveform Monitor, provides a convenient means of relating various short-time distortions to their subjective picture impairments. The graticule indicates maximum deviations from the correct response for 2% and 5% K-Factors. See Fig. 1-17 for examples.

Distortion on fast pulse and bar signals, due to slew-rate limiting and other nonlinearities, may be evaluated by comparing full- and half-amplitude response.

Some equipment, such as video enhancers with coring, have a controlled change in transient response as a function of luminance level. This change may be observed by using the TSG15 to compare transient response to half amplitude pulse and bar with and without pedestal.

An alternate method of observing changes in transient response as a function of luminance level, is to use the reduced pedestal. Reduced pedestal is generated by the TSG15 whenever LUMINANCE, FULL AMPL, and PEDESTAL are selected simultaneously. Reduced pedestal is factory set to 0 V but may be set anywhere between that value and 175 mV by internal adjustment. This signal is useful for evaluating headroom, because the pulse and bar peaks extend well above 700 mV.

Line-Time Distortion

Equipment with ac-coupled circuits or thermal time-constant problems may exhibit line-time distortion. The luminance bar can be used to measure line-time distortions, as shown in Fig. 1-18. 2T, T, or T/2 bar can be selected as appropriate for the system under test.

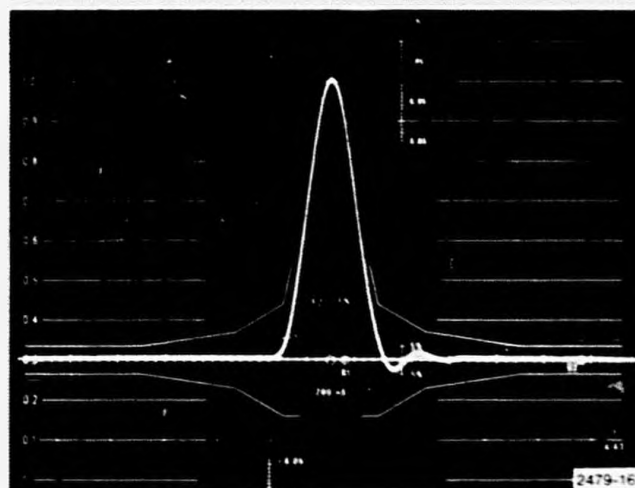


Fig. 1-17. K-factor measurement of 2T pulse.

Operating Instructions—TSG15

The first 1 μ s of the line should not be included in line-time distortion because it is considered in the measurement of short-time distortions.

Although not normally much of a problem, line-time distortions often occur in the chrominance channel. The modulated bar or chrominance bar can be used to measure these distortions.

Line-time distortions are normally measured with a 26 μ s bar, which is the factory-programmed bar for the TSG15. The use of a shorter bar is preferred in some cases; an internally-selectable option allows the TSG15 to generate 18 μ s bars.

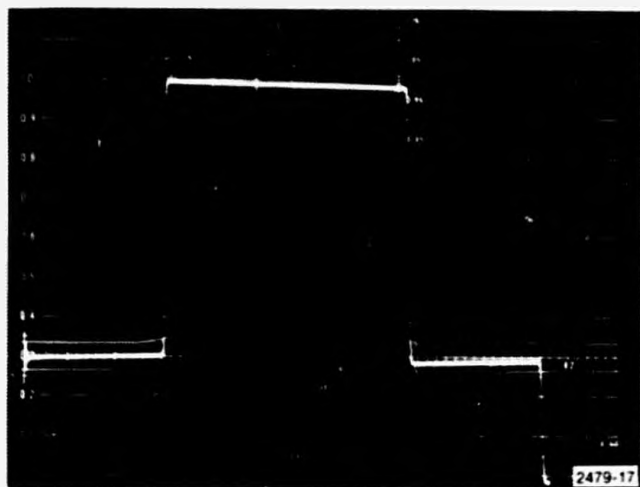


Fig. 1-18. Line-time distortion measured with the luminance bar.



Fig. 1-19. Field-time distortion measured with the TSG15 Field Squarewave.

Field-Time Distortion

Inadequate low-frequency response and clamping problems may cause field-time distortions which can be measured with a field square wave or window. See Fig. 1-19.

The field square wave consists of lines at 100% amplitude for 52 μ s during the white parts of the field square wave and 0% amplitude otherwise. The window consists of line bars, 26 (or 18) μ s wide, which are gated with field timing to generate a window. The field square wave, having a larger APL change than the window, provides increased sensitivity to field-time distortions. For this reason, the field square wave permits more critical evaluation of field-time distortion, or tilt. The window signal, however, permits a luminance pulse and modulated pulse to be included so that other measurements, in addition to field-time distortion, may be made with the same test signal. The window signal is also useful for checking picture monitor smearing.

In the TSG15, field square wave is generated with 250 ns risetime, full amplitude and no setup. The line bars used in window have front-panel selectable risetime, amplitude, and pedestal.

Pulse-to-Bar Ratio

High-frequency information in the video signal corresponds to fine horizontal detail in the television picture. Low-frequency information is related to overall brightness or to vertical detail. Pulse-to-bar ratio is an indication of the correct frequency- and phase-response at high frequencies, compared to low frequencies. Errors in either frequency response or phase response, at high frequencies, may result in incorrect pulse amplitude compared to bar amplitude.

The 2T pulse confines the pulse energy to 5 MHz bandwidth, and is well suited to transmitter testing. The 10 MHz bandwidth of the T pulse provides a more rigorous test of wideband video amplifiers, switchers, and video processors.

The T/2 pulse, with its 20 MHz bandwidth, provides a very critical test of wide bandwidth, high quality video equipment. The T/2 pulse can show subtle changes in frequency- or phase-response which might not be obvious with slower pulses, but which might degrade the video signal when cascaded. For example, slight errors at each point in a distribution system can result in a large combined error.

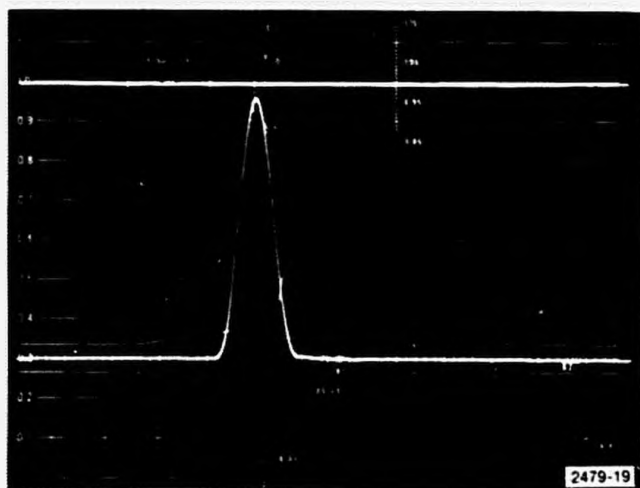


Fig. 1-20. Measuring pulse-to-bar ratio with the overlay signal.

To facilitate pulse-to-bar ratio measurements, the TSG15 includes an overlay mode. In the overlay mode, the timing of pulse and bar is altered so that reference bar amplitude is displayed overlaying the luminance and/or modulated pulses. See Fig. 1-20.

Bar amplitude is defined at the center of the bar, equidistant from the leading and trailing edge. In the presence of line tilt, the base of the pulse may be referenced to the wrong level due to the baseline tilt. This should be taken into account when measuring pulse-to-bar ratio in the presence of significant line tilt, regardless of whether the TSG15 or 1480-Series overlay feature is used. Unlike some other overlay methods, the TSG15 can maintain a constant APL from line to line even in the overlay operating mode.

NOTE

Pulse-to-bar ratio for each pulse has been factory-set on a 100 MHz bandwidth oscilloscope. Waveform Monitors, such as the 1481, are designed to be extremely flat over a 5 MHz bandwidth, but roll off outside that bandwidth. As a result, the pulse-to-bar ratio and pulse transient-response of any waveform monitor may not be adequate for use with T and T/2 pulses. Critical evaluation of wideband video equipment should be done with a wideband oscilloscope, such as a TEKTRONIX 7603 and 7A26 or a waveform monitor modified to obtain the required bandwidth and phase accuracy.

Quadrature Distortion

A vestigial sideband transmission received by an envelope detector results in a distortion known as quadrature distortion. When quadrature distortion is present, high-frequency information experiences a non-linear level shift. The result is a widening of black vertical lines and narrowing of white vertical lines.

The TSG15 includes a positive-going 2T pulse and an inverted 2T pulse in a T bar. These two pulses can be used to measure quadrature distortion by comparing the half-amplitude duration (HAD) of the two pulses. The pulses may be set to full amplitude for maximum sensitivity to quadrature distortion, or may be set to half amplitude with a high APL (pedestal on) or low APL (pedestal off). See Fig. 1-21.

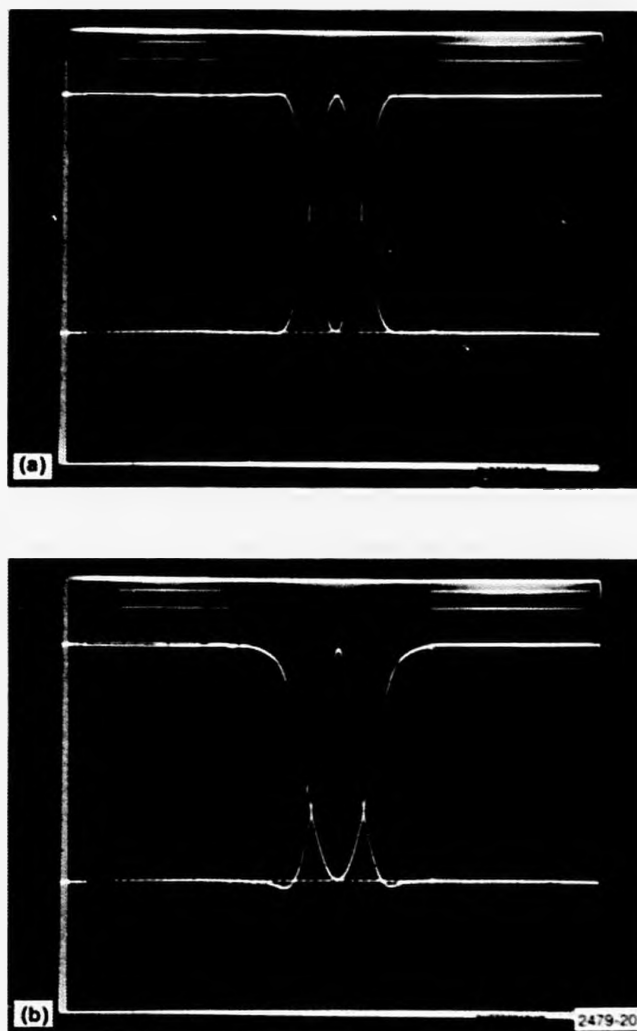


Fig. 1-21. Comparison of the two 2T Pulse HADs. a. Synchronous detector. b. Envelope detector.

The chrominance pulse and chrominance bar on a pedestal may also be used to see quadrature distortion causing chrominance/luminance intermodulation. See Fig. 1-22.

As mentioned in the discussion of chrominance/luminance gain errors, quadrature distortion may interfere with the relative chrominance-gain measurement. The nonlinear shift of the luminance component of the modulated pulse and modulated bar results in an incorrect reference for the chrominance amplitude. See Fig. 1-23.

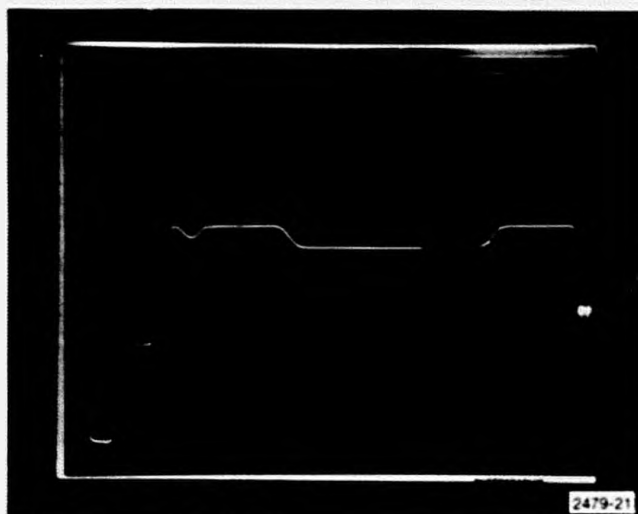


Fig. 1-22. Chrominance/luminance intermodulation as a result of quadrature distortion.

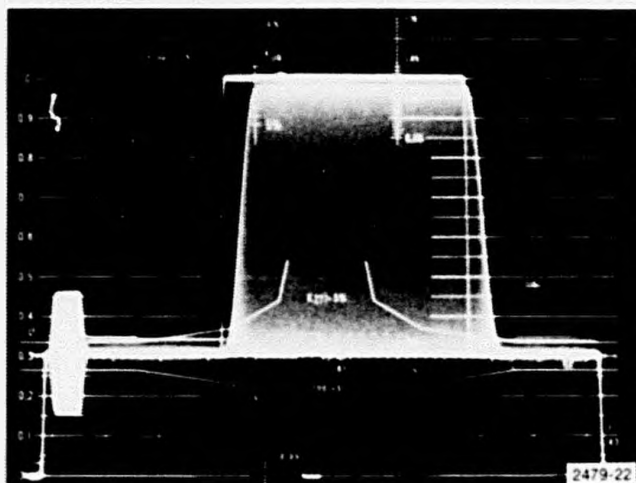


Fig. 1-23. Nonlinear shift of the luminance component of the modulated bar caused by quadrature distortion.

Envelope detectors inherently display quadrature distortion. Therefore, a good synchronous detector should be used to determine transmitter chrominance/luminance gain error.

In a vestigial sideband transmission system, the sum of the sideband vectors is not necessarily in phase with the carrier vector. In an AM transmission with both sidebands present, the sum of the sidebands will be in phase with the carrier. See Fig. 1-24.

In Fig. 1-25, the dashed lines indicate the result of envelope detection. The dotted lines indicate the result of proper synchronous demodulation. Note that the quadrature components in vestigial sideband cause the envelope detector to give too large an output at 50% modulation. This corresponds to grays being compressed toward black on high-frequency (horizontal detail) signals.

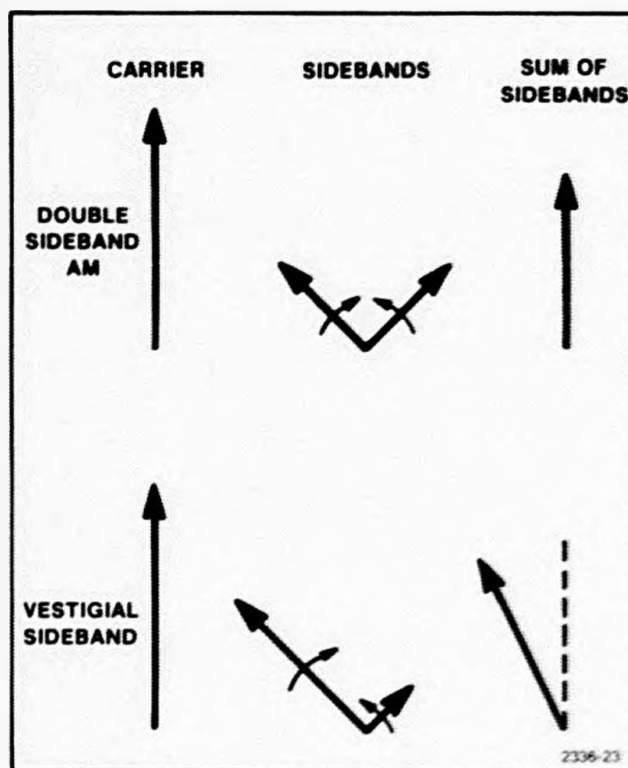


Fig. 1-24. Vector addition of sidebands. a. Double sideband AM. b. Vestigial sideband.

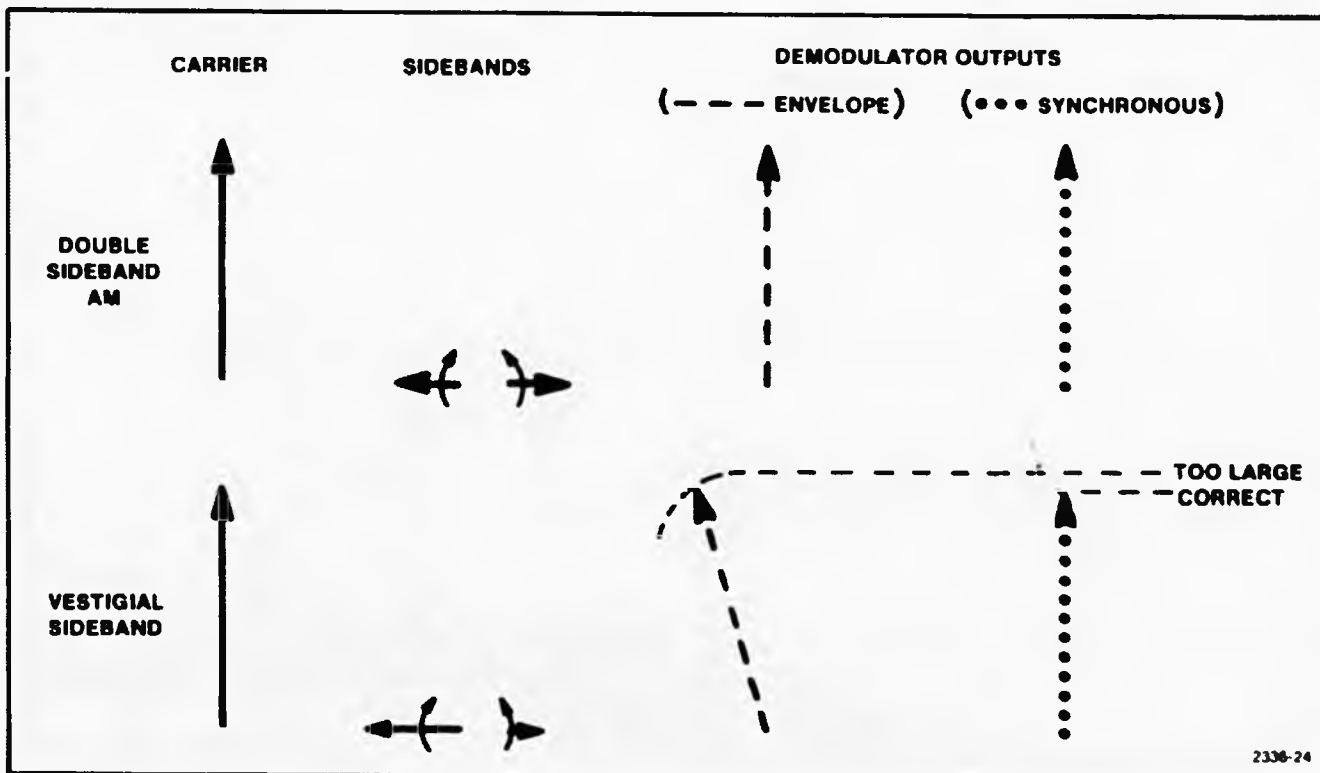


Fig. 1-25. Vector results of envelope and synchronous detection of double sideband am and vestigial sideband transmissions.

Insertion Test Signals (ITS)

Any of the TSG15 pulse and bar test-signals may be enabled on desired lines of the vertical interval. The ITS components are selected from the TSG15 front panel and the ITS line determined by the ITS key signal. The ITS key, to generate IT Signal(s) on selected line(s) can come from any one of three sources:

1. Remotely through the 1411, J41.
2. Internal Black Burst.
3. A Test Signal Switcher.

The ITS will always be the pulse and bar signal selected from the front panel. Window and Field Square Wave are full field test signals and not available as ITS. The overlay can be a part of the ITS, if selected by an internal jumper. Normal factory-set position has the overlay disabled.

See the 1411 Instruction Manual for more information on the ITS enable.

Other User Options

Sync Disable. Sync may be deleted through 1411 J41 connector.

SPECIFICATION

ELECTRICAL CHARACTERISTICS

The following electrical performance requirements are valid only if the instrument has been calibrated at an ambient temperature between $+20^{\circ}\text{C}$ and $+30^{\circ}\text{C}$, and the instrument is operating at an ambient temperature between 0°C and $+50^{\circ}\text{C}$. The instrument must have a warm-up period of at least 20 minutes before checking specification.

Characteristic Instants

Signals shown in Fig. 2-1 are defined in characteristic instants. A characteristic instant equals approximately $2\ \mu\text{s}$ (or $64\ \mu\text{s}/32$). T_{m} is at the leading edge of sync.

Characteristics

Characteristics are the properties of the TSG15.

Performance Requirement

Items listed in the Performance Requirement column of the Electrical Characteristics are quantitative terms of performance, usually stated in terms of tolerance limits, and are guaranteed by Tektronix, Inc. Items in this column that are verified in the Performance Check Procedure,

which follows this Specification, are indicated by a Performance Check Step number. Items not verified in the Performance Check Procedure may be one or more of the following:

- Parameters that are dependent upon other 1411 system components (i.e., an SPG unit), where they are verified in the appropriate Instruction Manual.
- Parameters that are fixed by the nature of their function, such as range settings of switches, etc.
- Characteristics that are digitally derived from a timing reference that is checked (for example, from a crystal-controlled oscillator whose frequency is checked).
- Characteristics of a very stable nature that would necessitate an exhaustive procedure and require test equipment that is not commonly available in order to perform a check.

Supplemental Information

Items listed in the Supplemental Information column are not verified in this manual; they are either explanatory notes or characteristics for which no limits are specified.

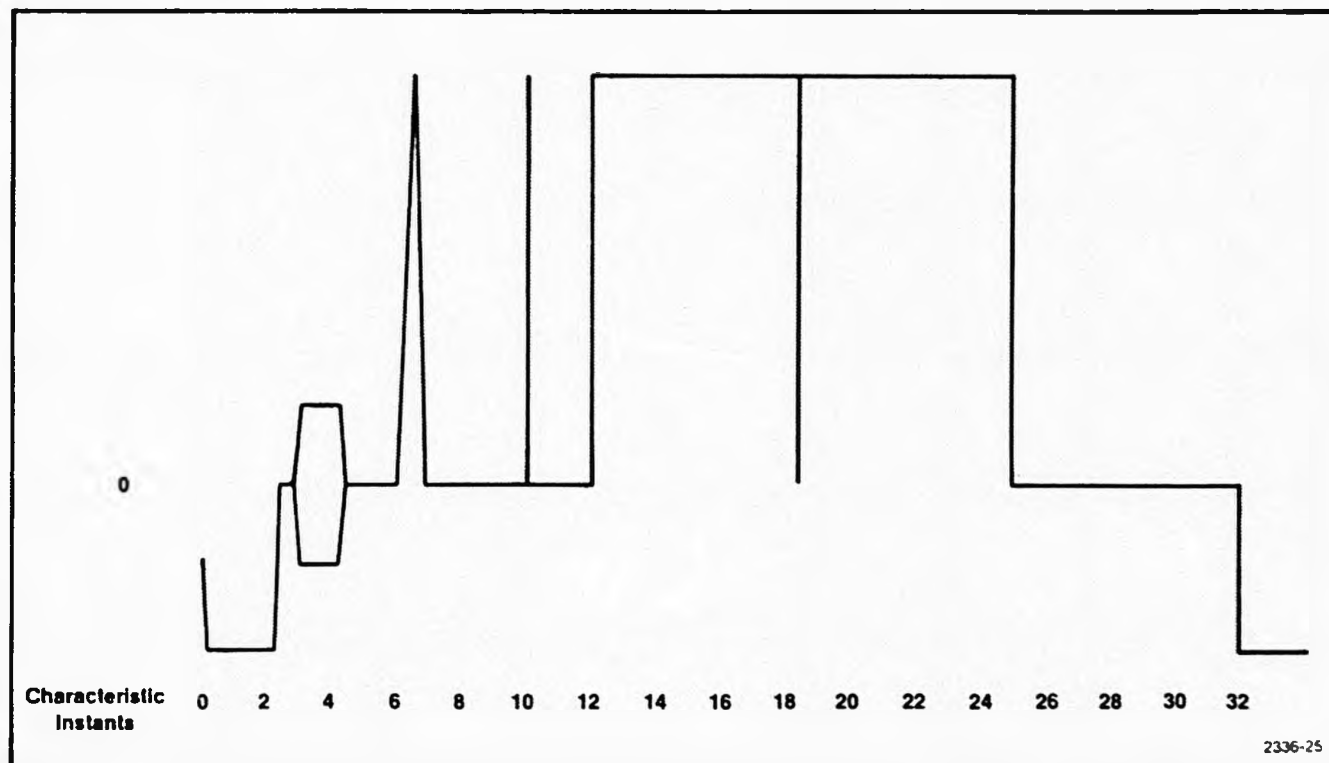


Fig. 2-1. Characteristic instants plotted against the TSG15 Pulse and Bar signal with inverted 2T pulse.

Table 2-1
ELECTRICAL SPECIFICATION

Characteristic	Performance Requirement		Supplemental Information	Check Step
Luminance Bar Amplitude	Full Ampl. 700 mV $\pm$ 7.0 mV	Half Ampl. 350 mV $\pm$ 5.3 mV		5
Modulated Bar Amplitude	Full Ampl. 350 mV $\pm$ 3.5 mV	Half Ampl. 175 mV $\pm$ 1.8 mV		5
Luminance				
Chrominance peak-to-peak	700 mV $\pm$ 14.0 mV	350 mV $\pm$ 7.0 mV		5
Pulse Amplitude	Full Ampl.	Half Ampl.		
2T	700 mV $\pm$ 7.0 mV	350 mV $\pm$ 5.3 mV		4
2T Inverted	700 mV $\pm$ 10.5 mV	350 mV $\pm$ 7.0 mV		4
T	700 mV $\pm$ 10.5 mV	350 mV $\pm$ 7.0 mV		4
T/2	700 mV $\pm$ 14.0 mV	350 mV $\pm$ 14.0 mV		4
Modulated	700 mV $\pm$ 7.0 mV	350 mV $\pm$ 5.3 mV		4
Pulse to Bar Ratio	Full Ampl.	Half Ampl.		
2T	1:1 $\pm$ 0.005	1:1 $\pm$ 0.01		7
2T Inverted	1:1 $\pm$ 0.01	1:1 $\pm$ 0.02		7
T	1:1 $\pm$ 0.01	1:1 $\pm$ 0.02		7
T/2	1:1 $\pm$ 0.02	1:1 $\pm$ 0.04		7
Modulated Pulse to Luminance Bar	Full Ampl. 1:1 $\pm$ 0.005	Half Ampl. 1:1 $\pm$ 0.01		7
Modulated Pulse to Modulated Bar	Full Ampl. 1:1 $\pm$ 0.01	Half Ampl. 1:1 $\pm$ 0.02		7
Modulated Pulse & Bar Chrominance/Luminance Gain	3.5 mV Max.		Peak luminance to peak chrominance	14
Delay	5.0 ns Max.		Chrominance/luminance gain and delay can be set for less than 3.5 mV combined baseline perturbation with a 1480-Series Waveform Monitor.	14
Residual Subcarrier	2.0 mV Max.			3

Table 2-1 (cont)

Characteristic	Performance Requirement		Supplemental Information	Check Step
Other Baseline Errors	3.5 mV Max.			14
Subcarrier Phase Factory Set	Fixed Phase 60° typical.			
Internal Option	Phase Mod. 360° at field rate.			
Subcarrier Harmonics	>40 dB down.			
Luminance Pulse Timing	Normal	Inverted		
Normal Start ^a	10t	18.5t (16.5t) ^h		
Overlay Start ^a	22.5t (18.5t)	10t ^h		
Detail	HAD	Overshoot		
2T	200 ns ± 5%	0.5%		12
T	100 ns ± 5%	1.0%		12
T/2	50 ns ± 10%	2.0%		12
Modulated Pulse Timing				
Normal Start	6t			
Overlay Start	25t (19t) ^h			
Detail	HAD	Overshoot		
10T	1000 ns ± 5%	1.0%		12
20T	2000 ns ± 5%	1.0%		12
Bar Timing				
Normal	12t to 25t (12t to 21t) ^h			
Overlay	5.5t to 18.5t (5.5t to 14.5t) ^h			
Detail	Risetime	Overshoot		
2T	192.9 ns ± 10%	0.5 %		8
T	96.4 ns ± 10%	1.0%		8
T/2	48.2 ns ± 15%	2.0%		8
10T	964 ns ± 10%	1.0%	Modulated Bar Envelope	8
20T	1929 ns ± 10%	1.0%	Modulated Bar Envelope	8

^aTiming given in characteristic instants; 32 instants equal one full line, including sync. T ≈ 2 μs (or 64 μs/32). T is the leading edge of sync.

^hTiming for use with short bar shown in parentheses.

Table 2-1 (cont)

Characteristic	Performance Requirement	Supplemental Information	Check Step
Line Tilt	0.5% Max.		16
Window Composition		2T, 1, T/2, or Modulated Bars	
Line Timing	Bar Timing		
Field Timing		White lines from 65 to 270 each field.	
Field Tilt	0.5% Max		15
Field Square Wave Amplitude	700 mV $\pm$ 14.0 mV	Fixed Full Amplitude	1
Line Timing		From 1411 Horizontal Blanking	
Field Timing		50% duty cycle. White lines from 65 to 221 each field.	
Line Tilt	0.5% Max.		16
Field Tilt	0.5% Max.		15
Pedestal Amplitude			
Luminance Off or Half Ampl.	350 mV $\pm$ 7.0 mV		6
Luminance on or Full Ampl. Factory Set	0	Pedestal Off	
Internal Option	0 to 175 mV	Adjustable	
Timing		From 1411 Horizontal & Vertical Blanking	
Risetime	250 ns. $\pm$ 50 ns		11
Blanking DC Level	0 $\pm$ 50 mV		2
Timing		From 1411 Horizontal & Vertical Blanking	
Front Porch	1.4 μ s min.		
Sync Amplitude	300 mV $\pm$ 6.0 mV		6
Timing		From 1411 Composite Sync	

Table 2-1 (cont)

Characteristic	Performance Requirement	Supplemental Information	Check Step
Risetime	250 ns, ± 50 ns		9
HAD	4.8 μ s $\pm$ 0.01 μ s		9
Burst Amplitude	300 mV $\pm$ 9.0 mV (peak-to-peak)		13
Phase Angle between Mean of Bursts	Bursts $90^\circ \pm 1^\circ$ Adjustable $\pm 10^\circ$	Bruch Sequence From 1411 Subcarrier Reference	
Timing		From 1411 TSG Burst Flag.	
Risetime	350 ns $\pm$ 50 ns		10
HAD	2.25 μ s $\pm$ 0.17 μ s	9 subcarrier cycles	10
Delay	5.6 μ s $\pm$ 0.1 μ s	From leading edge of sync. (25 subcarrier cycles)	
Breezeway	900 ns	Typical	
Output Impedance	75 ohms nominal.		
Return Loss	30 dB or more to 5 MHz.		17
Isolation	40 dB or more.	Calibration Procedure	

PERFORMANCE CHECK PROCEDURE

The procedure in this section serves as a guide to check the instrument performance. It will check the key Performance Requirements listed in the Electrical Characteristics that immediately precede it.

Limits and tolerances appearing in this procedure are not Performance Requirements unless listed, as such, in the preceding Specification Table (Table 2-1). The TSG15 front-panel control names are fully capitalized; for example, FIELD SQ WAVE. Control and connector names on test equipment have only the first letter capitalized; for example, test oscilloscope Vertical Position.

Test Equipment

Test equipment capabilities described here are the minimum required to check or calibrate the instrument. Examples represent test equipment used in developing this procedure.

1. Waveform Monitor. Capable of viewing line-rate and field-rate signals. For example, A TEKTRONIX 1481 Waveform Monitor.

2. Test Oscilloscope. Bandwidth dc to 100 MHz; minimum deflection, 1 mV/div; two input channels with provisions for independent or differential operation. For example, a TEKTRONIX 7603 oscilloscope with 7A26, (used to evaluate pulse shaping), 7A13, and 7B53A plug-in units. A 7L12 may be required to check subcarrier harmonics. Its use is optional and no directions appear here.

3. Sinewave Signal Generator. Minimum output, 500 mV with a frequency range of 50 kHz, fixed, variable from 1 MHz to 5 MHz. For example, a TEKTRONIX SG 503 in a Tektronix TM 500-Series mainframe.

4. Return Loss Bridge. Tektronix part number, 015-0149-00.

5. Vectorscope. A TEKTRONIX 521A PAL Vectorscope.

6. Variable Amplitude Calibrator. Calibration fixture, see Fig. 2-2.

7. 50- to 75-ohm Minimum Loss Attenuator. Tektronix part number, 011-0057-00.

8. 75-ohm Cable, 42 inches long, 6 each, Tektronix part number, 012-0074-00.

9. 75-ohm End-Line Termination, 3 each, Tektronix part number 011-0102-00.

10. 75-ohm In-Line Termination. Tektronix part number, 011-0103-00.

11. 75-ohm 10X Attenuator. Tektronix part number, 011-0061-00.

Calibration Fixture

The Variable Amplitude Calibrator and the 1481 provide a source of calibration levels that is variable. Voltages can be read directly from the 10-turn dial. A schematic diagram and parts list for this fixture are shown in Fig. 2-2. Whenever the switch labeled S1 is in the Fixed position, the 1481 internal calibrator works as normal.

With P9034 connected to J9034 on the 1481 rear panel, and the switch in the Fixed position, calibrate the 1481 for a 1-V sync tip-to-peak white signal equivalent. Then set the dial to 1000, change the switch to Variable, and adjust R3 for the same amplitude displayed when the switch was in the Fixed position. This calibrates the dial for a range of 100 mV/360° rotation. Set dial to 0 and check that 0 = 0 V.

Measurements

The signal to be measured must be fed into the 1481 A Input, and the Operate, Calibrate, and Sync Tip buttons pushed. To check an amplitude within a given tolerance, adjust the Amplitude dial while watching the waveform monitor display and when the level being measured overlays the blanking level, read the amplitude directly from the dial.

PROCEDURE

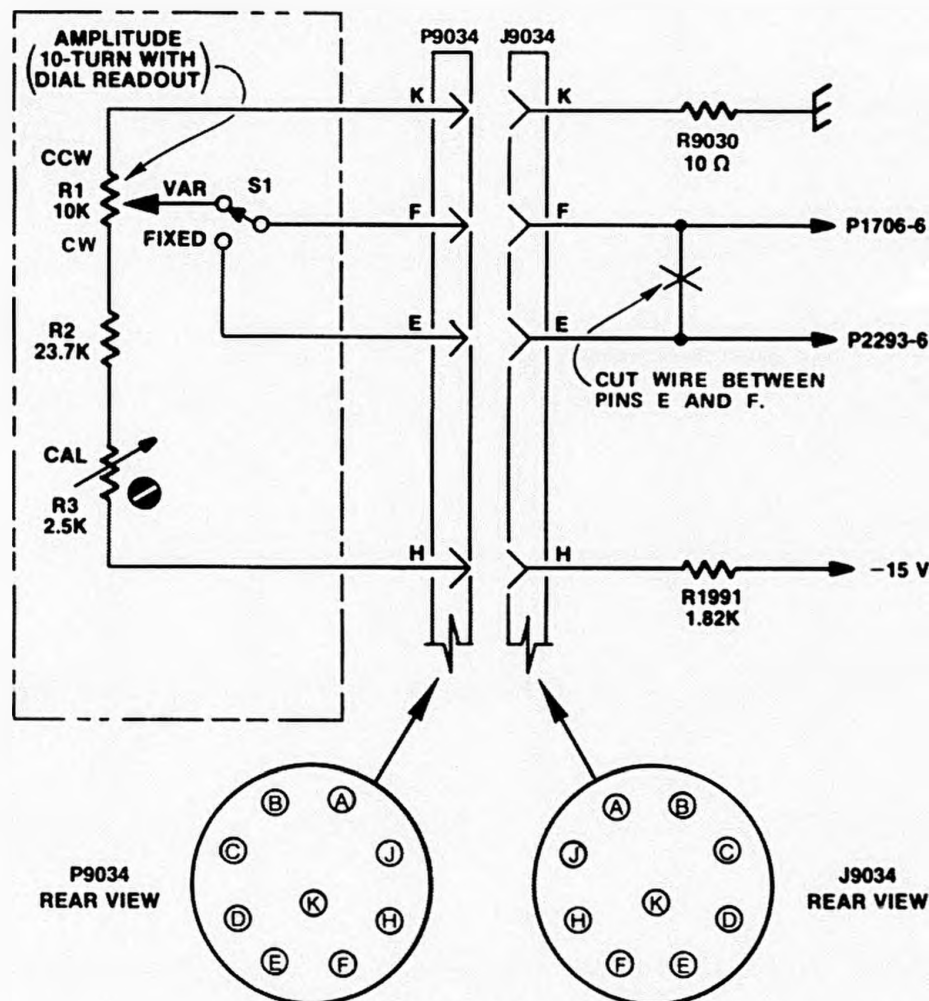
Preliminary Connection

Connect the output of the TSG15 through its appropriate 1411 rear-panel Module Output, and a 75-ohm coaxial cable to the A Input of the 1481 Waveform Monitor. Terminate the remaining side of the 1481 A Input loop-through connector with a 75-ohm termination.

Connect the Variable Amplitude Calibrator to J9034 on the 1481 rear panel.

CALIBRATION FIXTURE

1480-SERIES REAR PANEL



Parts List

Ckt No.	Description	Tektronix Part No.
P9034	9-pin Amphenol #165-13	134-0049-00
R1	10 k Ω $\pm$ 5%, Ind. Lin. $\pm$ 0.1%, 10-turn, prec. var.	311-1729-00
R2	23.7 k Ω $\pm$ 1%, 1/8 W, metal film	321-0325-00
R3	2.5 k Ω $\pm$ 20%, variable	311-0086-00
S1	SPDT toggle switch 10-turn dial for R1, Kilo-dial Mod. 461-S-41	200-0613-00 331-0139-00
Misc.	Approx. 3 ft. of 4-conductor cable and a small metal or plastic enclosure or case.	

Set R1 fully counterclockwise (ccw) and the dial at 0 when installing the dial.

2329-04

Fig. 2-2. Variable Amplitude Calibrator.

Procedure

1. Check Luminance Amplitude

a. Select the TSG15 FIELD SQ WAVE.

b. Check for a peak luminance level of 700 mV $\pm$ 7.0 mV with the Variable Amplitude Calibrator. See Fig 2-3.

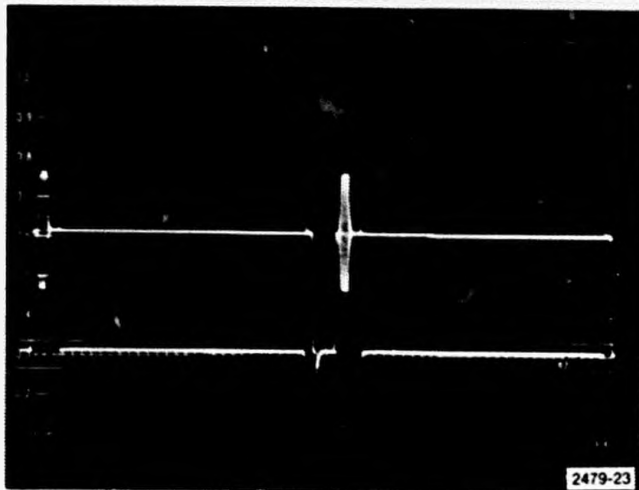


Fig. 2-3. Measuring luminance or bar amplitudes.

2. Check Dc Level

a. Connect the TSG15 Module Output to the test oscilloscope.

b. Set the test oscilloscope for a 0 V dc level.

c. Check that the blanking level is at 0 V $\pm$ 50 mV.

3. Check Residual Subcarrier

a. With the blanking level still at graticule center, test oscilloscope vertical bandwidth at 5 MHz, check that residual subcarrier (trace width) is less than 2.0 mV.

4. Check Pulse Amplitude

a. Select 2T, PULSE.

b. Check for a pulse amplitude of 700 mV $\pm$ 7.0 mV with the Variable Amplitude Calibrator.

c. Using the waveform monitor and the Variable Amplitude Calibrator check the amplitude of each pulse according to Table 2-2.

Table 2-2
PULSE AMPLITUDE

Pulse	Amplitude	Check for
2T	Full	700 mV $\pm$ 7.0 mV
2T	Half	350 mV $\pm$ 5.3 mV
2T Inverted	Half	350 mV $\pm$ 7.0 mV
2T Inverted	Full	700 mV $\pm$ 10.5 mV
T	Full	700 mV $\pm$ 10.5 mV
T	Half	350 mV $\pm$ 7.0 mV
T/2	Half	350 mV $\pm$ 14.0 mV
T/2	Full	700 mV $\pm$ 14.0 mV
MOD	Full	700 mV $\pm$ 7.0 mV
MOD	Half	350 mV $\pm$ 5.3 mV

5. Check Bar Amplitude

a. Select HALF AMPL and MOD BAR.

b. Release the LUMINANCE pushbutton.

c. Check for P-P amplitude of 350 mV $\pm$ 7.0 mV (from blanking to bar top) using the Variable Amplitude Calibrator. (Bar amplitude is defined at a point equidistant from the bars leading and trailing edges.)

d. Check bar amplitudes, using the Variable Amplitude Calibrator, according to Table 2-3 and Fig. 2-3.

Table 2-3
BAR AMPLITUDES

Bar	Amplitude	Check for
MOD (chrominance)	Half	350 mV $\pm$ 7.0 mV
MOD (chrominance)	Full	700 mV $\pm$ 14.0 mV
Turn on LUMINANCE	Set Waveform Monitor to Lowpass	
MOD (luminance)	Full	350 mV $\pm$ 3.5 mV
MOD (luminance)	Half	175 mV $\pm$ 1.8 mV
2T	Half	350 mV $\pm$ 3.5 mV
2T	Full	700 mV $\pm$ 7.0 mV
T	Full	700 mV $\pm$ 7.0 mV
T	Half	350 mV $\pm$ 3.5 mV
T/2	Half	350 mV $\pm$ 3.5 mV
T/2	Full	700 mV $\pm$ 7.0 mV

6. Check Sync and Pedestal Amplitude

- All TSG15 buttons except PEDESTAL should be out.
- Use the Variable Amplitude Calibrator to check sync amplitude, $300 \text{ mV} \pm 6.0 \text{ mV}$.
- Select PEDESTAL.
- Measure pedestal amplitude, $350 \text{ mV} \pm 7.0 \text{ mV}$.

7. Check Pulse to Bar Ratio

- Set the TSG15 for FULL AMPL, LUMNANCE, 2T PULSE and T BAR.
- Use the 0.2 Volts Full Scale on the 1481 and check the difference in amplitude between the pulse and bar according to Table 2-4. Check at both FULL and HALF AMPL.

Table 2-4
PULSE/BAR RATIO

Pulse	Bar	Full Amplitude Max Difference	Half Amplitude Max Difference
2T	T	3.5 mV	3.5 mV
Invert 2T	T	7.0 mV	7.0 mV
T	T	7.0 mV	7.0 mV
T/2	T	14.0 mV	14.0 mV
MOD	T	3.5 mV	3.5 mV
MOD	MOD	7.0 mV	7.0 mV

- Release the MOD PULSE pushbutton by pushing all the way in and releasing.

- While holding in the T BAR pushbutton, push the other BAR pushbuttons one at a time and check that the amplitudes of the other bars are equal to that of the T bar. See Fig. 2-4. (This uses the TSG15 Bar Calibrate Mode, and assumes that the enabling internal jumper is in the factory-set position.)

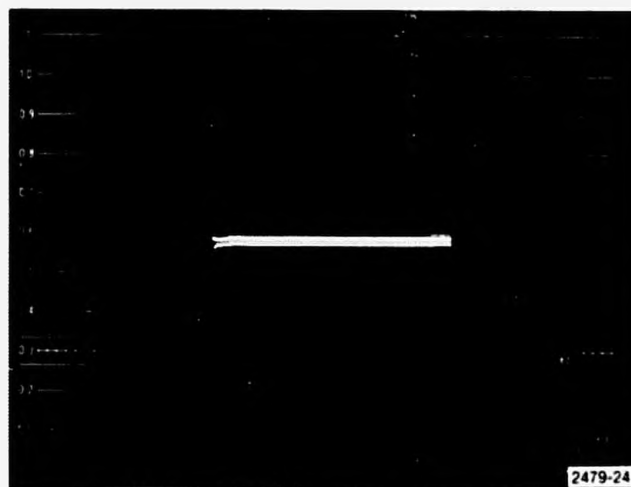


Fig. 2-4. T bar and 2T bars overlaid to compare bar amplitude (factory programmed option, activated by selecting two bars simultaneously).

8. Check Bar Risetimes and Overshoot

NOTE

For the timebase accuracies required in this measurement it will be necessary to verify the test oscilloscope timebase accuracy.

- Connect the TSG15 Module Output to the test oscilloscope, using an In-Line Termination at the input.
- Check overshoot as shown in Fig. 2-5a and listed in Table 2-5 for each bar.
- Check the risetime of each bar listed in Table 2-5 as shown in Fig. 2-5b.

Table 2-5
BAR RISETIME & OVERSHOOT

Bar	Max Overshoot	Risetime
2T	3.50 mV	192.9 ns $\pm$ 19.3 ns
T	7.0 mV	96.4 ns $\pm$ 9.6 ns
T/2	14.0 mV	48.2 ns $\pm$ 7.2 ns
20T	7.0 mV	1929 ns $\pm$ 192.9 ns

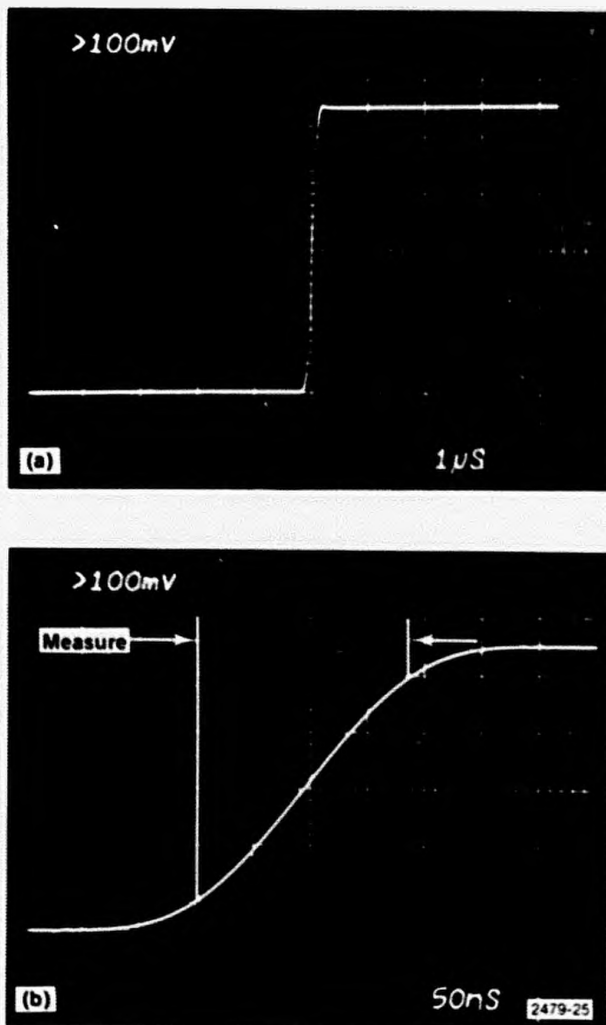


Fig. 2-5. Checking bar. a. Overshoot. b. Risetime.

9. Check Sync Risetime

- Set the test oscilloscope to measure sync pulse risetime.
- Check sync pulse risetime for $250\text{ ns} \pm 50\text{ ns}$. See Fig. 2-5b.

10. Check Burst Risetime

- Select BURST.
- Measure burst risetime, $350\text{ ns} \pm 50\text{ ns}$. See Fig. 2-5b.

11. Check Pedestal Risetimes

- All TSG15 pushbuttons, except PEDESTAL should be out.
- Measure pedestal risetime, see Fig. 2-5b, $250\text{ ns} \pm 50\text{ ns}$.

12. Check Pulse Overshoot and HAD

NOTE

For the timebase accuracies required in this measurement it will be necessary to verify the test oscilloscope timebase accuracy.

- Select LUMINANCE, FULL AMPL. and 2T PULSE.
- Set the test oscilloscope Vertical Bandwidth (7A26) to 20 MHz .
- Check 2T pulse for overshoot, see Fig. 2-6.
- Set the pulse mid-point (half amplitude) to the graticule center and measure the duration of the pulse see Fig. 2-6.
- Use Table 2-6 and check overshoot and half-amplitude duration for each pulse as outlined in steps c and d.

Table 2-6
PULSE OVERSHOOT & HAD

Pulse	Maximum Overshoot	HAD
2T	3.50 mV	$200\text{ ns} \pm 10.0\text{ ns}$
Inverted 2T	3.50 mV	$200\text{ ns} \pm 10.0\text{ ns}$
T	7.00 mV	$100\text{ ns} \pm 5.0\text{ ns}$
T/2	14.00 mV	$50\text{ ns} \pm 5.0\text{ ns}$
20T	7.00 mV	$2000\text{ ns} \pm 100.0\text{ ns}$

13. Check Burst Amplitude

- Reconnect the TSG15 Module Output to the 1481, A Input.
- Select BURST.
- Check for total burst amplitude of $300\text{ mV} \pm 9.0\text{ mV}$ with the Variable Amplitude Calibrator.

14. Check Chrominance/Luminance Gain and Delay Errors

- Set the TSG15 for a 20T MOD PULSE, SETUP OFF.
- Set the waveform monitor for 1 μ s/Div and 0.2 Volts Full Scale.
- Check the amplitude of the baseline error beneath the pulse. Maximum error due to chrominance/luminance gain and delay errors is 3.5 mV, other baseline errors 3.5 mV or less. Fig. 2-7 shows chrominance/luminance errors, a. gain, b. delay.

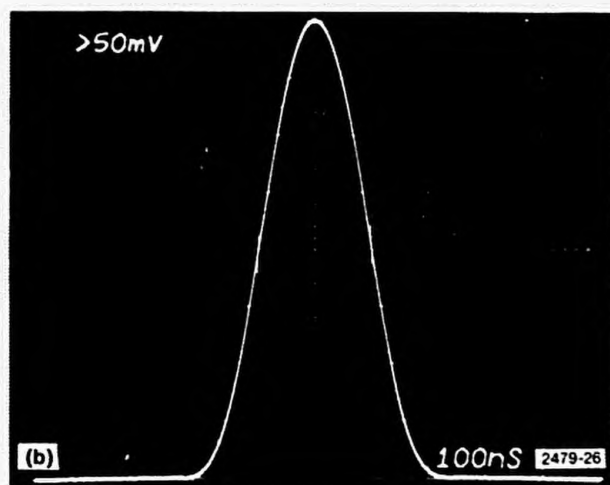
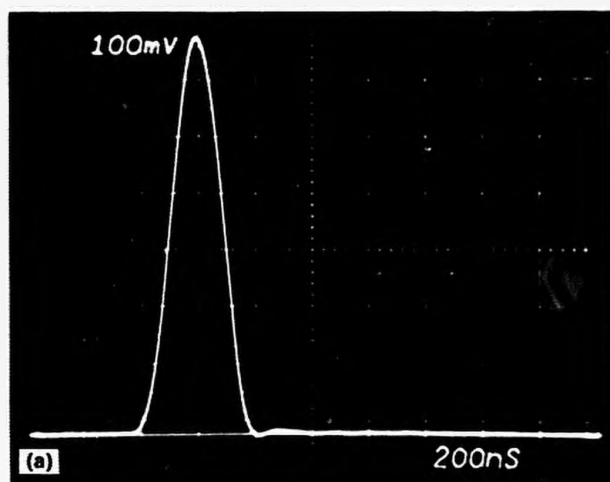


Fig. 2-6. Checking 2T pulse. a. Overshoot. b. HAD.

15. Check Field Tilt

- Select LUMINANCE, FULL AMPL, PEDESTAL OFF, and WINDOW.
- Set the 1481 to display a 2 Field Signal.
- Check for 0.5% (3.5 mV) or less tilt to the top of the bar.
- Select the FIELD SQ WAVE.
- Check for 0.5% (3.5 mV) or less tilt to the top of the bar.

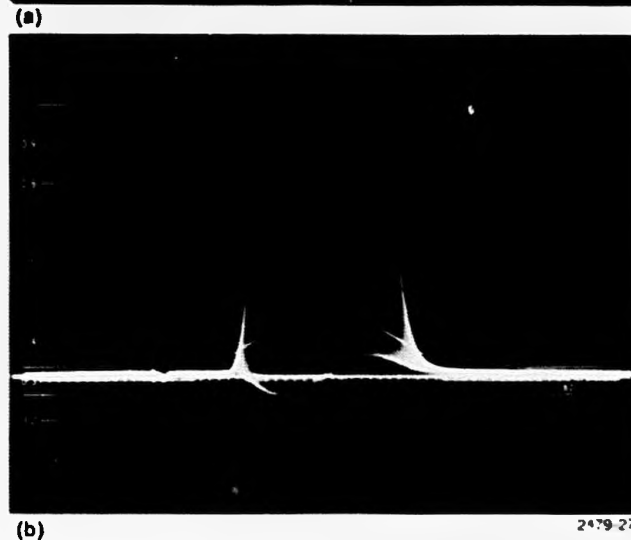
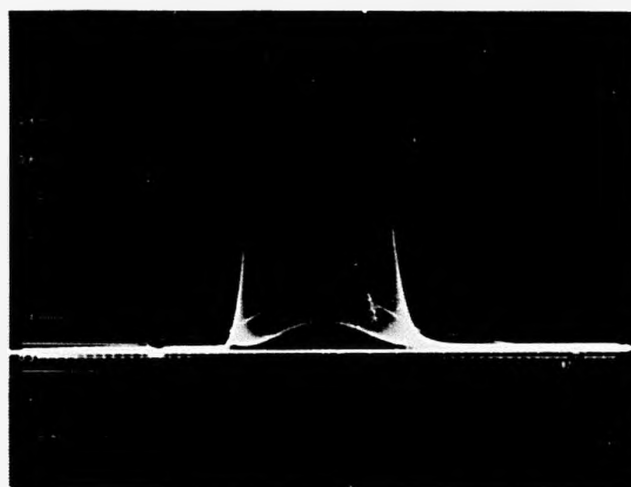


Fig. 2-7. Checking chrominance/luminance errors. a. Gain. b. Delay.

Specification—TSG15

16. Check Line Tilt

- a. Use 2T BAR
- b. Set the 1481 to 5 μ s Div sweep rate
- c. Check for 0.5% (3.5 mV) or less tilt to the top of the T, 2T, and T/2 bars

17. Check Return Loss

NOTE

The instruction manual supplied with the Return Loss Bridge gives detailed information on making return loss measurements for those that are unfamiliar with the technique. The following procedure should give sufficient information for those who are familiar with return loss measurements.

- a. Connect the Return Loss Bridge to the test oscilloscope vertical inputs. Set the vertical mode to added
- b. Connect the leveled sine wave generator output to the Return Loss Bridge. Set the generator amplitude for a 250 mV output signal
- c. Balance the bridge with both terminations in place
- d. Connect the unknown arm of the bridge without termination to the Module Output on the 1411 rear panel
- e. CHECK for return loss of 30 dB or more from 50 kHz to 5 MHz

Checking isolation requires the removal of the 1411's top cover and this check should not be performed at this time. The isolation check is included as a part of the Adjustment Procedure found in Section 4, Calibration.

WARNING

THE FOLLOWING SERVICING INSTRUCTIONS ARE FOR USE BY QUALIFIED PERSONNEL ONLY. TO AVOID PERSONAL INJURY, DO NOT PERFORM ANY SERVICING OTHER THAN THAT CONTAINED IN OPERATING INSTRUCTIONS UNLESS YOU ARE QUALIFIED TO DO SO. REFER TO OPERATORS SAFETY SUMMARY AND SERVICE SAFETY SUMMARY PRIOR TO PERFORMING ANY SERVICE.

PART II

SERVICE INFORMATION

INSTALLATION

INSTALLING IN THE MAINFRAME

Use the following steps to install the TSG15 Module in the 1411 Mainframe.

CAUTION

Turn the Mainframe POWER switch off before starting installation.

1. Select one of the available 1411 Module locations for installation: use locations 2 through 6 when a dual-width SPG is installed, or locations 1 through 6 with a single-width SPG installed (see Fig. 3-1).

2. Position the shield board over the shield mounting pins, located between the circuit board pins on the Interface board. See Fig. 3-2. Seat the shield board firmly on the Interface board.

CAUTION

Be sure that all of the pins on the Interface etched circuit board are properly aligned with the connectors on the TSG15. Improper alignment of the pins and connectors can permanently damage the TSG15 connectors.

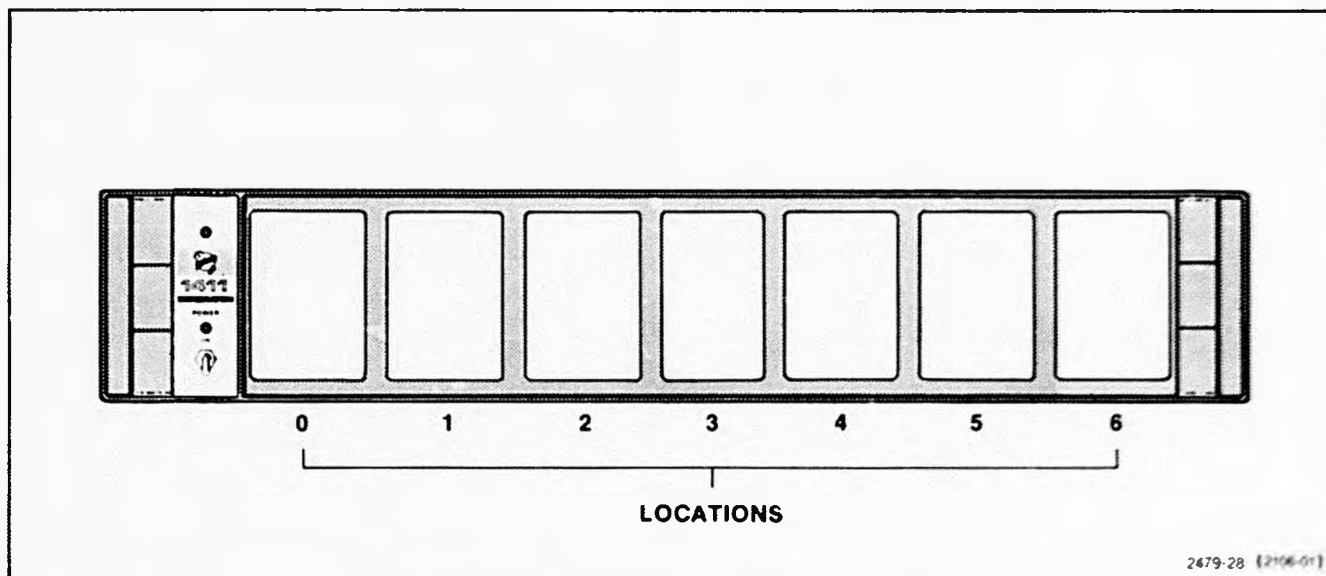


Fig. 3-1. 1411 module locations.

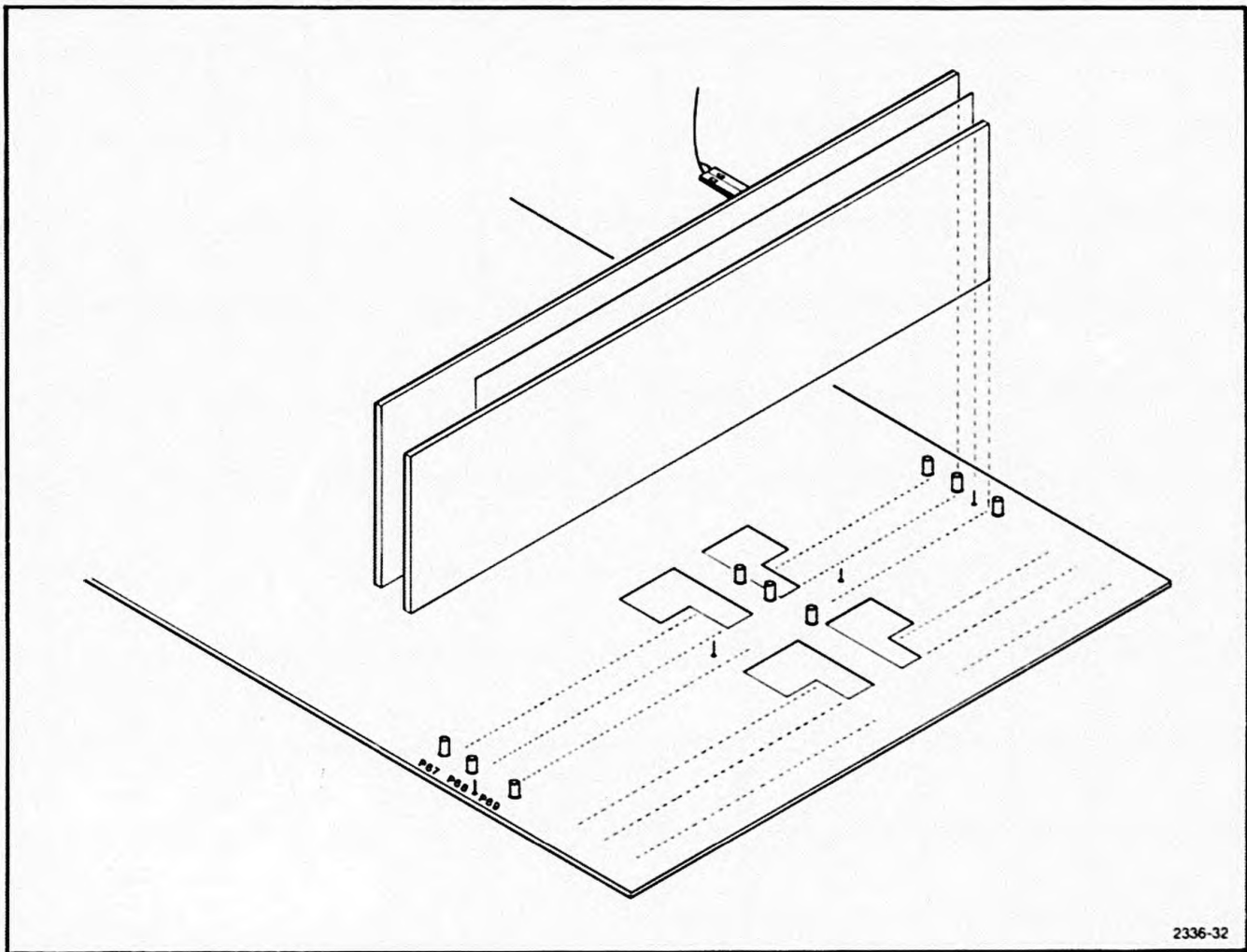


Fig. 3-2. Installation of the TSG15 in a 1411.

3. Position the A51-1 Pulse & Bar Output board over the middle row of Interface pins; use the plastic guides for proper pin alignment. See Fig. 3-2. Seat the board firmly on the Interface board. The Pulse & Bar Output board should be mounted on P56 (single-width SPG only), P59, P62, P65, P68, or P71.

4. Position the A50-1 Pulse & Bar Logic board over the right set of Interface pins. See Fig. 3-2. Align the board pins, using the plastic guides, and seat firmly on the Interface board. The Pulse & Bar Logic board, mounted adjacent to the Pulse & Bar Output board, occupies P57 (single-width SPG only), P60, P63, P66, P69, or P72 depending on the placement of the Pulse & Bar Output board.

5. Install the plastic pushbutton extenders on the board pushbutton shafts.

6. Position the TSG15 front panel over the pushbutton extenders and secure the front panel to the 1411 front casting with the self-tapping screws provided.

7. Connect the coaxial cable from the appropriate rear-panel MODULE Output connector to the board output connector, P428. See Fig. 3-3.

INSTALLATION ADJUSTMENT

When a TSG15 is added to the 1411, it is necessary to make some adjustment to mate the plug-in module to the mainframe. The number of these adjustments are kept to a minimum. The Readjustment Procedure, Section 4, has a more detailed procedure; if in doubt consult it. The equipment required for the installation adjustments is listed with a detailed description in the introduction to the Readjustment Procedure. A short list of equipment, without description, follows:

1. A TEKTRONIX 1481 Waveform Monitor.
2. A TEKTRONIX 521A PAL Vectorscope.
3. A TEKTRONIX 7603 Oscilloscope with 7A13, 7A26, and 7B53A plug-ins.

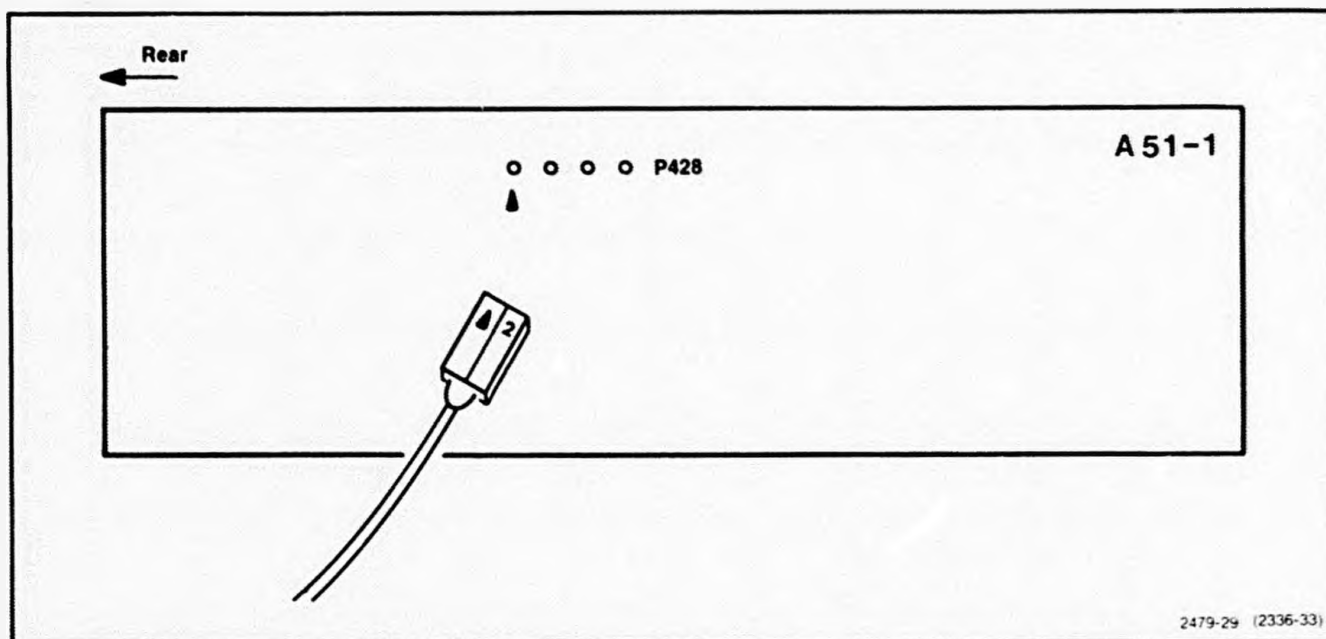


Fig. 3-3. Module output connector and P428.

4. A Variable Amplitude Calibrator. (Special test fixture built by user.)

5. A 75-ohm 10X Attenuator.

6. Two 75-ohm end-line terminations.

7. Three equal lengths of 75-ohm coaxial cable, with bnc connectors.

Procedure

1. Use the Variable Amplitude Calibrator along with the 1481 Waveform Monitor to set the Master Gain, R429, for a peak field square wave amplitude of 700 mV (± 7.0 mV).

2. Use the test oscilloscope to set the TSG15 DC Level, R455, for blanking level at 0 V (± 50 mV).

3. Use the test oscilloscope to touch up the Residual Subcarrier, R379 (and C347, if necessary) for minimum residual subcarrier.

4. Use the 1481 to set the Chrominance Gain, R369, for flattest baseline on the modulated-pulse and modulated-bar signal.

If Chrominance Gain is adjusted, it will be necessary to check modulated-pulse and modulated-bar amplitude.

5. Use the 521A and the 10X attenuator to set the Master Phase, C324 and/or L331. Match the mean of the TSG15 burst phase to the 1411 subcarrier phase reference, $180^\circ (\pm 10^\circ)$.

6. It may be desirable to set pulse amplitudes for pulse-to-bar ratio. See Procedure in Section 4. The ratio is often as much a function of the Waveform Monitor or oscilloscope as the new mainframe.

NOTE

Pulse-to-bar ratio for each pulse has been factory-set on a 100 MHz bandwidth oscilloscope. Waveform monitors, such as the 1481, are designed to be extremely flat over a 5 MHz bandwidth, but roll off outside that bandwidth. As a result, the pulse-to-bar ratio and pulse transient-response of any waveform monitor may not be adequate for use with T and T/2 pulses. Critical evaluation of wideband video equipment should be done with a wideband oscilloscope (TEKTRONIX 7603 and 7A26) or a waveform monitor modified to obtain the required bandwidth.

OPERATING MODE SELECTION

The TSG15 contains a number of internal jumper options to select operating modes. Altering an operating mode is accomplished by moving the appropriate plug jumpers. The following is a list of plug-jumper selections, that the user may wish to change to obtain operations other than those programmed at the factory. To locate plug jumpers, consult the A50-1 Pulse & Bar Logic board and A51-1 Pulse & Bar Output board Service Illustrations located on the pull-out pages at the rear of this manual.

Table 3-1 outlines the TSG15 internally-programmable options. Factory settings of the plug jumpers are identified.

Table 3-1
PROGRAMMABLE OPTIONS

Jumper(s)	Position	For
10T/20T		
P223	1-2	20T
P375	1-2	20T (Factory-Set)
P423	1-2	20T
P223	2-3	10T Readjust C225
P375	2-3	10T Mod Pulse Amplitude
P423	2-3	10T
Modulated Pulse Phase		
P158	2-3	Fixed Phase, R179, (Factory-Set)
P158	1-2	Modulated Phase, R169
Bar Timing		
P154	1-2	26 μ s Bar (Factory-Set)
P154	1-2	18 μ s Bar
Window Timing ¹		
P142	1-2	Lines 65-270 (Factory-Set)
P142	2-3	Lines 65-221
Field Square Wave Timing ¹		
P141	2-3	Lines 65-221 (Factory-Set)
P141	1-2	Lines 65-270
Bar Cal		
P134	2-3	On (Factory-Set)
P134	1-2	Off
P134	2-4	On in Overlay mode

¹See the SPG Manual for further timing details.

Table 3-1 (cont)

Jumper(s)	Position	For
Overlay Rate		
P133	2-3	H/4 (Factory-Set)
P133	1-2	H/2
P133	2-4	V/2
Inverted Pulse		
P175	2-3	On with normal 2T Pulse (Factory-Set)
P175	1-2	On
P175	2-4	Off
ITS Mode		
P319	1-2	Overlay Disabled (Factory-Set)
P319	2-3	Full Front Panel

ADDITIONAL PROGRAM CONSIDERATIONS

W124 is a wired connection for the PAL Bruch burst sequence.

W312, Setup off, is factory installed for PAL.

W191 connects R191 to +5 V dc, in PAL, for correct pedestal and no setup.

W191, W312, and W124 are factory-connected for PAL operation and should not be moved.

W242 is the inverted pulse drive to the 2T Filter.

W187 causes the inverted pulse to occur with the T Bar only.

Whenever FULL AMPL, LUMINANCE ON, and PEDESTAL are selected, a reduced pedestal is generated. The pedestal is factory set to 0 mV, but may be adjusted to any point between 0 and 175 mV. The pedestal amplitude adjustment is R186.

ITS KEY

Normally, the TSG15 uses the vertical blanking signal from the 1411 Mainframe to ensure that no test signals are generated during the vertical interval. If a pulse and bar ITS is desired, the TSG15 can be enabled by grounding the TSG15's Interface line 19 during the active portion of the appropriate tv line. See Table 3-2 and the 1411 Instruction Manual for more detail.

REMOTE DISABLE OPTIONS

Sync can be disabled by applying a ground closure to 1411 Interface line 55, through J41 (REMOTE). Table 3-2 provides the programming information required to remotely disable sync.

Interface line 56 to the TSG15 is a spare remote-control line. It is brought out to a spare pin on P211 and a 2 k Ω pull-up resistor. Interface line 57 is not connected to TSG15 circuitry.

Table 3-2
1411 REMOTE INPUTS

TSG15 In Module Location	J41 (REMOTE) connector pins to 1411 Interface Line			
	55 Sync Disable	56 (Spare)	57 (Not Used)	19 ITS Key
1	16	17	18	24
2	13	14	15	23
3	10	11	12	22
4	7	8	9	21
5	4	5	6	20
6	1	2	3	19

For example: If the TSG15 is in 1411 Location 4; pin 7 REMOTE (J41) connector, when grounded, would ground 1411 Interface line 55, and TSG15 sync would be disabled. Grounding pin 21 on J41, for the duration of tv line 17, would ground Interface line 55 and the TSG15 would generate a pulse and bar signal during line 17 of the vertical interval.

CALIBRATION

This section contains the Calibration (adjustment) Procedure. It is intended to return the TSG15 to operation within its written specification. With the exception of the Isolation Check, no performance checks are included. In order to evaluate instrument performance, it will be necessary to use the Performance Checkout Procedure in Section 2. Limits and tolerances appearing in this procedure are not Performance Requirements unless listed, as such, in Section 2, Specification.

Calibration at 20°C to 30°C, with a 20-minute warm-up period, is required to attain stated accuracies.

Calibration steps are for the TSG15 as shipped from the factory. Changes to operating parameters will need to be accounted for by the technician performing this procedure.

The TSG15 front-panel control names in the text are fully capitalized; for example, FIELD SQ WAVE. Control and connector names on test equipment have only the first letter capitalized; for example, test oscilloscope Vertical Position.

The Short Form Calibration Procedure may be used as an index to the regular Calibration Procedure, or as a guide for those familiar with the adjustment of the TSG15.

Test Equipment

Test equipment capabilities described here are the minimum required to calibrate the instrument. Examples represent test equipment used in developing these procedures.

1. Waveform Monitor. Capable of viewing line-rate and field-rate signals. For example, a TEKTRONIX 1481 Waveform Monitor.

2. Test Oscilloscope. Bandwidth, dc to 100 MHz; minimum deflection, 1 mV/div.; two input channels with provisions for independent or differential operation. For example, a TEKTRONIX 7603 Oscilloscope with 7A13, 7A26, and 7B53A plug-in units. A 7L12 spectrum analyzer plug-in unit will be necessary if the chrominance band-pass filter is to be adjusted.

3. Sinewave Signal Generator. Minimum output, 500 mV with a frequency range of 50 kHz fixed, variable from 1 MHz to 5 MHz. For example, a TEKTRONIX SG 503 in a Tektronix TM 500 Series mainframe.

4. Return Loss Bridge. Tektronix part number 015-0149-00.

5. Vectorscope. For example a TEKTRONIX 521A PAL Vectorscope.

6. Variable Amplitude Calibrator. Calibration fixture, see Fig. 2-2.

7. 50 ohm to 75 ohm Minimum Loss Attenuator. Tektronix part number 011-0057-00.

8. 75 ohm Cable (6 required) 42 inches long. Tektronix part number 012-0074-00.

9. 75 ohm End-Line Termination (3 required). Tektronix part number 011-0102-00.

10. 75 ohm In-Line Termination. Tektronix part number 011-0103-00.

11. 75 ohm 10X Attenuator. Tektronix part number 011-0061-00.

Calibration Fixture

The Variable Amplitude Calibrator and the 1481 provide a source of calibration levels that is variable. Voltages can be read directly from the 10-turn dial. A schematic diagram and parts list for this fixture are shown in Fig. 2-2. Whenever the switch labeled S1 is in the Fixed position, the 1481 internal calibrator works as normal.

With P9034 connected to J9034 on the 1481 rear panel and the switch in the Fixed position, calibrate the 1481 for a 1 V sync tip to peak white signal equivalent. Then set the dial to 1000, change the switch to Variable, and adjust R3 for the same amplitude displayed when the switch was in the Fixed position. This calibrates the dial for a range of 100 mV/360° rotation. Set the dial to 0.00 and verify that 0 V occurs at dial 0.

Measurements

The signal to be measured must be fed into the 1481 A Input, and both the Operate and Calibrate buttons pushed. To check an amplitude within a given tolerance, adjust the Amplitude dial while watching the waveform monitor display. When the level being measured overlays the blanking level, read the amplitude directly from the dial.

SHORT FORM PROCEDURE

The Short Form Procedure serves as an index to the main Calibration Procedure or as a quick outline to recalibration for those familiar with the TSG15. Step numbers and adjustment numbers listed in the Short Form Procedure are identical to those in the main Calibration Procedure.

Table 4-1

Step	Parameter	Required	Adjustment
1	Master Luminance Gain (Field Sq Wave)	700 mV $\pm$ 7.0 mV	R429
2	DC Level	0 V $\pm$ 50 mV	R455
3	Residual Subcarrier	<2.0 mV	R379 & C347
4	250 ns Filter	Optimum Sq Wave	L432 & L445
5	200 ns Filter (2T)	<0.5% Overshoot Optimum Pulse Sym.	C441, L451, L455, L465 & L468
6	100 ns Filter (T)	<1% Overshoot Optimum Pulse Sym.	C461, L472, L474, L476 & L479
7	50 ns Filter (T/2)	<2% Overshoot Optimum Pulse Sym.	C481, L491, L494, L496 & L499
8	Chrominance Bandpass	_____	L379 & L389
9	Burst Filter	Best Burst Shape	L335
10	10T Filter	<1% Overshoot Min. Ringing	L415 & L425, L395 & L405
10	20T Filter	<1% Overshoot Min. Ringing	L402 & L412, L342 & L391
11	Chrominance Gain & Delay	<3.5 mV	R369 & L408
12	Master Phase	Mean of Bursts $180^\circ \pm 10^\circ$	L331 & C324
12	Burst Phase	Angle between bursts $90^\circ \pm 1^\circ$	L117
13	Reduced Pedestal	0	R186
14	2T Bar Amplitude	700.0 mV $\pm$ 7.0 mV	R259
14	T Bar Amplitude	700.0 mV $\pm$ 7.0 mV	R277
14	T/2 Bar Amplitude	700.0 mV $\pm$ 7.0 mV	R298
14	MOD Bar Amplitude	700.0 mV $\pm$ 7.0 mV	R229
15	Full 2T Pulse Ampl.	1:1 $\pm$ 3.5 mV Pulse to Bar Ratio	C255

Table 4-1 (cont)

Step	Parameter	Required	Adjustment
15	Full Inv. 2T Pulse Ampl.	1:1 $\pm$ 7.0 mV Pulse to Bar Ratio	C235
15	Full MOD Pulse Ampl.	1:1 $\pm$ 3.5 mV Pulse to Bar Ratio	C225
15	Full T Pulse Ampl.	1:1 $\pm$ 7.0 mV Pulse to Bar Ratio	C273
15	Full T/2 Pulse Ampl.	1:1 $\pm$ 14.0 mV Pulse to Bar Ratio	C293
16	Half-Ampl. MOD Pulse	1:1 $\pm$ 3.5 mV Pulse to Bar Ratio	R239
16	Half-Ampl. Inv. 2T Pulse	1:1 $\pm$ 7.0 mV Pulse to Bar Ratio	R258
16	Half-Ampl. 2T Pulse	1:1 $\pm$ 3.5 mV Pulse to Bar Ratio	R269
16	Half-Ampl. T/2 Pulse	1:1 $\pm$ 14.0 mV Pulse to Bar Ratio	R299
16	Half-Ampl. T Pulse	1:1 $\pm$ 14.0 mV Pulse to Bar Ratio	R289
17	Mod. Pulse Phase	60° Typ.	L408 & R179
18	Check Isolation	1% or less	Check only

CALIBRATION PROCEDURE

Preliminary

Before recalibrating a TSG15, refer to instructions for circuit board removal, Section 6.

Put the Pulse and Bar Output circuit board (A51-1) on the Circuit Board Extender.

NOTE

Compensation for the effects of the Circuit Board Extender capacitance is covered in a later step.

Connect the output of the TSG15 through its appropriate 1411 rear-panel Module Output, and a 75 ohm coaxial cable to the A Input of the 1481 Waveform Monitor. Terminate the remaining side of the 1481 A Input Loop-Thru connector with a 75 ohm termination.

Connect the Variable Amplitude Calibrator to J9034 on the 1481 rear panel.

1. Set Master Luminance Gain

- Select FIELD SQ WAVE.
- Adjust—R429, Luminance Gain, for peak luminance of 700.0 mV (± 7.0 mV), using the Variable Amplitude Calibrator. See Fig. 4-1.

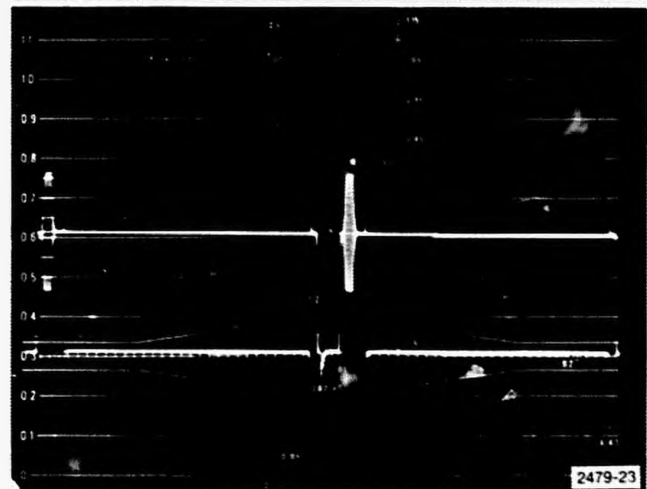


Fig. 4-1. Setting Luminance Gain with the Variable Amplitude Calibrator.

2. Set DC Level

- Connect the TSG15 output to the test oscilloscope Vertical Input through an In-Line termination.
- Set the test oscilloscope bandwidth to 5 MHz, Volts/Div to 1 mV, Input switch to Ground, and position the trace to graticule center.
- Change the test oscilloscope Vertical Input switch to DC.
- Adjust—R455, DC Level, for 0 V dc ± 50 mV Blanking level.

3. Balance Modulator (Residual Sub-carrier Adjustment)

NOTE

This assumes that the Chrominance Bandpass Filter is not completely misadjusted. If it is, harmonic distortion may interfere with residual subcarrier adjustment.

a. Check that the blanking level is still at graticule center on the test oscilloscope.

b. Adjust—R379, Modulator Bal, and C347, Residual Subcarrier, for minimum residual subcarrier (trace width) during blanking and active portions of the line for a Modulated Bar signal, (2.0 mV or less).

4. Adjust 250 ns Filter (Field Sq Wave)

a. Set the TSG15 to FIELD SQ WAVE.

b. Adjust—L432 and L445 for optimum transient response.

5. Adjust 200 ns Filter (2T)

Pulse shaping and pulse to bar ratio for T/2 pulse should be set on a 100 MHz bandwidth oscilloscope.

Waveform monitors, such as the TEKTRONIX 1481, are designed to be extremely flat over a 5 MHz bandwidth, but rolloff outside that bandwidth. As a result, the pulse to bar ratio and pulse transient response of any waveform monitor may not be adequate for use with T/2 pulses and possibly T pulses. Critical evaluation of wideband video equipment should be done with a wideband oscilloscope, such as a TEKTRONIX 7603 and 7A26 or a waveform monitor modified to obtain the required bandwidth and phase accuracy.

The 2T, T, and T/2 filters use a similar adjustment procedure; 2T filter adjustment will be used as an example. Each of the five adjustments in the 2T filter affects a general area on the trailing edge of the pulse as shown in Fig. 4-2. It may be easier to adjust the filter starting from the output, L468, and working back to the input, L451. Using this method of adjustment, interaction is minimized. Go over the adjustment procedure several times to eliminate the remaining interaction. See Fig. 4-3.

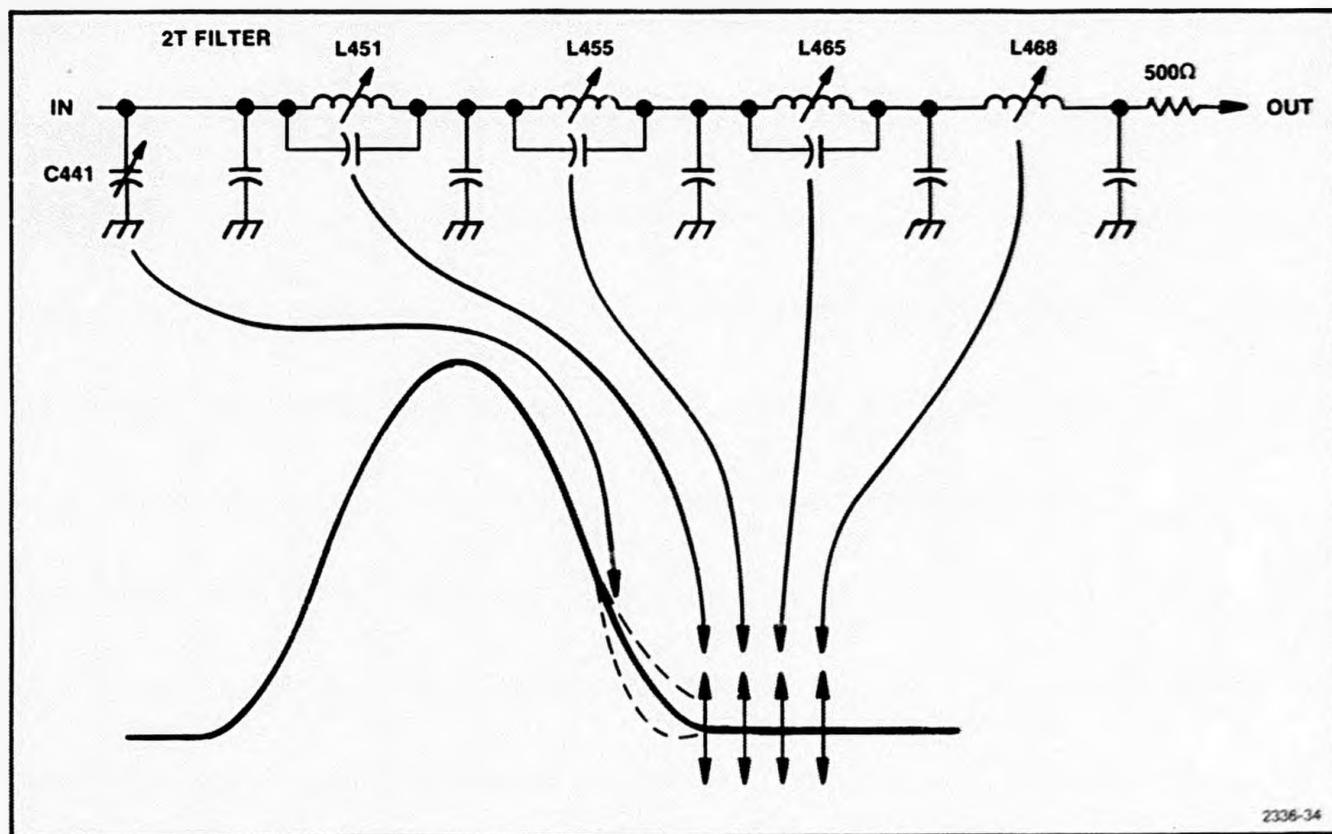


Fig. 4-2. Effects of adjustments on the pulse shape.

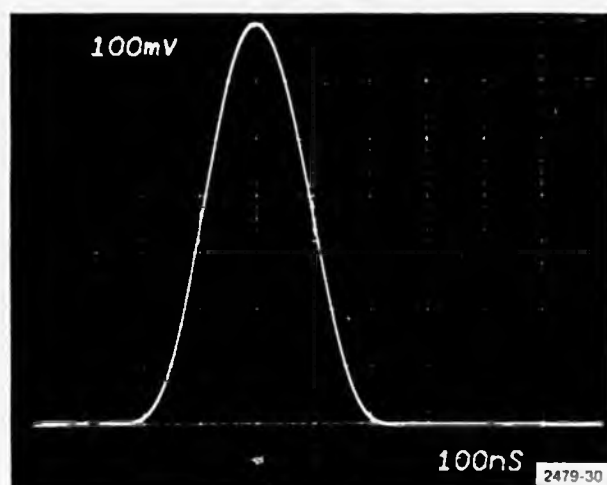


Fig. 4-3. Properly adjusted 2T filter will produce this pulse shape.

C441 sets pulse symmetry. If C441 is too large (plates meshed), it will be easy to minimize pulse ringing, but the pulse will not be symmetrical, see Fig. 4-4a. If C441 is too small, there will be excessive overshoot as shown in Fig. 4-4b. With C441 set properly, pulse symmetry will be good and it will be possible to set the coils for minimum ringing.

NOTE

C441 also compensates for the added capacitance of the Circuit Board Extender. After any filter adjustment on the Extender, C441, and its counterparts in other filters, will need to be adjusted after the board is re-installed in the instrument.

Filter adjustment affects pulse amplitude. Pulse amplitude adjustment is made later in the procedure.

- a. Select FULL AMPL, 2T PULSE, and 2T BAR.
- b. Set the 7A26 to approximately 10 mV/div.; do not overload.
- c. Adjust—C441 for optimum pulse symmetry and L451, L455, L465 and L468 for 0.5% or less overshoot and ringing.

NOTE

Filter adjustment will affect pulse amplitudes. Pulse amplitudes are adjusted later in this procedure.

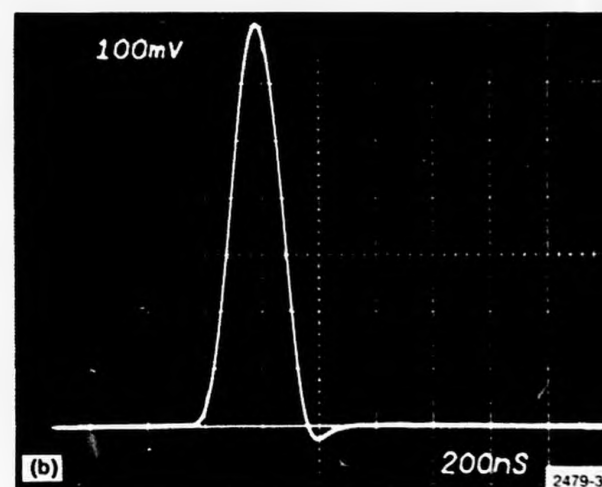
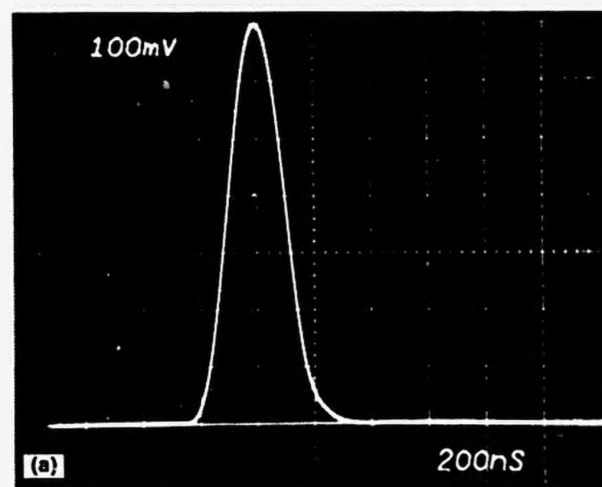


Fig. 4-4. Errors caused by incorrect setting of symmetry adjustment, a. Non-symmetrical pulse. b. Excessive overshoot.

6. Adjust 100 ns Filter (T)

- a. Select FULL AMPL, T PULSE, and T BAR.
- b. Set the 7A26 to approximately 20 mV/div.; do not overload.
- c. Adjust—C461 for optimum pulse symmetry and L472, L474, L476, and L479 for 1% or less overshoot and ringing.

7. Adjust 50 ns Filter (T/2)

- a. Select FULL AMPL, T/2 PULSE, and T/2 BAR
- b. Set the 7A26 to approximately 50 mV/div; do not overload.

Calibration—TSG15

c. Adjust—C481 for optimum pulse symmetry and L491, L494, L496, and L499 for 2% or less overshoot and ringing.

8. Adjust Chrominance Bandpass Filter

NOTE

This filter does not require readjustment unless components are replaced, or it has been accidentally mis-adjusted. Adjustment of the Chrominance Bandpass Filter should not be made without a spectrum analyzer.

a. Select FULL AMPL, MOD PULSE, and MOD BAR.

b. Connect Module Output to the vectorscope Input. Connect a second Module Output to the spectrum analyzer. (A second Module Output cable can be obtained by borrowing the output cable from the adjacent module and connecting it to the unused part of P428, Module Output plug.)

c. Adjust—L379 and L389, Bandpass Filter, for 40 dB harmonic attenuation on the spectrum analyzer, Fig. 4-5a, and the best phase transient response on the vectorscope, Fig. 4-5b.

9. Adjust Burst Shape

a. Connect the TSG15 Module Output to the 1481.

b. Adjust—Burst Shape, L335, for optimum burst shape. See Fig. 4-6.

10. Adjust 10T and 20T Filters

NOTE

If only one of these pulses is used, only that filter need be adjusted.

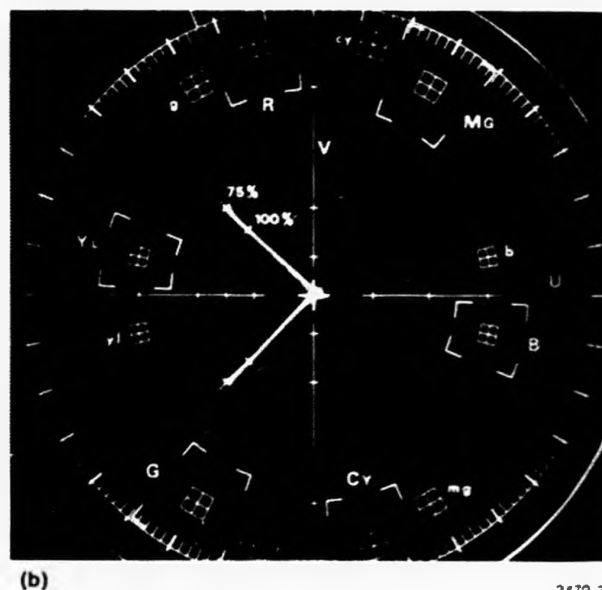
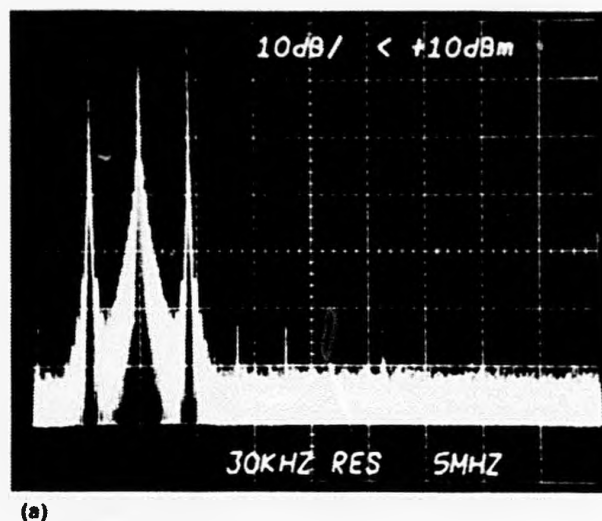
a. Select MOD PULSE, MOD BAR, and LUMINANCE ON.

b. Set the 1481 to 0.2 Volts Full Scale.

c. Adjust—filters according to Table 4-2.

NOTE

Be sure to finish calibration with the filter that will be used in normal operation.



2479-32

Fig. 4-5. Adjusting chrominance bandpass filter for: a. Optimum harmonic attenuation. b. Best phase transient response.

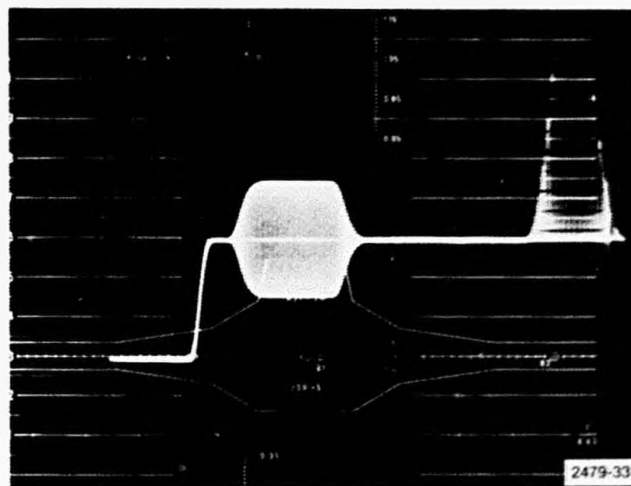


Fig. 4-6. Properly adjusted Burst Shape.

Table 4-2
10 AND 20T FILTERS

Filter	Adjust	for
10T	L415 & L425	minimum overshoot <1%
10T	L395 & L405	minimum ringing
20T	L402 & L412	minimum overshoot <1%
20T	L392 & L381	minimum ringing

11. Set Chrominance Gain and Delay

- Set the TSG15 to MOD PULSE (20T).
- Set the waveform monitor to 0.2 Volts Full Scale.
- Adjust—R369 and L408 for a flat baseline beneath the pulse, ± 3.5 mV combined chrominance/luminance gain and delay error. See Fig. 4-7.

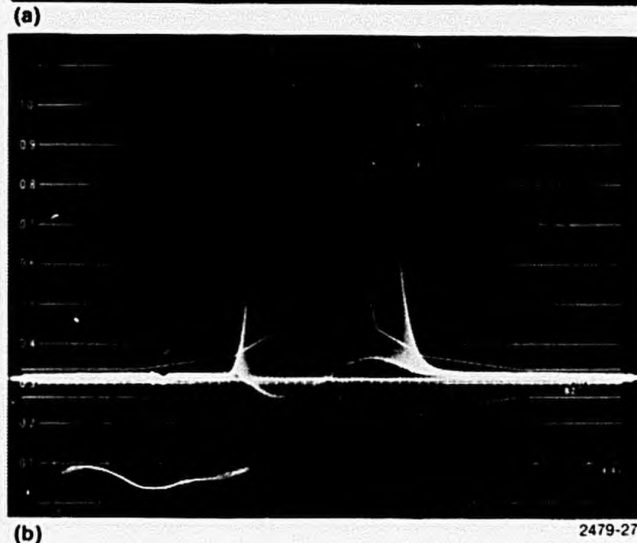
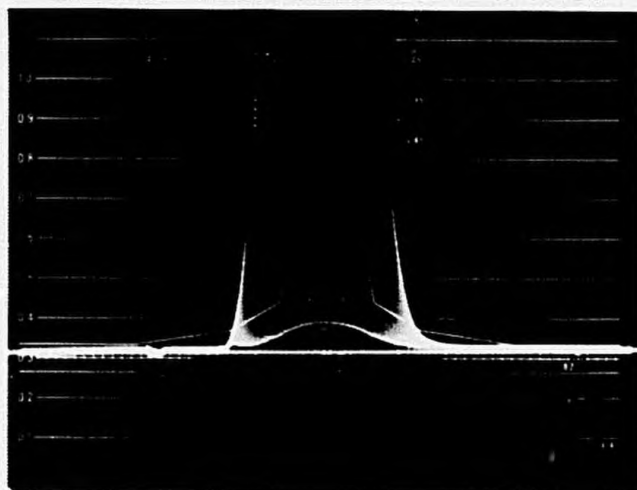


Fig. 4-7. Identifying chrominance/luminance errors. a. Gain. b. Delay.

12. Set Master Phase and Burst Phase

- Connect the 1411 to the vectorscope as shown in Fig. 4-8a.
- Set the vectorscope Phase to position the 1411 subcarrier reference at 180° , as shown in Fig. 4-9.
- Change connections as shown in Fig. 4-8b. Do not change vectorscope Phase control setting.
- Adjust—L331 and C324, Master Phase, set the -135° Burst phase to be -45° from the 1411 rear panel subcarrier reference. (Mean of bursts $180^\circ \pm 10^\circ$.) See Fig. 4-6a.
- Adjust—L117, Burst Phase, to set the angle between bursts to $90^\circ \pm 1^\circ$.

NOTE

No further coil adjustment required. Replace the Pulse and Bar Output board into the 1411. With neither circuit board out on extender, adjust C441, C461, and C481 for best 2T, T, and T/2 pulse shaping. These capacitors compensate for the capacitance of the Circuit Board Extender.

Most Pulse and Bar Logic board adjustments can be done without placing it on the Circuit Board Extender.

13. Set Reduced Pedestal

- Select FULL AMPL, LUMINANCE, and PEDESTAL.
- Set the waveform monitor Volts Full Scale to 1.
- Adjust—Reduced Pedestal, R186, for the desired reduced pedestal (factory-set to 0). Range of the adjustment is 0 to 175 mV.

14. Set Bar Amplitudes

NOTE

After adjustment of bar amplitudes, if pulse amplitudes are not adjusted, check pulse to bar ratio as outlined in the Performance Check in Section 2.

- Select FULL AMPL, LUMINANCE, and 2T BAR.
- Use the waveform monitor and Variable Amplitude Calibrator to set 2T Bar amplitude.

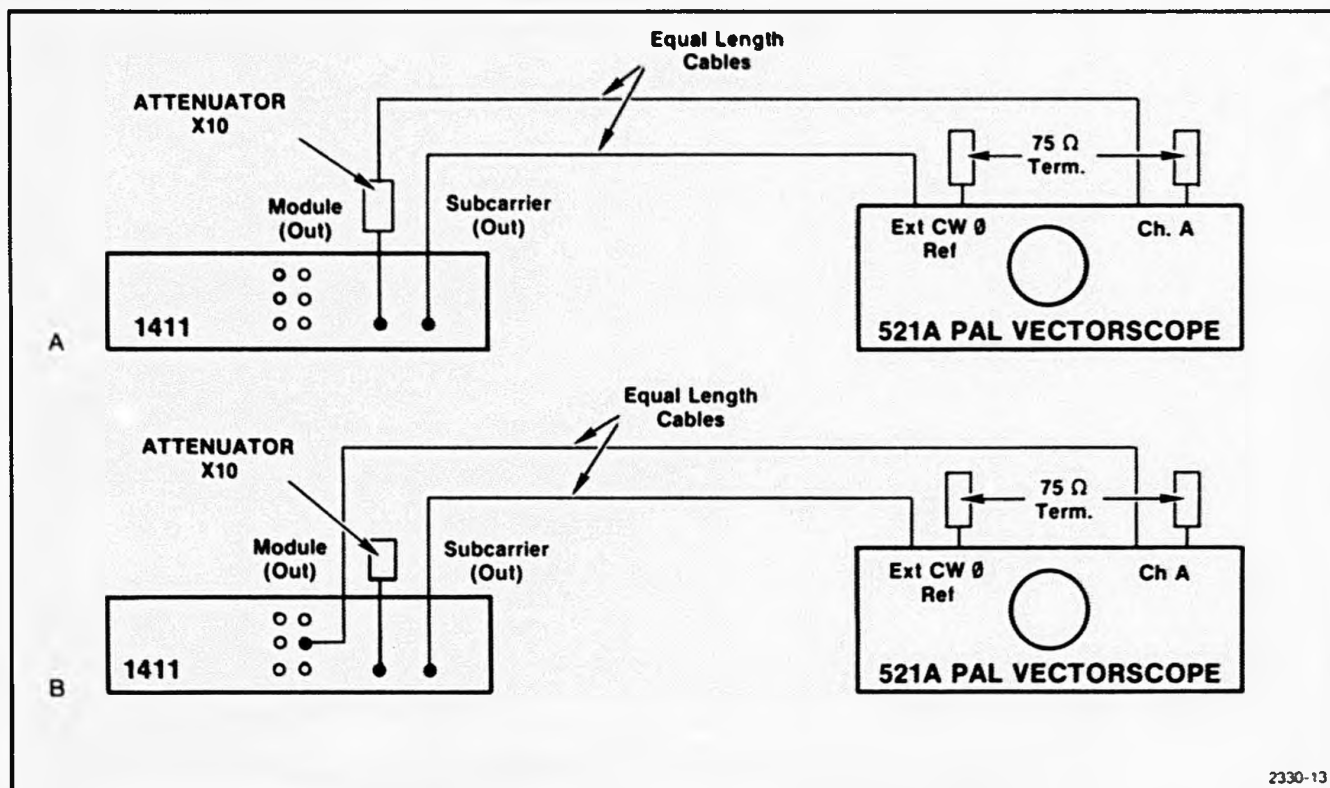


Fig. 4-8. Equipment connections for master phase adjustment. a. Establishing subcarrier reference vector. b. Setting master phase.



Fig. 4-9. Correct setting of subcarrier reference vector or master phase.

c. After 2T Bar amplitude is set, use bar calibrate mode to compare and set bar amplitudes; (factory set) any two BAR switches pushed. Use with Table 4-3.

Match all bars as closely as possible.

Table 4-3
BAR AMPLITUDES

Bar	Adjust	for Bar Amplitude (blanking to peak)
2T	R259	700.0 mV $\pm$ 7.0 mV
T	R279	700.0 mV $\pm$ 7.0 mV (matched to 2T)
T/2	R298	700.0 mV $\pm$ 7.0 mV (matched to 2T)
MOD	R229	700.0 mV $\pm$ 7.0 mV (matched to 2T)

15. Set Full Amplitude Pulse Amplitudes

a. Select FULL AMPL. LUMINANCE. 2T PULSE and 2T BAR.

b. Use the waveform monitor to match pulse amplitudes to bar amplitudes according to Table 4-4.

NOTE

Final pulse amplitude adjustments cannot be made with either circuit board on the Extender Board.

Table 4-4
FULL PULSE AMPLITUDES

Pulse	Adjust	for Pulse to Bar Ratio
2T	C255	1:1 $\pm$ 3.5 mV
INVERTED 2T	C235	1:1 $\pm$ 7.0 mV
MOD	C225	1:1 $\pm$ 3.5 mV
Use test oscilloscope with 7A26.		
T	C273	1:1 $\pm$ 7.0 mV
T/2	C293	1:1 $\pm$ 14.0 mV

16. Set Half-Amplitude Pulse Amplitudes

a. Select LUMINANCE, HALF AMPL, and MOD PULSE.

b. Use the waveform monitor and test oscilloscope to match pulse amplitudes to bar amplitudes according to Table 4-5.

NOTE

See note accompanying step 15.

Table 4-5
PULSE HALF-AMPLITUDES

Pulse	Adjust	for Pulse to Bar Ratio
MOD	R239	1:1 $\pm$ 7.0 mV
2T	R269	1:1 $\pm$ 3.5 mV
INVERTED 2T	R258	1:1 $\pm$ 7.0 mV
Use test oscilloscope with 7A26.		
T/2	R299	1:1 $\pm$ 14.0 mV
T	R289	1:1 $\pm$ 7.0 mV

17. Set Modulated Pulse Phase Range

a. Connect the TSG15 Module Output to the waveform monitor.

b. Select MOD PULSE.

c. Set the waveform monitor to view the modulated pulse.

d. L108 is set approximately to phase shift mid-range. Its adjustment is not critical.

e. Check that P158 is in the Mod Phase position.

f. Adjust R169, Modulated Phase Range, for approximately 360° Phase Modulation on the modulated pulse.

g. Move P158 to the Fixed Phase position. Adjust R179, Fixed Phase, for 60° Subcarrier on the modulated pulse.

Factory Set position for P158 is Fixed Phase.

18. Check Isolation

a. Connect the TSG15 Module Output to the test oscilloscope. Use an In-Line termination.

b. Use the TSG15 FIELD SQ WAVE signal.

c. Use an output cable from an adjacent test module output connector to provide a parallel rear-panel output from the TSG15.

d. Alternately terminate and un-terminate the unused parallel module output and check for 1% or less change in the viewed signal output. (An alternate method would be to short and unshort the unused parallel output and check for 1% or less change.)

e. Connect a terminated, 0.5 V, 4.43 MHz sine wave to the unused parallel module output connector.

f. Check—for 5 mV or less sine-wave signal on the viewed output signal.

g. Return the borrowed module output cable to its original plug.

After the completion of this procedure, check instrument performance with the Checkout Procedure in Section 2.

THEORY OF OPERATION

This section of the manual is divided into two parts. The first, an overview, provides a basic insight into the operation of the TSG15 by means of the block diagram. The second part provides detailed discussion of the individual circuits.

This circuit description is divided by diagram; circuits appearing on Diagram 1 are discussed before Diagram 2 & 3 circuits. The circuit diagrams are blocked off according to circuit function. Circuit block titles serve as topics in the circuit description.

The block diagram and schematic diagrams are located on the foldout pages at the rear of the manual. Refer to the appropriate diagram when reading the circuit description.

BLOCK DIAGRAM

Circuits in the TSG15 employ power supplies of the 1411 and use the subcarrier, sync, blanking, and other timing signals from the installed SPG. The TSG15 circuits can be grouped in two main categories: Logic (A50-1, located on diagrams 1 and 2), and Output (A51-1, diagrams 2 and 3). The logic circuits consist of timing and current drives for the output circuits. The output includes the modulator, filters and the output amplifier.

The Block Diagram includes parallel paths for chrominance and luminance signals. Mode Control and Timing combines the front-panel switching, internal programming and SPG timing signals to control the various luminance and chrominance signals of the TSG15.

Chrominance Logic and Burst Timing signals are supplied to the Subcarrier Phase Switching from the SPG. Subcarrier phase is shifted by the Subcarrier Phase Switching circuit. The subcarrier signal is then limited and shaped to provide the carrier drive to the Modulator. The Modulator output chrominance signal, whose amplitude is determined by the chrominance drive circuitry, is passed through a bandpass filter to the Output Amplifier.

Luminance signals are timed from the Mode Control circuits and take the form of pulses, bars, and pedestal.

The outputs of the Pulse and Bar Drives are applied to the Luminance Filters and the Output Amplifier. Chrominance and luminance signals are combined at the input of the Output Amplifier.

DIAGRAM 1

Pulse & Bar Timing

Timing & Control

Characteristic instants for the Pulse and Bar signals are derived from H, H*, 2H, 4H, 8H, and 16H signals from the SPG. These are in gray code form and drive U153, the timing PROM, inputs A0 through A5. This 256 x 4 PROM decodes the gray code into timing signals coincident with the 64 characteristic instants, at 1 μ s intervals.

U153 output; Q1 is the inverted pulse timing, Q2 is bar timing, Q3 is modulated pulse timing, and Q4 is luminance pulse timing. See Fig. 5-1.

Each of the outputs of U153, in addition to the front-panel WINDOW switch, is latched by U163. Outputs of U163 are coincident with the positive-going edge of the 1 MHz clock. The latch guarantees that transition glitches from U153 do not interfere with signal timing.

In the overlay mode, U153 alters the timing of pulse and bar to give the overlay. Whenever input A6 is low, the reverse timing for overlay is used. P133 selects the overlay rate.

P154, Bar Timing, selects yet another set of timing signals in U153. Whenever input A7 is grounded, a shortened (18 μ s) bar is generated.

U153 is disabled during blanking, field square-wave (pin 13), or in case of a video disable command from the SPG (pin 14).

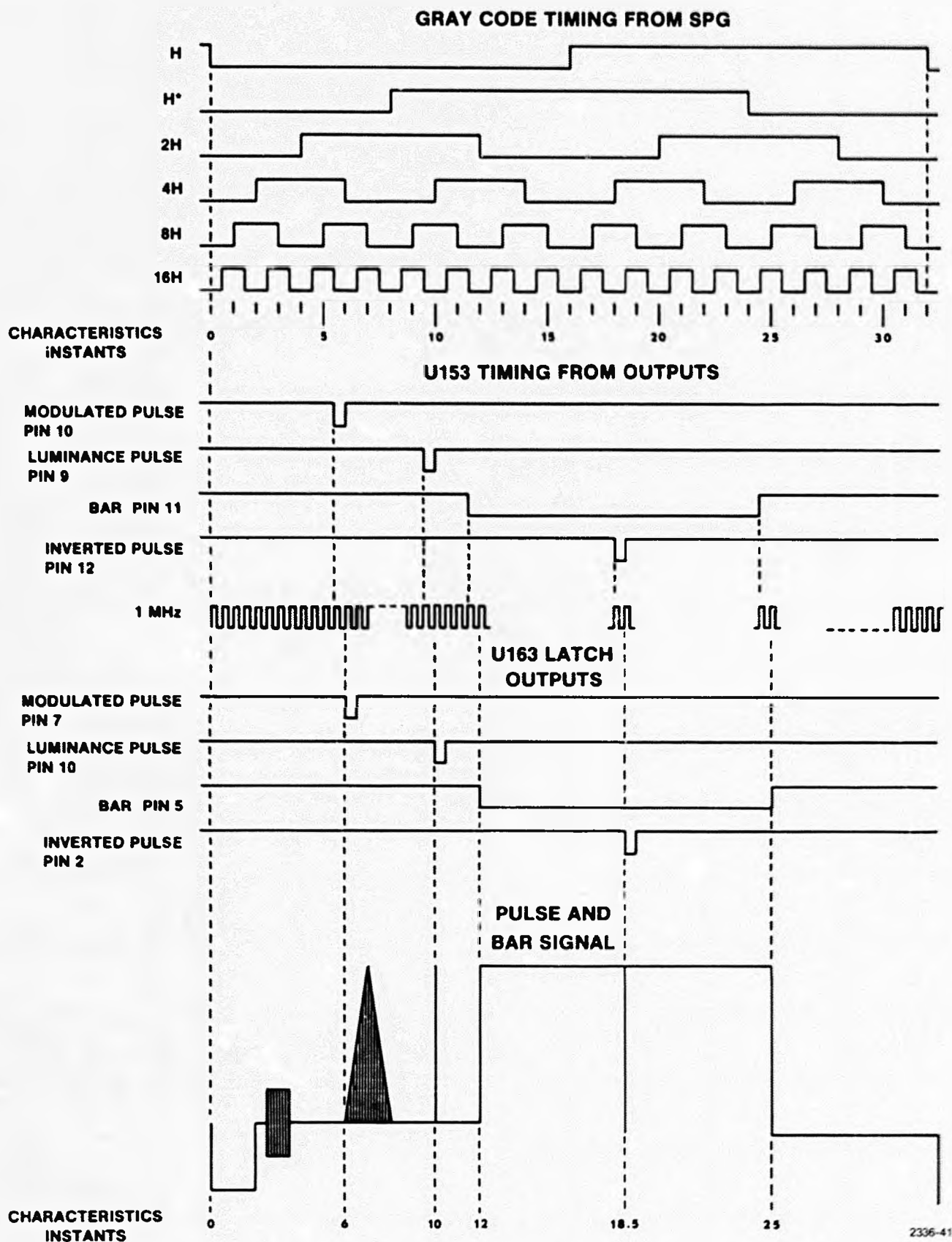


Fig. 5-1. Shows the operation of U153, starting with the gray code, U153 output, and the latched output from U163.

Sync, Pedestal, & Burst Drive

U133 is a 256 x 4 PROM with a stored program which is used to select the appropriate combination of Pulse and Bar, Field Square-Wave, Pedestal, ITS, and Blanking, as well as switching subcarrier phase with HBL* to provide the correct phase for burst and modulated test signals. See Fig. 5-2.

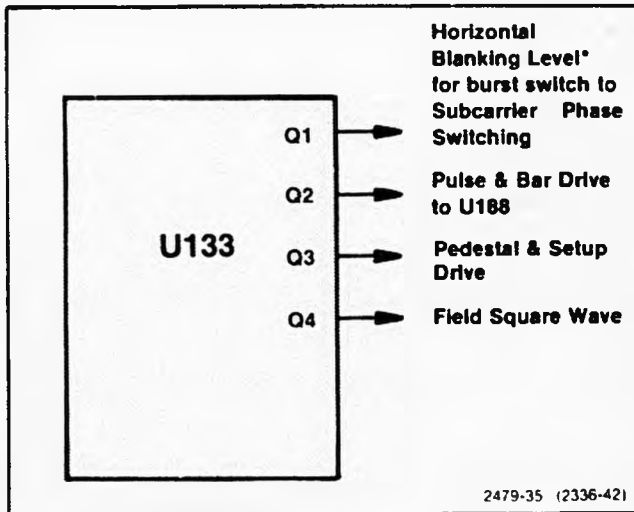


Fig. 5-2. Outputs of U133 and their destinations.

Vertical Drive (Input A0), Vertical Blanking (Input A4), and ITS key (Input C) are used to determine what signal should be generated during the vertical interval.

The field square-wave switch (Input A1) and field square-wave timing (Input A6) from P141 determine when a field square-wave or pulse and bar signal should be generated as the full field signal. A pulse and bar ITS may be selected even if the full-field signal is field square-wave. P141 selects a signal from the 1411 interface line 33 or line 39 as the vertical timing for field square-wave.

Pedestal is off during the vertical interval except for those lines where ITS is keyed.

U133 is disabled during horizontal blanking or in case of a video disable command from the TSG15. This inhibits all of the TSG15 test signals during horizontal blanking or in case of an SPG lock fault.

When pedestal is generated, U133 output (Q4) switches off Q184 and the pedestal driver, Q194, drives the 250 ns filter. Q194 is normally biased off through R191. When pedestal is selected, additional drive current is supplied through S306B from R183. If a reduced pedestal is used, current drive may be taken away by R186 to the +15 V supply.

In the field square-wave mode, pedestal drive is switched off and Q195 also switches off, providing field square-wave drive through Q205 to the 250 ns filter.

Q204 provides the sync current drive to the 250 ns filter. U113C and U122B switch Q204 off during sync.

U113D switches the burst current driver, Q203.

Subcarrier Phase Modulator

The subcarrier phase modulator, which includes Q135, Q137, Q138, Q118, Q157, and U137, takes the fixed-phase subcarrier from the 1411 mainframe and provides either a 360° phase-modulated sub-carrier or an adjustable phase subcarrier for use in the modulated pulse and modulated bar.

Q156 buffers the 2 volt peak-to-peak subcarrier from the 1411 mainframe.

Q135 amplifies and limits the subcarrier to approximately TTL levels to drive the Schmitt Trigger, input of U137.

U137 is a 74LS221 dual monostable multivibrator which operates as a controllable digital delay. The delay of each half of U137 is controlled by current sources Q137 and Q138. Each half of U137 may have a delay of 56 to 169 ns, i.e., a 180° range of subcarrier delay per monostable. See Fig. 5-3.

U137A output has a fixed leading edge and a variable delay trailing edge. U137B output has a delayed leading edge and adds an additional delay to the trailing edge.

At the trailing edge of the output from U137B, Q118 is driven by a short current-pulse. This excites the tank circuit in the collector of Q118. The tank circuit provides a small range of phase adjustment as well as good shaping of the adjustable-phase subcarrier which drives the sub-carrier switch.

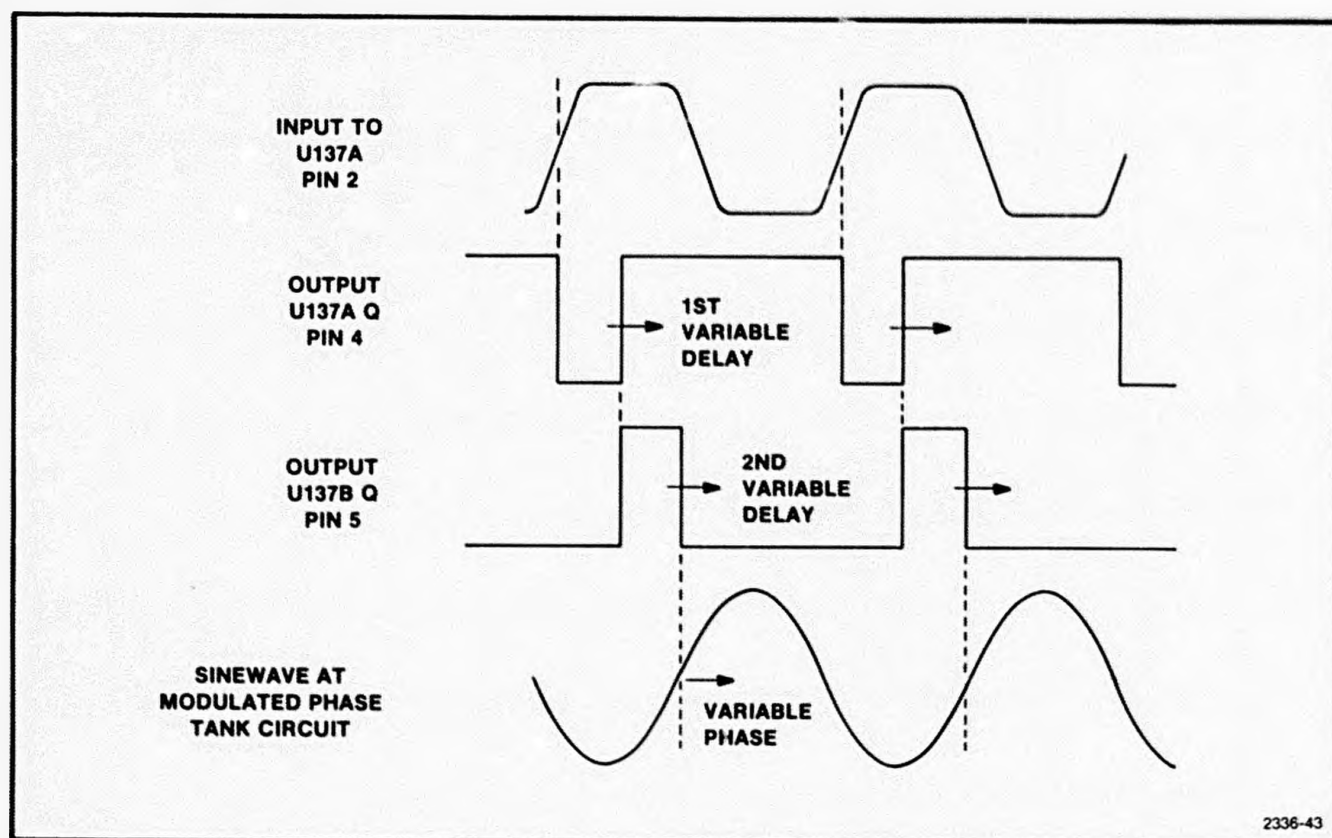


Fig. 5-3. Operation of U137, Subcarrier Digital Delay.

As mentioned earlier, the delay of U137A and U137B is determined by the current from current sources Q137 and Q138. In the Fixed Phase Mode (P158 pins 2 and 3), R179, Fixed Phase, sets the base voltage, hence output current, of Q137 and Q138. In the Modulated Phase Mode (P158 pins 1 and 2), Q157 discharges C157 at the beginning of every field. During each field, C157 charges asymptotically to a voltage set by R169, Modulated Phase Range. The charging curve of C157 and the delay characteristics of U137 combine to give a linear subcarrier phase sweep of approximately 360° every field.

Only one of the control signals (through CR111, CR113 or CR102) is low at a time, permitting one subcarrier phase at a time to drive Q110.

The various phases of subcarrier are amplified by Q110 and shaped by the Master Phase tank circuit on the Pulse and Bar Output circuit board.

DIAGRAM 2

Pulse & Bar Drivers

Subcarrier Phase Switching

A 2 V subcarrier signal at approximately 180° is applied to Q156, which operates as a buffer to drive C116 and R108 (a 45° lead network) L117 and R117 (a 45° lag network) and the subcarrier phase modulator. CR110 or CR112 is turned on during horizontal blanking, passing -135° burst phase subcarrier from Q114 to Q110, or $+135^\circ$ burst phase subcarrier from Q115 to Q110. CR101 is on during active line time, passing fixed adjustable phase subcarrier or modulated phase subcarrier from Q103 to Q110. W124 selects the correct H/2 signal to provide the proper Bruch burst sequence for PAL.

Bar Drive

Bar Logic U188 is a 256×4 PROM with a stored program especially for the control of the 2T, T, T/2 and Modulated Bars. This stored program is different from the programs in the other PROMs. Four of the PROM inputs are used to assign a priority to the four front-panel bar select switches. Normally, if more than one bar is selected (Inputs A0 through A3), the higher priority bar is generated. In the Bar Calibration Mode, the second highest priority bar is generated at times selected by the overlay-rate jumper (Input A4); thus, if 2T and Modulated Bars are selected (with the Bar Calibration Mode enabled), the 2T and Modulated Bars would alternate.

Whenever the front-panel WINDOW (Input A7) switch is enabled, a timing signal from the 1411 mainframe (as selected by P142) gates the bar (Input A5). This results in a window test signal. The timing for the bar from U153 and U163 is used to enable U188 (pin 13). The other enable (pin 14) disables the bar during blanking and when field square-wave is selected. Field square-wave overrides the other bars. A video disable command from the SPG may also disable U188 in case of an SPG lockup fault.

Each bar driver circuit includes two transistors operating in a current steering mode. See Fig. 5-4. Whenever the logic signal from the bar control PROM goes low, bar current switch Qa turns off. Emitter current I_e is thereby steered to bar current driver Qb. Qb is biased from a temperature compensating +0.6 volt dc reference, so the emitter voltage of Qb is approximately zero volts dc when the bar is on; therefore, the bar driver current is $-13 \text{ volts} \div R_{e1} + R_{e2}$.

$-V_c$ is a dc control voltage determined by the AMPLITUDE switch.

Whenever the logic signal from bar control PROM U188 goes high, bar current switch Qa turns on. The entire emitter current from Re1 and Re2 is steered through Qa, turning off the bar. Diodes CRa and CRb limit the excursion of the bar logic signal, improving the switching characteristics of Qa and Qb.

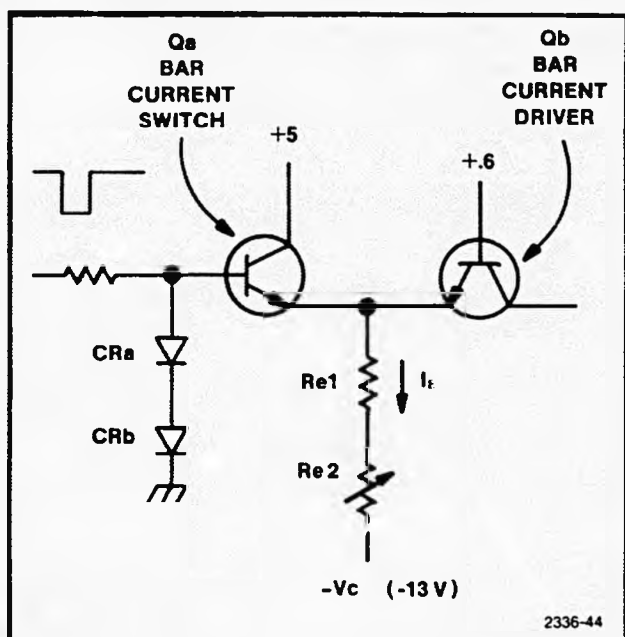


Fig. 5-4. A typical bar driver.

Pulse Drive

The 2T, T, T/2 and Modulated Pulse drivers are all similar in circuitry and function. The inverted 2T pulse driver is a complementary circuit that generates a pulse of the opposite polarity to the other pulse drivers.

For this discussion, the 2T pulse driver is chosen as an example. Luminance pulse timing from U153 and U163 is gated by U208. If a 2T pulse is selected, U208D will switch on Q256. Normally, Q256 (the pulse switch) and Q262 (the pulse current driver) are both off. For a full amplitude pulse, Q256 collector would switch rapidly from +13 volts dc to V_{ce} saturation, approximately 0.2 volt.

C254 and C255 discharge rapidly through R261. The current impulse drives the 2T shaping filter via Q262. See Fig. 5-5.

The 2T shaping filter gives a sine-squared shaped pulse output for a current-impulse input.

After $1 \mu s$, pulse switch Q256 turns off and the collector voltage rises slowly back to +13 volts. This again causes a current pulse, but this (undesired) signal current goes through CR271 instead of pulse current driver Q262.

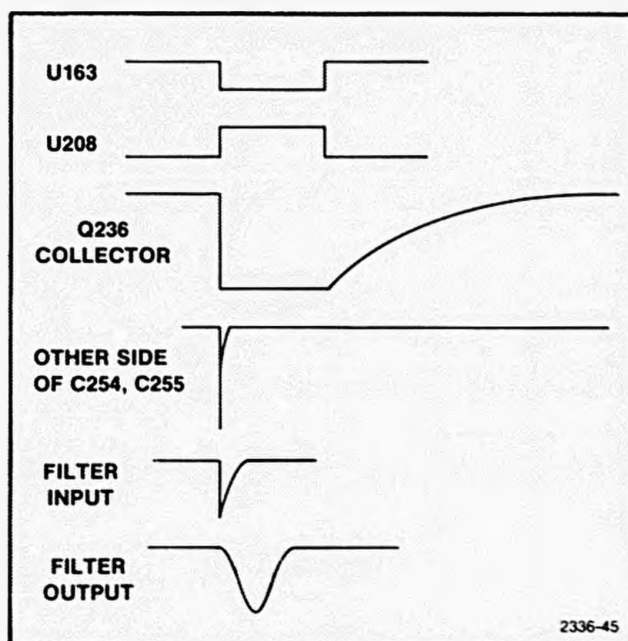


Fig. 5-5. 2T pulse drive from U163 to output amplifier input.

Theory of Operation—TSG15

Note that the above description assumed a full amplitude pulse so pulse amplitude control voltage +Vc was at 13 volts dc. The quiescent collector voltage of Q256 was already set at +13 volts dc by R266 and R264 so at full amplitude, R269 has no effect. C255 is used to adjust full amplitude 2T pulse.

The +Vc is not always 13 volts; however, it is a function of the AMPLITUDE switch.

At +Vc less than 13 volts dc, R269 lowers the quiescent collector voltage of Q256, thereby reducing pulse amplitude. Since full amplitude is set by C255 and half amplitude is set by R269, the pulse amplitude will be proportional to +Vc over all combinations of amplitude and pedestal.

Amplitude Selection

Amplitude selection consists of two voltage-follower circuits. The first, U353 and Q354, is non-inverting and its output is -Vc (full amplitude about -13 volts) used to set bar amplitudes. The second, U219 and Q214, inverts and is +Vc (full amplitude about +13 volts) for setting the pulse amplitudes.

S305 and S308B determine the reference for the voltage followers from voltage divider R317, R318, R309, and R319.

Table 5-1
AMPLITUDES OF Vc

Bar	
Full Amplitude	-13.0 V dc
Half Amplitude	-6.5 V dc
Pulse	
Full Amplitude	+13.0 V dc
Half Amplitude	+6.5 V dc

DIAGRAM 3

Pulse & Bar Output

Subcarrier Limiter & Shaper

The subcarrier limiter and shaper circuits ensure that the Modulator is driven by a constant amplitude subcarrier signal even though the incoming signal to this circuit may be varying in amplitude. Q344, Q334, Q335, Q325, and Q326 make up the Subcarrier Limiter & Shaper that drives the double-balanced modulator.

10T and 20T Filters

The modulated pulse and/or modulated bar current drive from the logic board is shaped by either the 10T or 20T filter. P423 and P375 select the desired filter. The 10T or 20T filter drives both the luminance delay and the modulator with the same waveform shaping.

At the end of odd fields, there is a half line immediately before an equalizing pulse. Q381 or Q392 abruptly terminates the modulated bar at the end of the half line to avoid spilling it into the equalizer, as would happen with a 10T or 20T falling edge on the modulated bar. See Fig. 5-6.

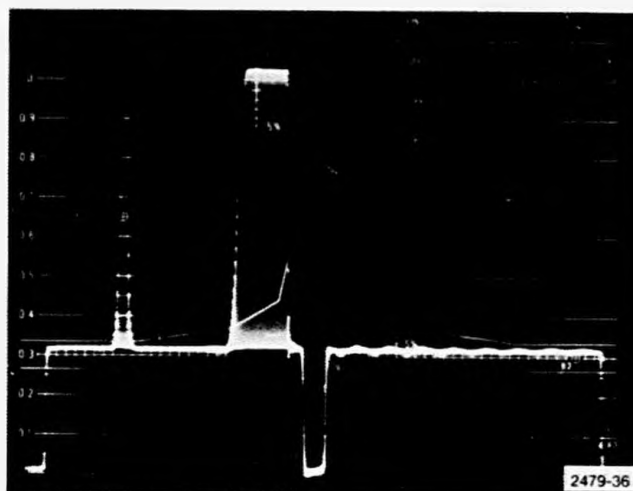


Fig. 5-6. Termination of modulated bar on half-lines.

Since Q381 and Q392 pull the filter output to +5 volts dc at vertical blanking, it is important that the normal quiescent filter output be exactly the same +5 volts dc.

R368 is selected to offset U349 and U375 bias currents in order to meet this condition.

The 10T or 20T filter drives modulator U349 and luminance buffer U375 with exactly the same waveform. The modulator and chrominance bandpass filter introduce a small delay. To match the luminance and chrominance components driving the output amplifier, it is necessary that the luminance be correspondingly delayed. Luminance buffer U375 has a high input-impedance and low output-impedance to drive luminance delay network C398, L408, and C408. The modulated pulse and modulated bar luminance components are switched by the LUMINANCE switch, S437B.

Modulator

The subcarrier for burst, modulated pulse, and modulated bar is controlled by U349, a double-balanced modulator. Referring to Fig. 5-7, the upper four transistors (Q1 through Q4) operate in a switching mode at a subcarrier rate. Q5 and Q6 in Fig. 5-7 operate in a linear mode. Normally, the collector currents for Q5 and Q6 are balanced, resulting in no net subcarrier output. R379, Modulator Balance, and C347, Residual Subcarrier, are adjusted for minimum residual subcarrier.

During burst or chrominance, Q5 and Q6 are unbalanced by the signal from Burst Shaping Filter, L335, C355, and C357 or from the 10T or 20T filter, Q1-Q4 then create an approximate square wave of subcarrier from T358. See Fig. 5-7E.

T358 drives the Chrominance Bandpass Filter, which passes the subcarrier fundamental while rejecting the harmonics. The result is a subcarrier sine wave to drive the Chrominance Output Amplifier. See Fig. 5-7F.

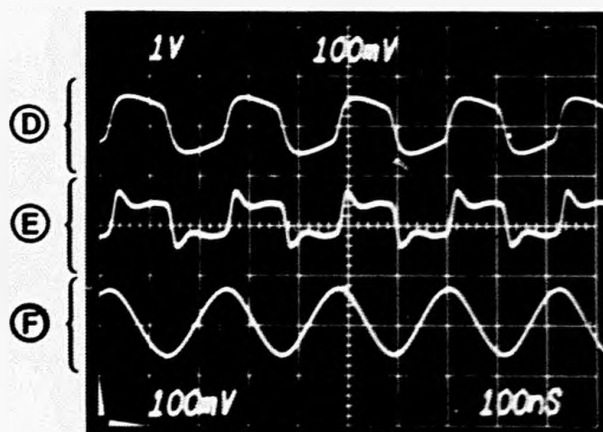
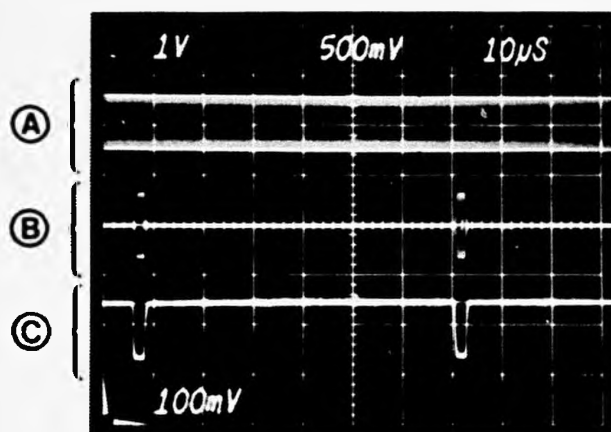
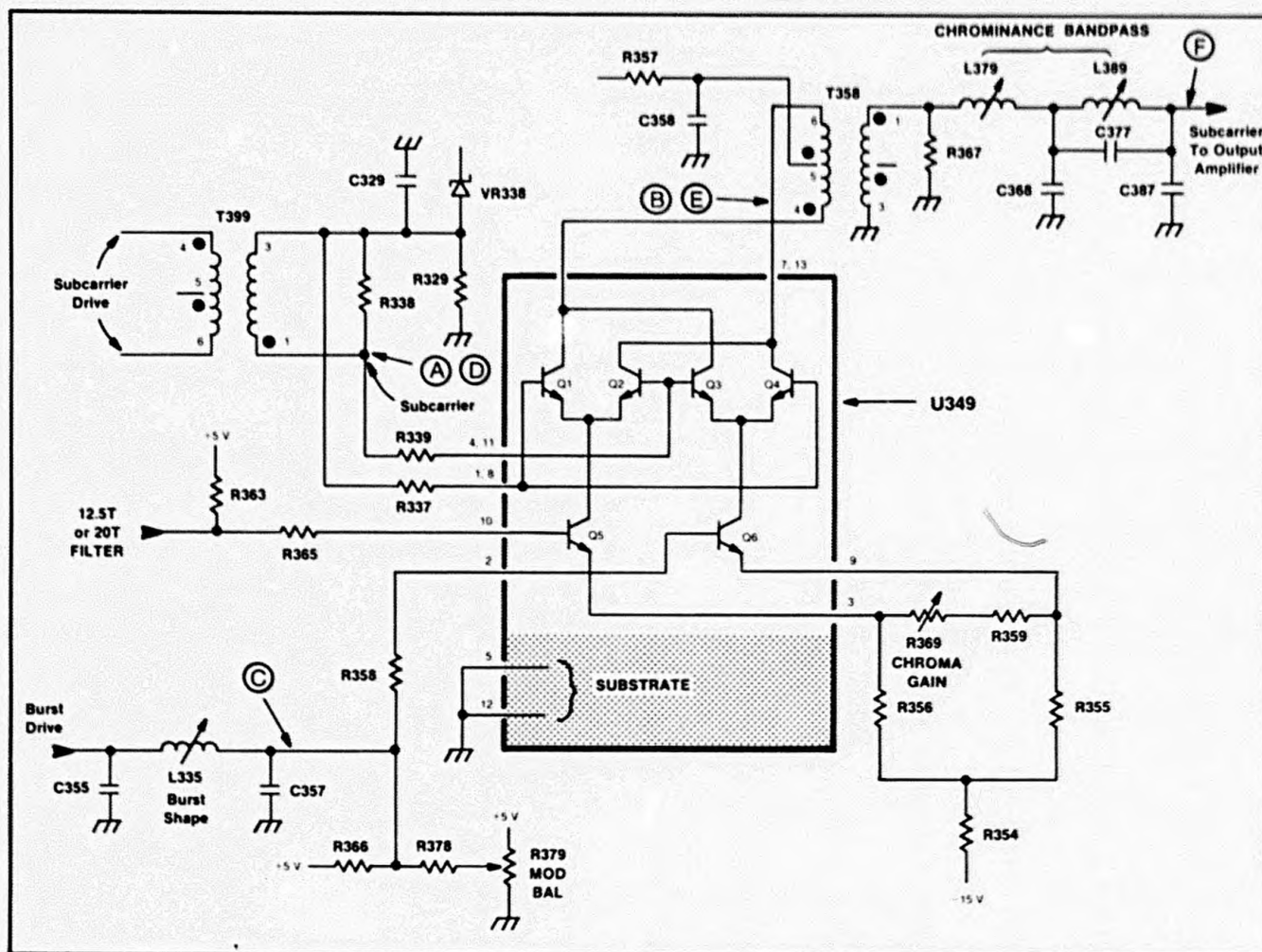
Luminance Filters

Sync, pedestal, and field square-wave all drive the 250 ns filter, L432, L445, and associated components on the output board. The filter is driven by current sources on the logic board, A50-1, and provides a five-pole approximation to sine-squared shaping with a 250 ns risetime. The filter terminates with 500 ohms into the output amplifier. None of the signals through this filter are switched by Luminance Switch S437B.

Luminance pulses and luminance bars are shaped by the 2T, T, and T/2 sine-squared shaping filters on the output board. The fast-rise pulse and bar current drives one of the nine-pole sine-squared shaping filters to drive the output amplifier with the desired shape pulse and bar. S437B switches the luminance pulses and bars off when released.

Output Amplifier

Q449, Q448, Q418, Q438, and Q439 form a high open-loop gain, transimpedance amplifier with near zero input and output impedance. The closed loop gain, V_{out}/I_{in} , is set by R427 and R429. The output of the Output Amplifier is the combination of luminance, chrominance, pedestal, and sync signal, providing a composite video signal to the module's output connector on the 1411 rear panel. The Output Amplifier is capable of driving two 75-ohm loads simultaneously.



NOTE: Test Oscilloscope Bandwidth 20 MHz maximum for all waveforms.

2338-47

Fig. 5-7. Operation of U349, the Double Balanced Modulator.

MAINTENANCE

This section is divided into three parts: Preventive Maintenance, Troubleshooting, and Repair (Corrective Maintenance).

Preventive Maintenance includes inspection, cleaning, and recalibration. Troubleshooting contains information for isolating a problem. Repair provides procedures for removing and replacing components.

PREVENTIVE MAINTENANCE

A regular schedule of preventive maintenance improves instrument reliability. The interval should be determined by the severity of the operating environment.

Cleaning

WARNING

Turn off the instrument power and remove the power cord before cleaning.

Dust accumulating on the circuit boards acts as an insulating blanket, preventing efficient heat dissipation, and causing overheating and possible component breakdown. A layer of dust can also provide an electrical conduction path, especially under high humidity conditions.

CAUTION

Avoid the use of chemical agents that might damage the plastics used in this instrument, including chemicals that contain benzene, freon, toluene, xylene, or similar solvents.

The best way to remove heavy accumulation of dust is to blow it off with dry, low-velocity air. After heaviest accumulation of dust has been removed, the remaining dust can be removed with a small brush followed by a soft cloth dampened in a mild detergent and water solution.

Visual Inspection

Visually inspect the circuit boards during the preventive maintenance routine. Make sure that the boards are properly seated in the mainframe and components are properly seated on the etched-circuit boards. Boards and shields should be held firmly by the plastic retainer bars provided for this purpose.

The corrective procedure for most visible defects is obvious; however, care must be taken to determine and correct the cause of heat-damaged components. Heat damage is sometimes an indication of trouble elsewhere in the instrument.

Periodic transistor and integrated-circuit checks are not recommended. The best performance check for these devices is actual operation of the instrument. Performance of the circuit is thoroughly checked during a performance check or recalibration. Substandard transistors or integrated-circuits are usually detected at that time.

Recalibration

The length of time between recalibration depends on the amount of use the instrument receives, the nature of the environment, or changes in performance when components are replaced.

In general, a partial recalibration is necessary if components are replaced that affect calibration. Complete recalibration is recommended if the board or boards are not operating to their full capability. To ensure correct and accurate operation, performance should be checked at regular intervals; for example, after 1,000 hours of operation if used continuously, or every six months if used infrequently.

A Performance Check is provided in Section 2 and a Calibration Procedure in Section 4.

TROUBLESHOOTING

This part of the section contains general troubleshooting information not contained in Section 6 of the 1411 Instruction Manual. It also contains information on components and applications that are peculiar to the TSG15.

Diagrams

Circuit diagrams are shown on the foldout pages in Section 9. The circuit number and electrical value of each component are shown on the diagrams. Important waveforms are identified and shown; waveform pictures appear on the apron, or leader, of the appropriate schematic.

Circuit Boards

The circuit boards are outlined on the schematic diagrams. Circuit board illustrations are provided on the back of the foldout pages and precede the relevant diagrams. The assembly number assigned to each circuit board is an abbreviated method for identifying the board. The TSG15 contains two etched-circuit board assemblies: A50-1, Pulse & Bar Logic and A51-1, Pulse & Bar Output, and a shield.

When troubleshooting circuit boards in the instrument, the use of an Extender board facilitates access to the board connections and components. Placing the suspected circuit board on the extender will save time in looking for faults. Carefully align the board pin connectors to ensure good electrical contact.

CAUTION

Be sure that pins on the Interface board and/or the Extender board are not bent. Forcing a bent pin into a connector may permanently damage the connector.

Circuit numbers are assigned on a grid system to facilitate component location. Low numbers start at the lower-front corner of the board increasing to the rear and top.

Multi-Pin Connectors

Board output signals are fed to the rear-panel connectors through coaxial cables and multi-pin connectors. A triangular key symbol is also located on the circuit board to identify pin No. 1 of the plug jumper (see Fig. 6-1) so that the connector can be properly oriented.

Troubleshooting Hints

Before going into a full-scale circuit by circuit investigation, check control settings. Incorrect control settings or wrong internal-jumper positions can indicate a trouble that does not exist. If there is any question about the correct function of any control or jumper, refer to Operating Instructions or Installation.

Before troubleshooting a circuit board, check that the correct input signals and power supplies are present. Make sure that other boards on the Interface board are not defective. For example, check the SPG before looking for a problem in the TSG15.

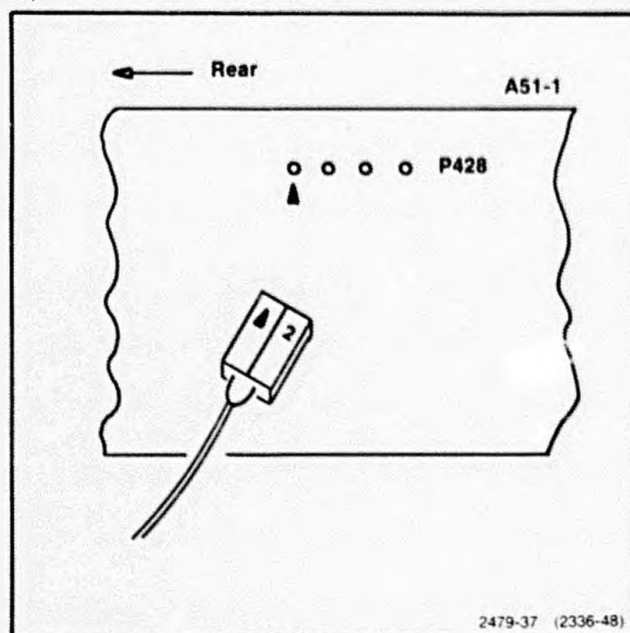


Fig. 6-1. Output connector pin numbering.

Symptoms will often identify the circuit in which the trouble is located. Incorrect operation of all circuits may mean trouble in the power supply. Consider this possibility if voltages are incorrect. Make sure that all board pin-connectors are making good contact before proceeding with trouble isolation.

Checking for correct voltages and waveforms usually pinpoints the stage or component that has failed. Typical waveforms are given to the left of the diagrams, where applicable.

CAUTION

Due to component density on the circuit boards, special care should be exercised when using meter or oscilloscope leads and probe tips. Accidental shorts can cause abnormal voltages or transients that may destroy components.

WARNING

"Ground lugs" are not always at ground potential. Check the diagrams before using such connections as ground for meter or oscilloscope.

CORRECTIVE MAINTENANCE

Corrective maintenance consists of component replacement and circuit board repair. Special techniques required to replace components in this instrument are given here.

Parts Location

The exploded-view drawings associated with the Replaceable Mechanical Parts list (at the rear of the manual) are helpful in the removal or disassembly of components or subassemblies. Circuit-board illustrations are provided on the backs of foldout pages in the Diagrams section of this manual.

Selected Components

Four resistors in the TSG15 are selected for optimum circuit performance. R445 (2T), R446 (T), and R447 (T/2) are selected for pulse half-amplitude duration (HAD). These resistors only need to be selected if filter components are changed. R368 is selected to match residual subcarrier at the beginning of vertical blanking following the modulated bar with residual subcarrier on active lines. R368 need be selected only if the modulator, U349, is replaced. With the old R368 removed, select a modulated bar with window and set Mod Bal, R379, for a minimum residual subcarrier on the active lines. Measure the amount of residual subcarrier at the beginning of vertical blanking. The value of R368 typically is within a range of 200 k Ω to 1 M Ω . Formula for selection of R368 (in kilohms) is:

$$\frac{2000}{\text{Residual Subcarrier (in mV)}}$$

Install new R368 and readjust R379, Mod Bal, for minimum residual subcarrier.

Circuit Board Replacement

If a circuit board is damaged beyond repair, the entire assembly, including all components, can be replaced. Tektronix part numbers are given in the Replaceable Electrical Parts list.

Circuit Board Removal

In order to work on one of the TSG15's circuit boards, it is necessary to extend it above the 1411 Mainframe. A Circuit-Board Extender is provided with the 1411 for this purpose. It will allow access to components while retaining connection to the 1411 power supplies and signal interface lines.

Board Removal

1. Remove the two Phillips-head screws mounting the TSG15 front panel.
2. Disconnect the signal output connector from the board (A51-1 only). Note the location of this connector so it can be correctly reconnected.
3. Grasp the board at both ends and pull straight up from the Interface board, until physical connection between Interface pins and the TSG15 is broken. Remove the board by tilting the rear of the board upward and pulling up and back.

When one of the TSG15 circuit boards is to be placed on an extender, both boards should be mounted so that there is an empty row of pins between. See Fig. 6-2.

It is not normally necessary to connect the remote plug (Interface lines 55-56-57) to the TSG15 when using the Extender board. Only the Module Output coaxial cable need be reconnected when troubleshooting the TSG15.

To replace the board, reverse the order of removal. Use the mating plastic guides to align pin connectors. Pin 1 of the plug is toward the rear of the connector on the board (A51-1); reconnect the module output cable. See Fig. 6-1.

Extracting Integrated Circuits

An extracting tool should be used to remove the 14- and 16-pin dual in-line packaged integrated circuits used in the TSG15. A tool for this purpose is available from Tektronix, Inc. Order Tektronix Part No. 003-0619-00. If an extracting tool is not available, use care to avoid damaging the pins. Pull slowly and evenly on both ends of the IC. Try to avoid having one end of the IC disengage from the socket before the other.

Pushbutton Switch Replacement

These instructions are valid for all pushbutton switches except the LUMINANCE switch. Before removing a pushbutton switch, disengage all pushbutton actuating arms so that they do not project beyond the rear of the switch. Remove the switch extender by pulling it straight off. Next, carefully pry back the plastic retainer clip at the rear of the switch with the tip of a small screwdriver (see Fig. 6-3). Remove the switch by lifting the switch body up and back from the front retainer clip.

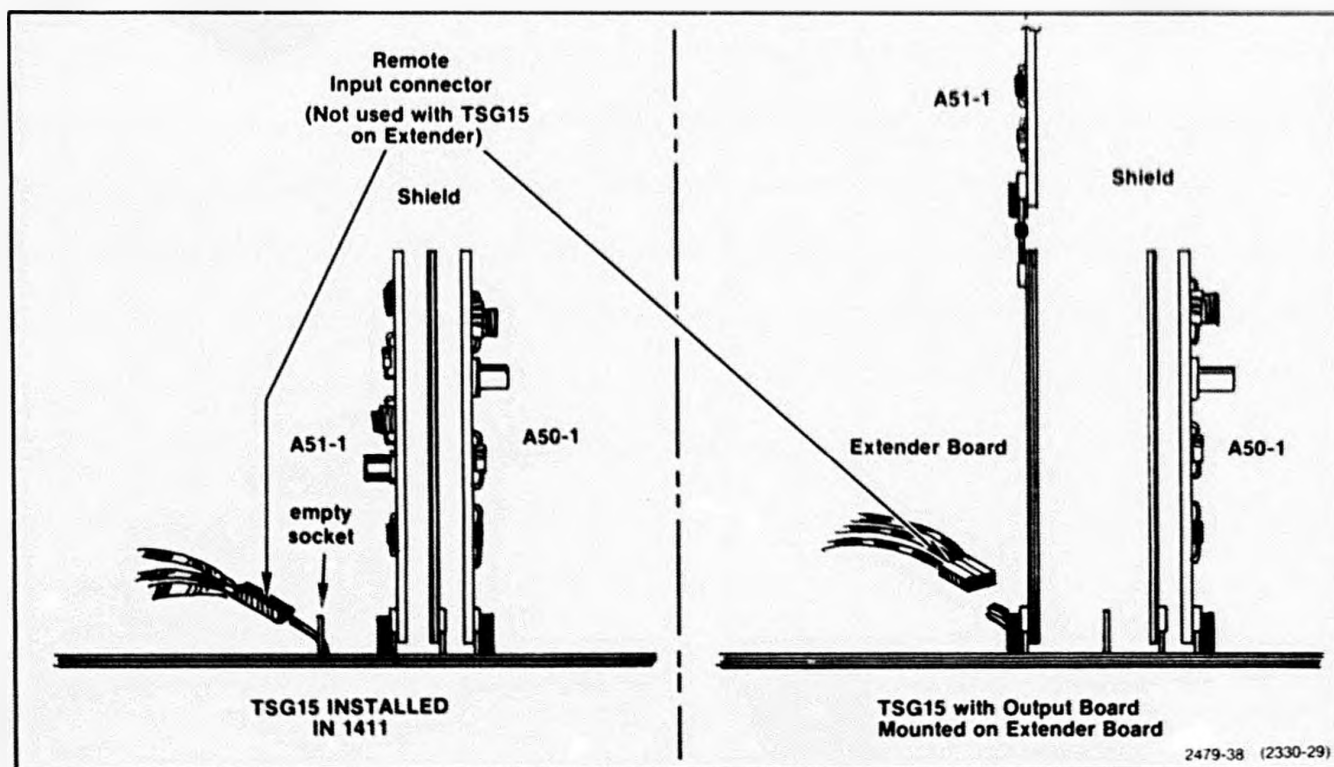


Fig. 6-2. TSG15 circuit boards shifted to accommodate Pulse and Bar Output circuit board (A51-1) on circuit board extender.

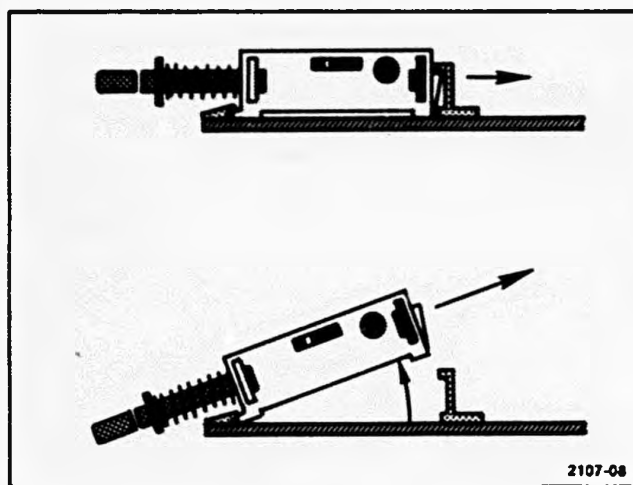


Fig. 6-3. Pushbutton switch removal.

Reverse the removal procedure to install the replacement switch.

LUMINANCE Switch

It is necessary to unsolder six mounting posts from the etched circuit board to remove the LUMINANCE switch.

After removing the switch-shaft extender, use a desoldering tool to remove as much solder as possible from the area of the mounting studs. Alternately heat each of the switch mounting posts while applying a slight amount of upward pressure to the switch body. The switch should be lifted straight up from the etched-circuit board.

When the switch has been removed from the etched-circuit board, use a desoldering tool to remove any remaining solder from the switch mounting holes.

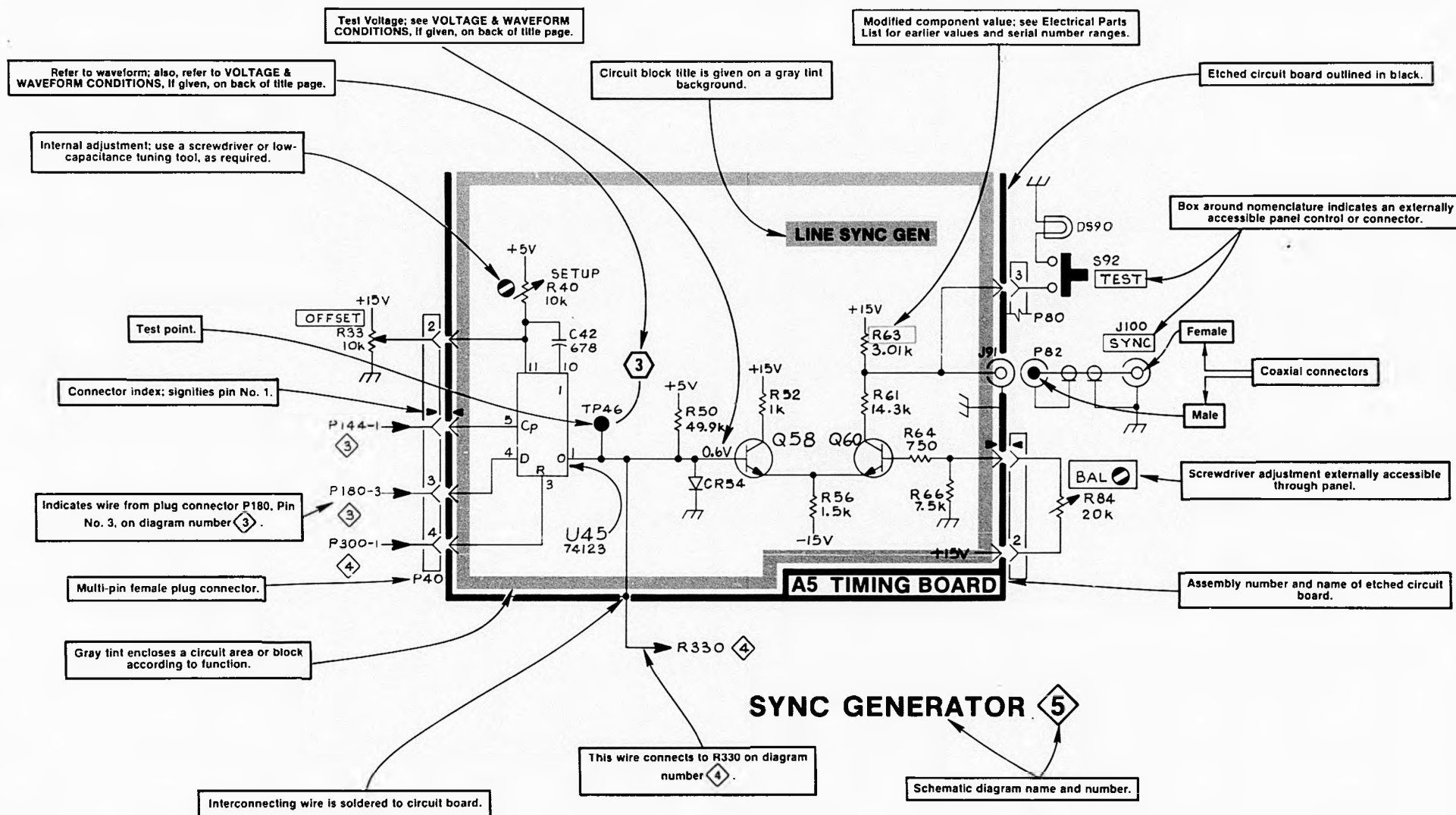
To install the LUMINANCE switch: First place the two plastic spacers on the switch mounting posts. Then position the switch on the circuit board. Make sure the switch body is properly positioned before soldering the contacts. See the 1411 Instruction Manual (Section 6) for additional information under Soldering Techniques.

The final step is to reinstall the switch-shaft extender.

Information contained in this section serves as an aid to the service technician who performs the calibration, maintenance, and troubleshooting procedures. Included are illustrations for each circuit board showing the adjustment and jumper locations.

2479-39A

REV A NOV 1980



SCHEMATIC EXAMPLE

(2234)2238-50

DIAGRAMS & CIRCUIT BOARD ILLUSTRATIONS

This section of the manual contains block and schematic diagrams with waveforms, and etched circuit board illustrations.

Symbols

Symbols used on the diagrams are based on ANSI Y32.2-1975 and IEEE No. 315 March 1971. Logic symbology is based on ANSI Y32.14-1973 (IEEE Std. 91-1973). Logic symbols depict the logic function performed and may differ from the manufacturer's data.

Component values

Electrical components shown on the diagrams are in the following units unless noted otherwise:

Capacitors = Values one or greater are in picofarads (pF).
Values less than one are in microfarads (μ F).

Resistors = Ohms (Ω).

Semiconductor Types

Refer to the Replaceable Electrical Parts list.

Reference Designators

The following letters are used as reference designators to identify components or assemblies on Tektronix, Inc. schematic diagrams.

A	Assembly, separable or repairable (circuit board, etc.)	LR	Inductor/resistor combination
AT	Attenuator, fixed or variable	M	Meter
B	Motor	P	Connector, movable portion
BT	Battery	Q	Transistor, silicon-controlled rectifier, or program-mable unijunction transistor
C	Capacitor, fixed or variable	R	Resistor, fixed or variable
CR	Diode, signal or rectifier	RT	Thermistors
DH	Decoupling Hybrid	S	Switch
DL	Delay Line	T	Transformer
DS	Indicating device (lamp)	TC	Thermocouple
E	Spark Gap	TP	Test Point
F	Fuse	U	Assembly, inseparable or non-repairable (integrated circuit, etc.)
FL	Filter	V	Electron tube
H	Heat dissipating device (heat sink, heat radiator, etc.)	VR	Voltage regulator (zener diode, etc.)
HR	Heater	Y	Crystal
J	Connector, stationary portion		
K	Relay		
L	Inductor, fixed or variable		

Partial Schematic Diagram With Explanations

The partial diagram at the left is an example of the various symbols and other information provided on Tektronix, Inc. diagrams.

TSG15

WAVEFORM CONDITIONS

Waveform photographs in this section were taken with a TEKTRONIX C-27 Oscilloscope Camera, mounted on a TEKTRONIX 7603 Oscilloscope with 7A13 Differential Comparator, 7A26 Dual-Trace Amplifier and 7B53A Time Base plug-in units.

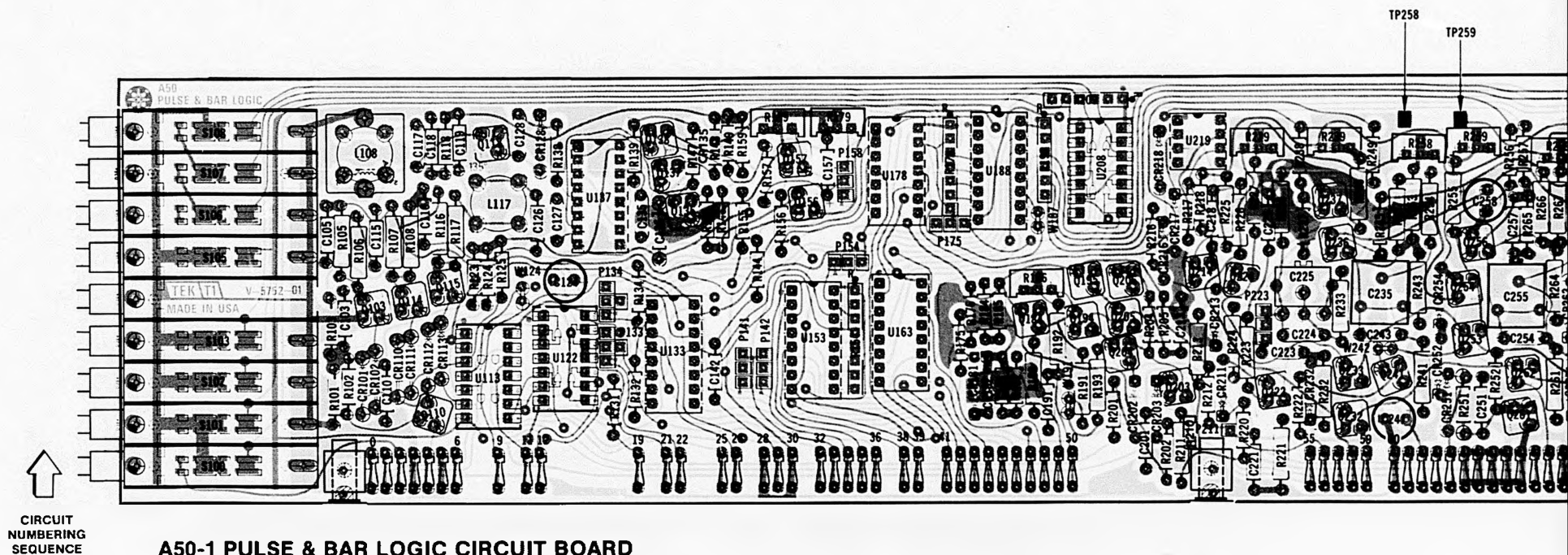
All Waveforms in this section were taken as shown. The front-panel switches were set for the appropriate signal. Triggering can be internal or from the 1410 Sync, Subcarrier, or H or V Drive depending on signal repetition rates.

Unless displaying other specific signals, such as T/2 pulse in the pulse drive, the waveforms were taken using 2T and 12.5T (modulated) pulses and the T bar.

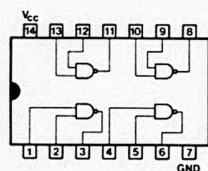
All program jumpers are assumed to be in their factory-set positions.

NOTE

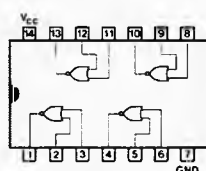
All waveforms and dc levels are not absolute, and may vary between instruments because of component and calibration tolerances.



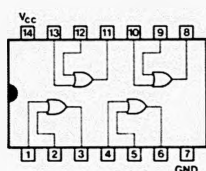
A50-1 PULSE & BAR LOGIC CIRCUIT BOARD



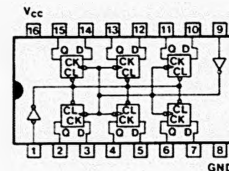
74LS00



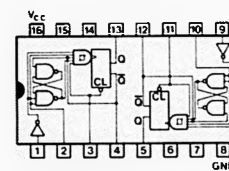
74LS02



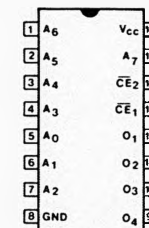
74LS32



74LS174

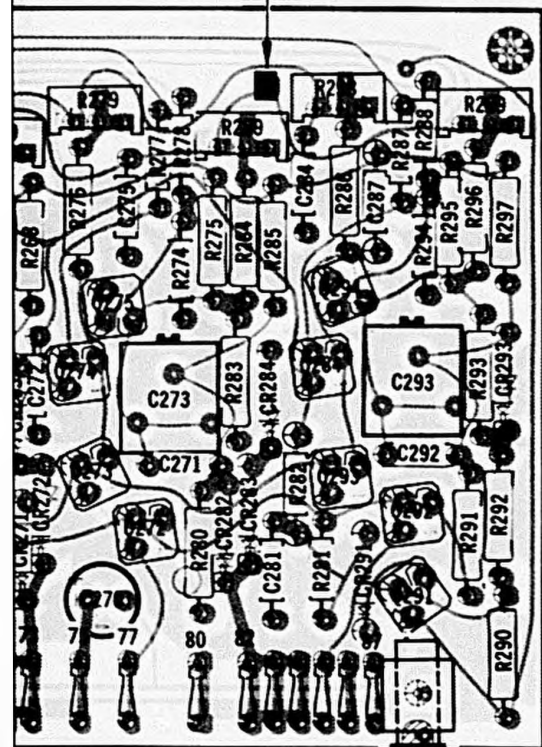


74LS221

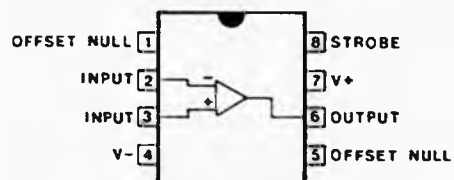


93427

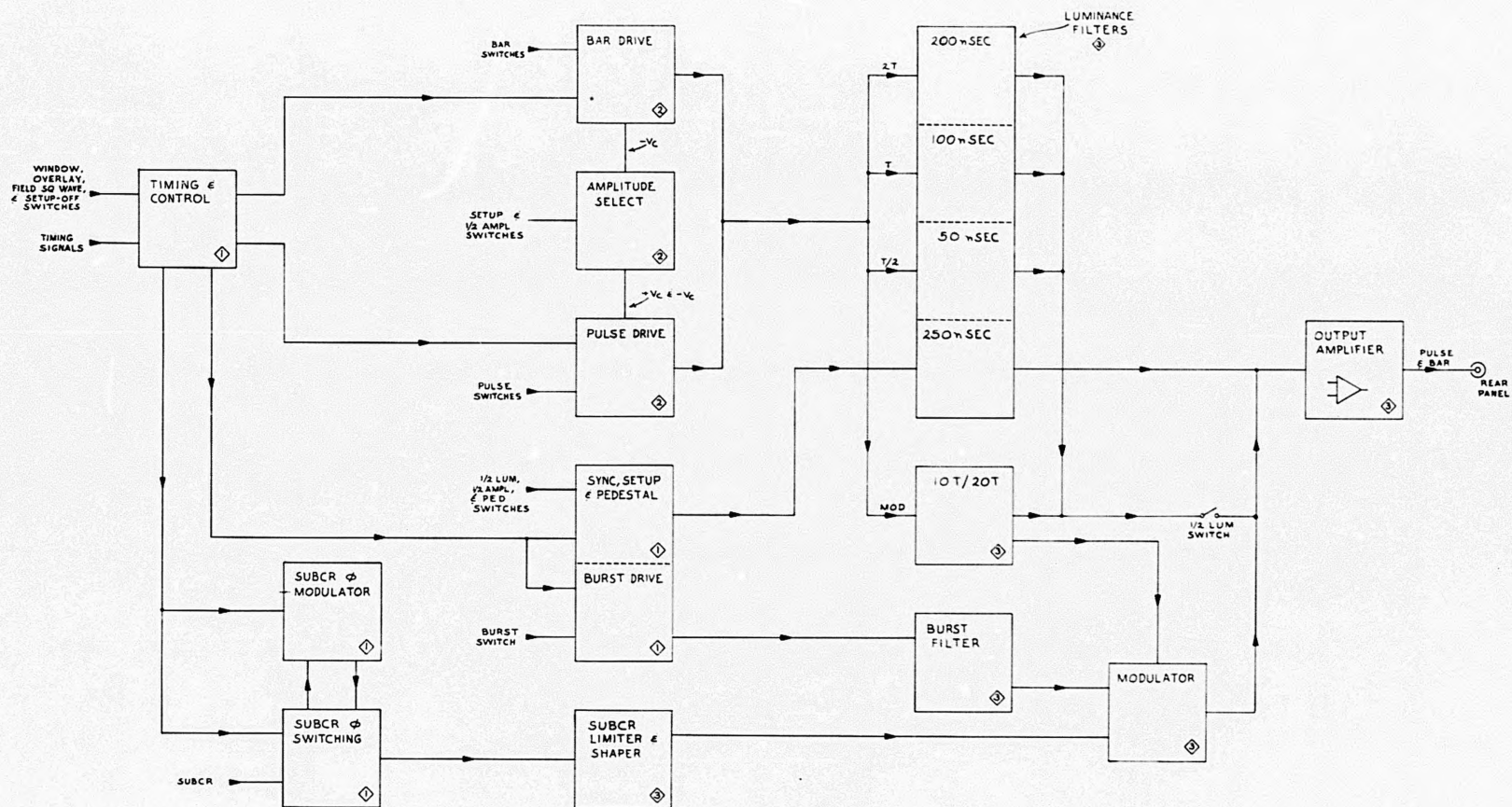
TP289



2479-41A



CA3160E

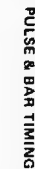


TSG 15

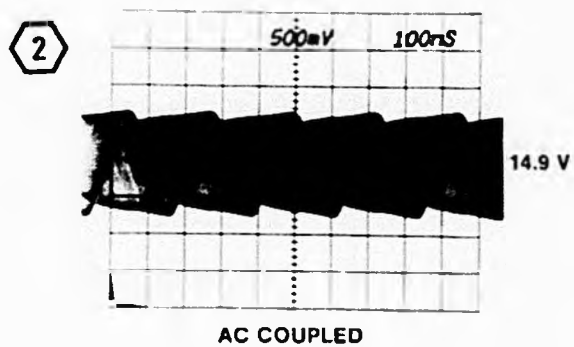
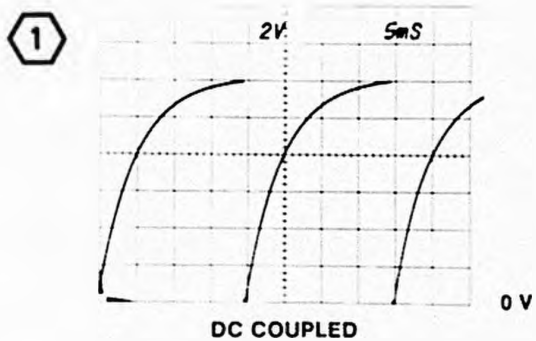
© 1968 2/78

BLOCK DIAGRAM

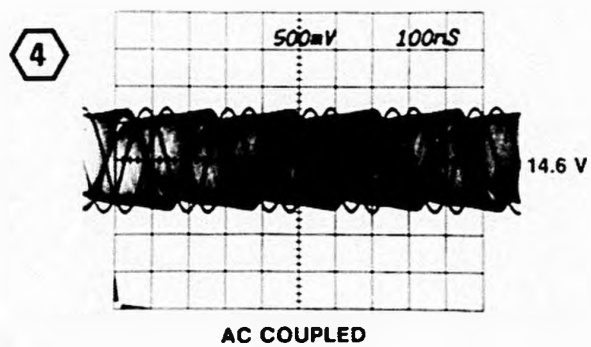
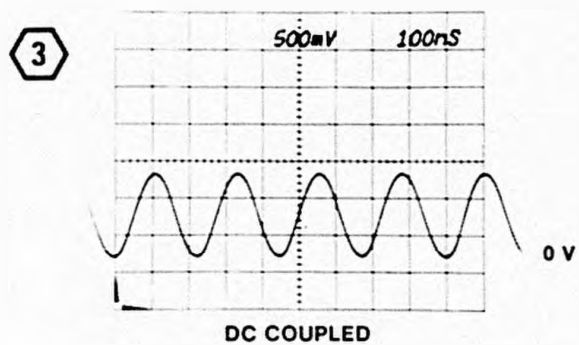
BLOCK DIAGRAM



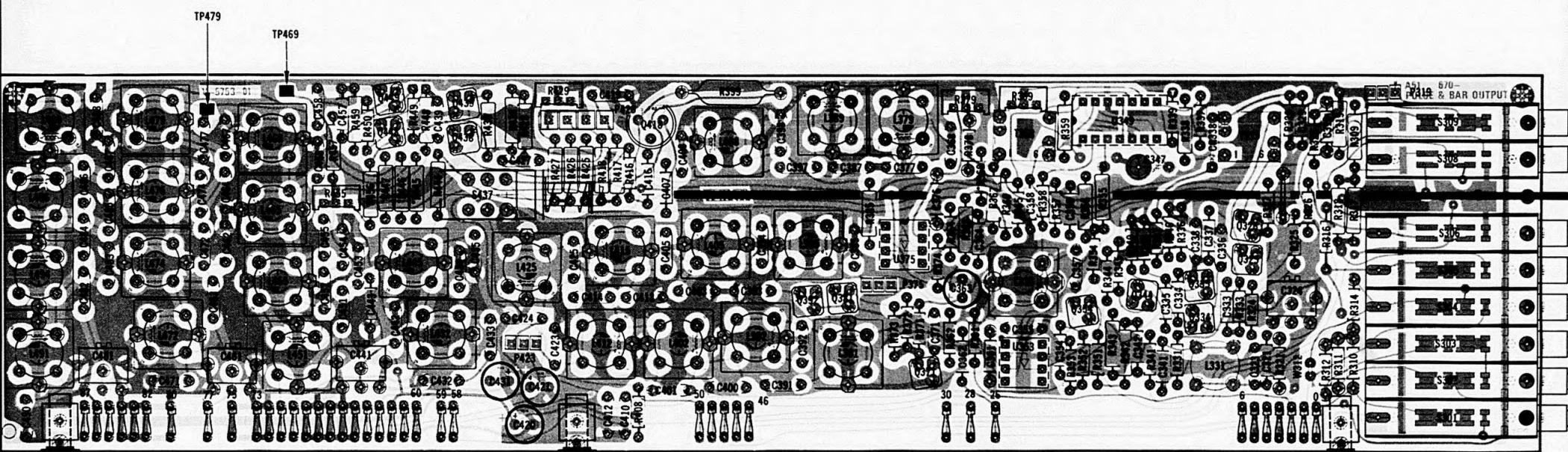
1
WAVEFORMS



1
**A51 PULSE & BAR OUTPUT
WAVEFORMS**



@



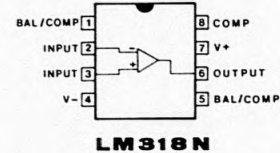
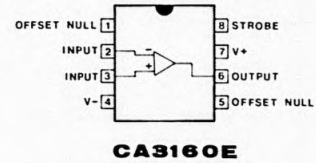
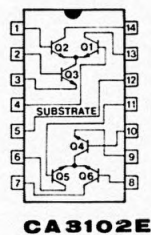
*Selected

A51-1 PULSE & BAR OUTPUT CIRCUIT BOARD

BOTTOM

←
↑
**CIRCUIT
NUMBERING
SEQUENCE**

2479-42



PULSE & BAR TIMING PARTS LOCATING CHART

C103	D5
C105	C4
C110	C5
C115	C5
C116	D4
C117	B4
C118	B4
C119	B4
C127	A5
C128	B5
C135	A5
C136	B4
C142	D4
C157	A4
C181	D3
C191	C3
C201	D3
C213	D2

CR101 C5
CR102 C5
CR110 D5
CR111 D5
CR112 D5
CR113 D5
CR128 B5
CR135* B4
CR181 C2
CR182 C2
CR202 D2
CR203 D2
CR213 D2

L108	85
L118	D5

P133	A1
P134	A2
P154	B1
P158	A4
P319	B1

Q103	C5
Q110	D5
Q114	D5
Q118	B5
Q135	B5

Q137	A5
Q138	A5
Q156	D4
Q157	A4
Q184	C2
Q194	C3
Q195	C2
Q203	D3
Q204	D3
Q205	C3
Q206	D2

R101	D5
R102	D5
R103	D5
R105	C5
R106	C5
R107	D5
R108	D5
R116	D5
R117	D5
R118	B5
R123	C5
R124	C5
R125	C5
R131	C2
R132	C2
R134	A2
R138	B5
R139	B5
R144	D4
R145	A4
R146	D4
R147	A5
R148	A5
R149	A5
R154A	B2
R154B	B2
R154D	B2
R154E	B2
R154F	B2
R154G	B2
R155	D4
R156	A4
R157	A5
R159	A5
R169	A5
R175	C2
R176	C2

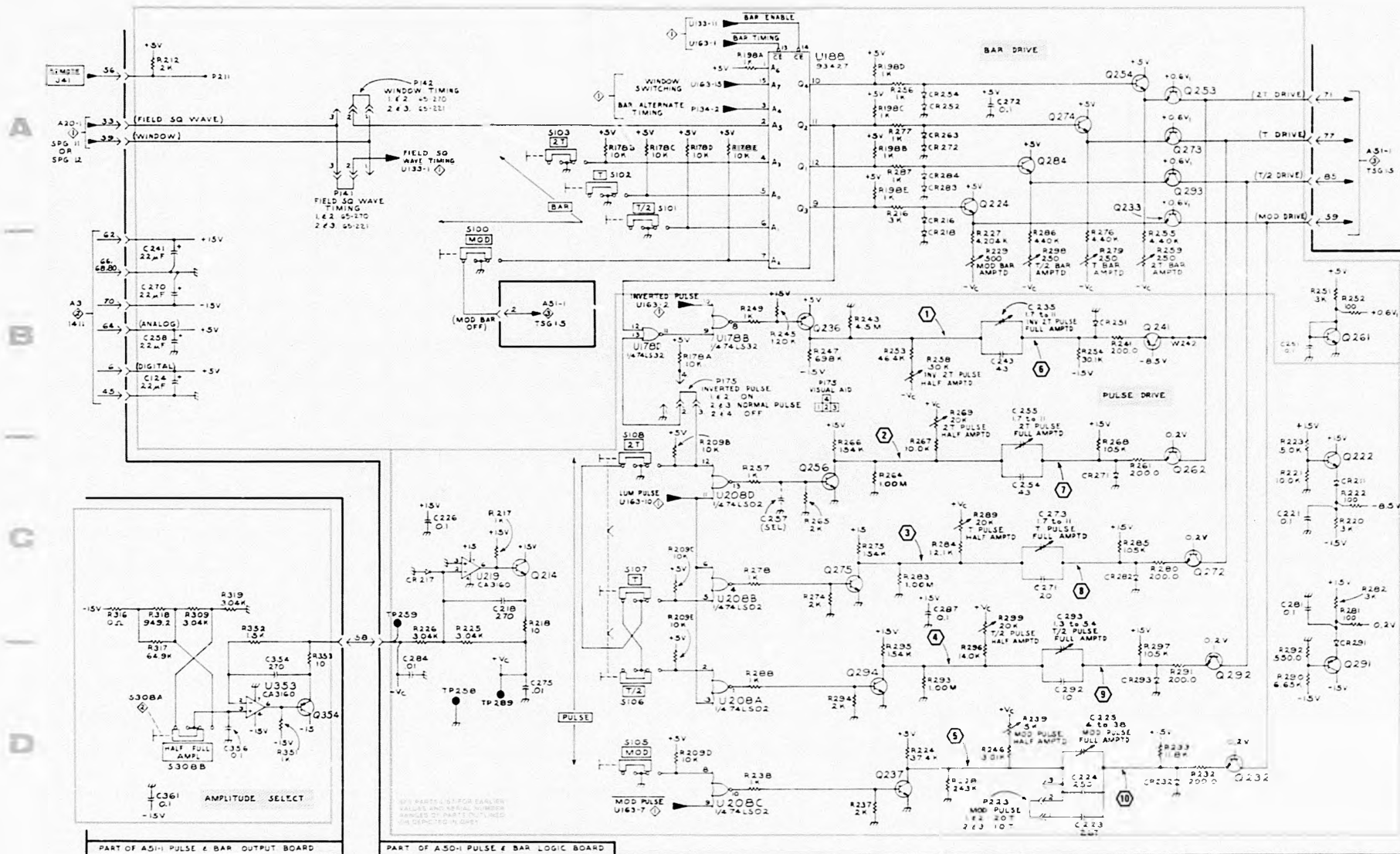
R179	A4
R182	C3
R183	C3
R184	C2
R185	C2
R186	C2
R191	C2
R192	C3
R193	C2
R201	D2
R202	D1
R204	D2
R205	D2
R210*	D2
R211	D2
R213	D2
R310	D1
R311	C1
R312	A1
R314	C1
R321	A1
R408	B3

S301	D1
S302	C1
S303	A1
S306A	C3
S308B	C3
S309	A1
S437A	C3

U113A C5
U113B C5
U113C D2
U113D D2
U122A C4
U122B D2
U122D C4
U133 C2
U137A B5
U137B B5
U153 B2
U163 A2
U178A A4
U178B A1

W124 C4
W191 C2

*See Parts List for serial number ranges.



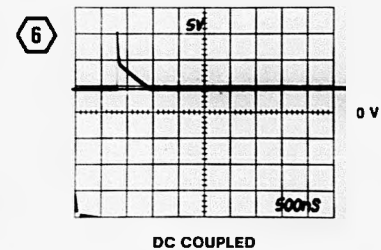
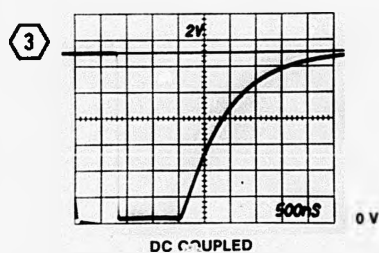
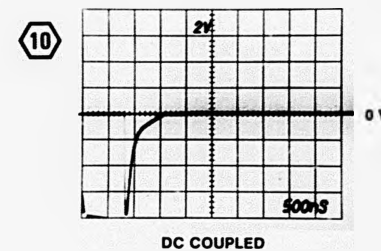
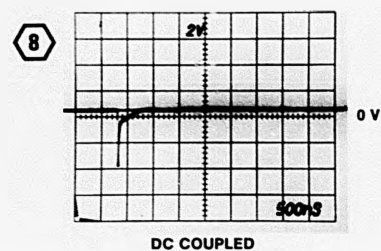
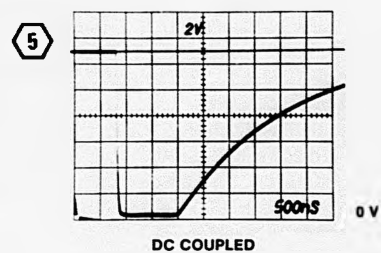
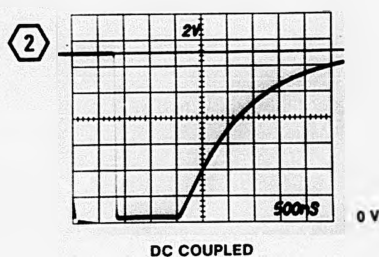
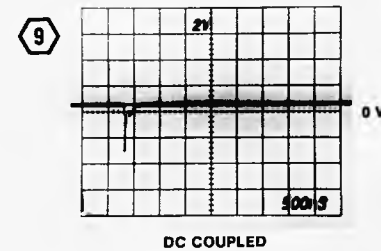
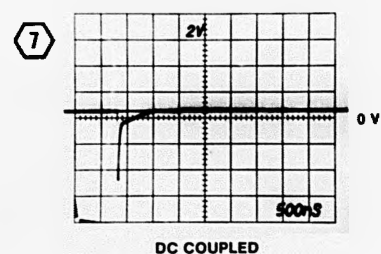
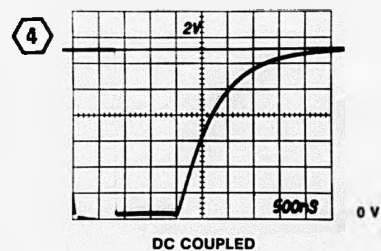
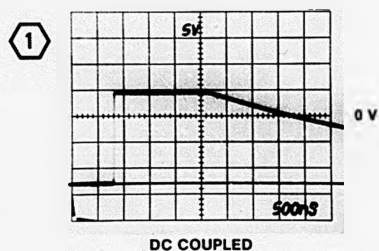
TSG 15

REV. JUL. 1985

PULSE & BAR DRIVERS

2

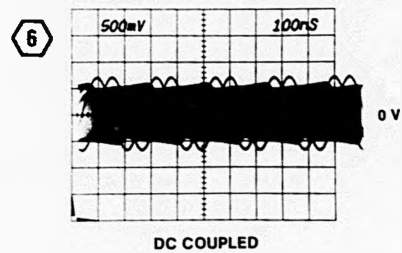
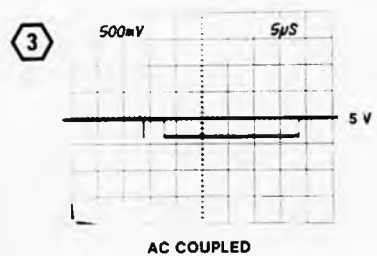
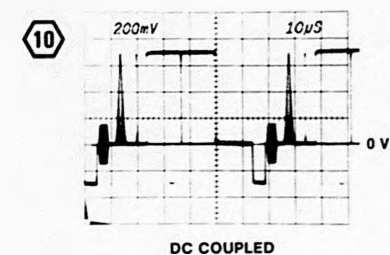
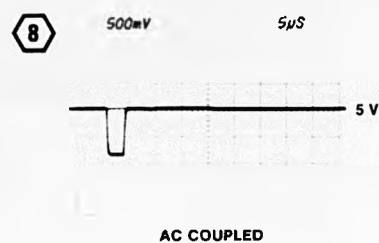
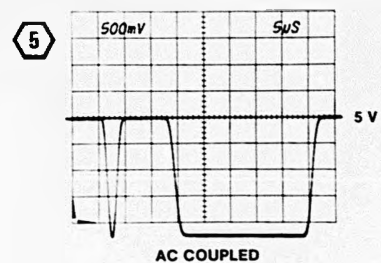
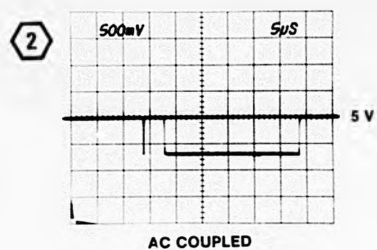
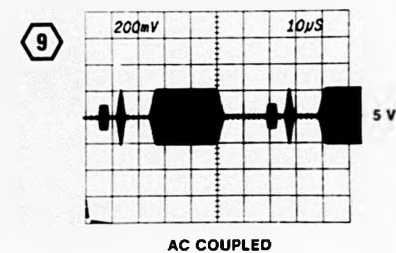
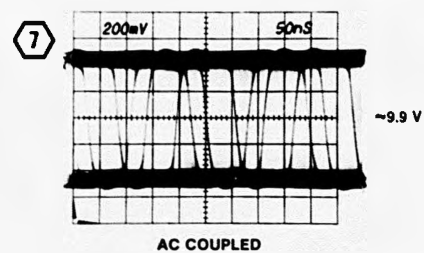
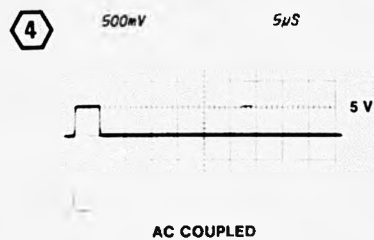
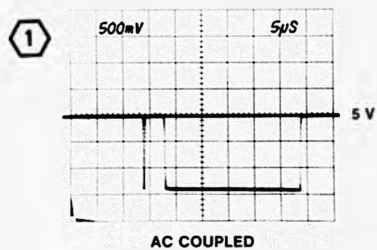
2 WAVEFORMS



PULSE & BAR DRIVERS **PARTS LOCATING CHART**

C124 B1	P175 B3	R224 D4	R293 D4
C218 C2	P223 D4	R225 C2	R294 D3
C221 C5		R226 C2	R295 D4
C223 D4		R227 B4	R296 D4
C224 D4		R228 D4	R297 D5
C225 D4	Q214 C2	R229 B4	R298 B4
C226 C2	Q222 C5	R232 D5	R299 C4
C235 B4	Q224 A4	R233 D5	R309 C1
C241 B1	Q232 D5	R237 D4	R316 C1
C243 B4	Q233 A5	R238 D3	R317 D1
C251 B5	Q236 B3	R239 D4	R318 C1
C254 C4	Q237 D4	R241 B5	R319 C1
C255 B4	Q241 D5	R243 B4	R351 D1
C257 C3	Q253 A5	R245 B3	R352 C1
C258 B1	Q254 A5	R246 D4	R353 D2
C270 B1	Q256 C3	R247 B3	
C271 C4	Q261 B5	R249 B3	
C272 A4	Q262 C5	R251 B5	S100 B2
C273 C4	Q272 C5	R252 B5	S101 A3
C275 D2	Q273 A5	R253 B4	S102 A3
C281 C5	Q274 A4	R254 B4	S103 A2
C284 D2	Q275 C4	R255 B5	S105 D3
C287 C4	Q284 A4	R256 A4	S106 D3
C292 D4	Q291 D5	R257 C3	S107 C3
C293 C4	Q292 D5	R258 B4	S108 C3
C354 D1	Q293 A5	R259 B5	S308B D1
C356 D1	Q294 D4	R261 C5	
C361 D1	Q354 D2	R264 C4	
		R265 C3	
		R266 C3	
		R267 C4	TP258 D2
		R268 B4	TP259 C2
		R269 B4	TP289 D2
		R274 C3	
		R275 C4	
		R276 B4	U178B B3
		R277 B4	U178D B3
		R278 C3	U188 A3
		R279 B4	U208A D3
		R280 C5	U208B C3
		R281 C5	U208C D3
		R282 C5	U208D C3
		R283 C4	U219 C2
		R284 C4	U353 D1
		R285 C5	
		R286 B4	
		R287 A4	
		R288 D3	W242 B5
		R289 C4	
		R290 D5	
		R291 D5	
		R292 D5	
CR211 C5	R178A B3		
CR216 A4	R178B A3		
CR217 C2	R178C A3		
CR218 A4	R178D A3		
CR232 D5	R178E A3		
CR251 B4	R198A A3		
CR252 A4	R198B A4		
CR254 A4	R198C A4		
CR263 A4	R198D A4		
CR271 C4	R209B C3		
CR272 A4	R209C C3		
CR282 C5	R209D D3		
CR283 A4	R209E C3		
CR284 A4	R212 A1		
CR291 D5	R216 A4		
CR293 D5	R217 C2		
	R218 C2		
	R220 C5		
	R221 C5		
	R222 C5		
	R223 C5		
P141 A2			
P142 A2			

3 WAVEFORMS



PULSE & BAR OUTPUT PARTS LOCATING CHART

C319 D2	C431 D5	L432 B2	R341 D1	R458 A4
C321 D1	C432 B2	L445 B2	R342 D1	R459 A4
C323 D1	C433 B2	L451 A1	R343 D1	
C324 D1	C434 B2	L455 A2	R344 D1	
C329 D3	C435 B2	L465 A2	R345 C1	
C333 C2	C439 A4	L468 A2	R346 D2	S347B B4
C334 D2	C441 A1	L472 A1	R347 D2	
C335 D2	C442 A1	L472 A2	R348 D2	
C336 C2	C443 B2	L476 A2	R349 D2	
C337 D2	C451 A1	L479 A2	R354 D3	T339 D2
C338 D2	C452 A2	L491 A1	R355 D3	T358 D3
C341 C1	C453 A1	L494 A2	R356 D3	
C342 D1	C454 A2	L496 A2	R357 C3	
C347 C3	C455 A2	L499 A2	R358 D3	TP469 A2
C355 D1	C456 A2		R359 D3	TP479 A2
C357 D2	C457 A4		R361 C1	TP489 A2
C358 C3	C458 A2		R362 C1	
C359 C3	C461 A1	P375 B3	R363 D3	
C362 C1	C462 A1	P423 B1	R364 C3	
C363 D2	C463 A1	P428 B5	R365 D3	U349 D3
C364 D2	C464 A2		R366 D3	U375 B3
C368 D4	C467 A2		R367 D4	
C371 B3	C471 A1		R368 B3	
C376 C4	C472 A2	Q325 C2	R369 D3	
C377 D4	C474 A2	Q326 D2	R371 B1	VR338 C3
C385 C3	C477 A2	Q334 C2	R372 C2	
C387 D4	C481 A1	Q335 D2	R373 B2	
C391 D2	C482 A1	Q344 D1	R374 B3	
C392 B2	C483 A1	Q371 C1	R378 D2	
C393 C2	C484 A2	Q381 B2	R379 D2	
C394 C2	C485 A2	Q392 C2	R385 B3	
C397 D4	C486 A2	Q418 B5	R399 B5	
C398 C3	C487 A2	Q438 A5	R416 B5	
C400 B2	C488 A2	Q439 A5	R417 B5	
C401 B2	C490 A1	Q448 A4	R418 B5	
C403 C2		Q449 A4	R425 B4	
C405 C2			R426 D4	
C407 A5			R427 B4	
C408 B4	L331 D1		R428 A5	
C410 B2	L335 D2	R324 D2	R429 B5	
C412 B1	L379 D4	R325 D2	R438 A5	
C413 C2	L381 B3	R326 D2	R439 A5	
C414 B2	L389 D4	R327 D2	R444 B2	
C416 A4	L392 B2	R328 D2	R445 A2	
C419 B5	L395 C3	R329 C3	R446 A2	
C420 D5	L402 B2	R331 C1	R447 A2	
C421 D5	L405 C2	R333 C2	R448 A4	
C423 B1	L408 B4	R336 C2	R449 A4	
C424 B1	L412 B2	R337 D3	R455 B4	
C425 C2	L415 C2	R338 D2	R456 B4	
C427 B5	L425 C2	R339 D3	R457 A4	

REPLACEABLE ELECTRICAL PARTS

PARTS ORDERING INFORMATION

Replacement parts are available from or through your local Tektronix, Inc. Field Office or representative.

Changes to Tektronix instruments are sometimes made to accommodate improved components as they become available, and to give you the benefit of the latest circuit improvements developed in our engineering department. It is therefore important, when ordering parts, to include the following information in your order: Part number, instrument type or number, serial number, and modification number if applicable.

If a part you have ordered has been replaced with a new or improved part, your local Tektronix, Inc. Field Office or representative will contact you concerning any change in part number.

Change information, if any, is located at the rear of this manual.

ITEM NAME

In the Parts List, an Item Name is separated from the description by a colon (:). Because of space limitations, an Item Name may sometimes appear as incomplete. For further Item Name identification, the U.S. Federal Cataloging Handbook H6-1 can be utilized where possible.

ABBREVIATIONS

ACTR	ACTUATOR	PLSTC	PLASTIC
ASSY	ASSEMBLY	QTZ	QUARTZ
CAP	CAPACITOR	RECP	RECEPTACLE
CER	CERAMIC	RES	RESISTOR
CKT	CIRCUIT	RF	RADIO FREQUENCY
COMP	COMPOSITION	SEL	SELECTED
CONN	CONNECTOR	SEMICON	SEMICONDUCTOR
ELCTLT	ELECTROLYTIC	SENS	SENSITIVE
ELEC	ELECTRICAL	VAR	VARIABLE
INCAND	INCANDESCENT	WW	WIREWOUND
LED	LIGHT EMITTING DIODE	XFMR	TRANSFORMER
NONWIR	NON WIREWOUND	XTAL	CRYSTAL

CROSS INDEX - MFR. CODE NUMBER TO MANUFACTURER

Mfr. Code	Manufacturer	Address	State, Zip Code
00853	SANGAMO WESTON INC	SANGAMO RD	PICKENS SC 29671
	SANGAMO CAPACITOR DIV	P O BOX 128	
01121	ALLEN-BRADLEY CO	1201 SOUTH 2ND ST	MILWAUKEE WI 53204
01295	TEXAS INSTRUMENTS INC	13500 N CENTRAL EXPRESSWAY	DALLAS TX 75265
	SEMICONDUCTOR GROUP	P O BOX 225012 M/S 49	
01686	RCL ELECTRONICS INC	195 MCGREGOR ST	MANCHESTER NH 03102
02735	RCA CORP	ROUTE 202	SOMERVILLE NJ 08876
	SOLID STATE DIVISION		
03508	GENERAL ELECTRIC CO	W GENESEE ST	AUBURN NY 13021
	SEMI-CONDUCTOR PRODUCTS DEPT		
04222	AVX CERAMICS DIV OF AVX CORP	19TH AVE SOUTH	MYRTLE BEACH SC 29577
		P O BOX 867	
04713	MOTOROLA INC	5005 E MCDOWELL RD	PHOENIX AZ 85008
	SEMICONDUCTOR GROUP		
07716	TRW INC	2850 MT PLEASANT AVE	BURLINGTON IA 52601
	TRW ELECTRONICS COMPONENTS		
	TRW IRC FIXED RESISTORS/BURLINGTON		
11236	CTS OF BERNE INC	406 PARR ROAD	BERNE IN 46711
18324	SIGNETICS CORP	811 E ARQUES	SUNNYVALE CA 94086
19701	MEPCO/ELECTRA INC	P O BOX 760	MINERAL WELLS TX 76067
	A NORTH AMERICAN PHILIPS CO		
24226	GOWANDA ELECTRONICS CORP	179 BROADWAY RD	GOWANDA NY 14070
24546	CORNING GLASS WORKS	550 HIGH ST	BRADFORD PA 16701
31918	ITT SCHADOW INC	8081 WALLACE RD	EDEN PRAIRIE MN 55343
32997	BOURNS INC	1200 COLUMBIA AVE	RIVERSIDE CA 92507
	TRIMPOT DIV		
50434	HEWLETT-PACKARD CO OPTOELECTRONICS DIV	640 PAGE MILL RD	PALO ALTO CA 94304
52768	SPRAGUE-GOODMAN ELECTRONICS INC	134 FULTON AVE	GARDEN CITY PARK NY 11040
54473	MATSUSHITA ELECTRIC CORP OF AMERICA	ONE PANASONIC WAY	SECAUCUS NJ 07094
56289	SPRAGUE ELECTRIC CO	87 MARSHALL ST	NORTH ADAMS MA 01247
57668	ROHM CORP	16931 MILLIKEN AVE	IRVINE CA 92713
74970	JOHNSON E F CO	299 10TH AVE S W	WASECA MN 56093
80009	TEKTRONIX INC	4900 S W GRIFFITH DR	BEAVERTON OR 97077
		P O BOX 500	
91637	DALE ELECTRONICS INC	P O BOX 609	COLUMBUS NE 68601

Component No.	Tektronix Part No.	Serial/Assembly No. Effective Discont	Name & Description	Mfr. Code	Mfr. Part No.
A50-1	670-5239-00	8010100 8011220	CIRCUIT BD ASSY:PULSE & BAR LOGIC	80009	670-5239-00
A50-1	670-5239-02	8011221 8011317	CIRCUIT BD ASSY:PULSE & BAR LOGIC	80009	670-5239-02
A50-1	670-5239-03	8011318 8011715	CIRCUIT BD ASSY:PULSE & BAR LOGIC	80009	670-5239-03
A50-1	670-5239-04	8011716	CIRCUIT BD ASSY:PULSE & BAR LOGIC	80009	670-5239-04
A51-1	670-5240-00		CIRCUIT BD ASSY:PULSE & BAR OUTPUT	80009	670-5240-00
A100	670-4459-00		CIRCUIT BD ASSY:SHIELD	80009	670-4459-00
C103	281-0773-00		CAP,FXD,CER DI:0.01UF,10%,100V	04222	MA201C103KAA
C105	281-0775-00		CAP,FXD,CER DI:0.1UF,20%,50V	04222	MA205E104MAA
C110	281-0791-00		CAP,FXD,CER DI:270PF,10%,100V	04222	MA101C271KAA
C115	281-0812-00		CAP,FXD,CER DI:1000PF,10%,100V	04222	MA101C102KAA
C116	283-0636-00		CAP,FXD,MICA DI:36PF,1.4%,100V	00853	D155E360G0
C117	283-0644-00		CAP,FXD,MICA DI:15PF,1%,500V	00853	D155F151F0
C118	281-0773-00		CAP,FXD,CER DI:0.01UF,10%,100V	04222	MA201C103KAA
C119	281-0775-00		CAP,FXD,CER DI:0.1UF,20%,50V	04222	MA205E104MAA
C124	290-0745-00		CAP,FXD,ELCTLT:22UF,+50-10%,25V	54473	ECE-A25V22L
C126	281-0812-00		CAP,FXD,CER DI:1000PF,10%,100V	04222	MA101C102KAA
C127	283-0663-00		CAP,FXD,MICA DI:16.8PF,+/-0.5PF,500V	00853	D155C16R8D0
C128	283-0643-00		CAP,FXD,MICA DI:22PF,0.5%,500V	00853	D105E220D0
C135	283-0663-00		CAP,FXD,MICA DI:16.8PF,+/-0.5PF,500V	00853	D155C16R8D0
C136	281-0775-00		CAP,FXD,CER DI:0.1UF,20%,50V	04222	MA205E104MAA
C142	281-0812-00		CAP,FXD,CER DI:1000PF,10%,100V	04222	MA101C102KAA
C157	290-0244-00		CAP,FXD,ELCTLT:0.47UF,5%,35V	56289	173D474X5035U
C181	281-0773-00		CAP,FXD,CER DI:0.01UF,10%,100V	04222	MA201C103KAA
C191	281-0773-00		CAP,FXD,CER DI:0.01UF,10%,100V	04222	MA201C103KAA
C201	281-0775-00		CAP,FXD,CER DI:0.1UF,20%,50V	04222	MA205E104MAA
C213	281-0775-00		CAP,FXD,CER DI:0.1UF,20%,50V	04222	MA205E104MAA
C218	281-0791-00		CAP,FXD,CER DI:270PF,10%,100V	04222	MA101C271KAA
C221	281-0775-00		CAP,FXD,CER DI:0.1UF,20%,50V	04222	MA205E104MAA
C223	283-0785-00	8010100 8011317	CAP,FXD,MICA DI:250PF,1%,500V	00853	D155F251F0
C223	283-0788-00	8011318	CAP,FXD,MICA DI:267PF,1%,500V	00853	D155F267F0
C224	283-0785-00		CAP,FXD,MICA DI:250PF,1%,500V	00853	D155F251F0
C225	281-0226-00		CAP,VAR,PLASTIC:4-38PF,100V	52769	GXD38000
C226	281-0775-00		CAP,FXD,CER DI:0.1UF,20%,50V	04222	MA205E104MAA
C235	281-0198-00		CAP,VAR,AIR DI:1.7-11PF,250V	74970	187-0306-105
C241	290-0745-00		CAP,FXD,ELCTLT:22UF,+50-10%,25V	54473	ECE-A25V22L
C243	283-0600-00		CAP,FXD,MICA DI:43PF,5%,500V	00853	D105E430J0
C251	281-0775-00		CAP,FXD,CER DI:0.1UF,20%,50V	04222	MA205E104MAA
C254	283-0600-00		CAP,FXD,MICA DI:43PF,5%,500V	00853	D105E430J0
C255	281-0198-00		CAP,VAR,AIR DI:1.7-11PF,250V	74970	187-0306-105
C257	-----		(SELECTED)		
C258	290-0745-00		CAP,FXD,ELCTLT:22UF,+50-10%,25V	54473	ECE-A25V22L
C270	290-0745-00		CAP,FXD,ELCTLT:22UF,+50-10%,25V	54473	ECE-A25V22L
C271	283-0637-00		CAP,FXD,MICA DI:20PF,2.5%,500V	00853	D155E200D0
C272	281-0775-00		CAP,FXD,CER DI:0.1UF,20%,50V	04222	MA205E104MAA
C273	281-0198-00		CAP,VAR,AIR DI:1.7-11PF,250V	74970	187-0306-105
C275	281-0773-00		CAP,FXD,CER DI:0.01UF,10%,100V	04222	MA201C103KAA
C281	281-0775-00		CAP,FXD,CER DI:0.1UF,20%,50V	04222	MA205E104MAA
C284	281-0773-00		CAP,FXD,CER DI:0.01UF,10%,100V	04222	MA201C103KAA
C287	281-0775-00		CAP,FXD,CER DI:0.1UF,20%,50V	04222	MA205E104MAA
C292	283-0648-00		CAP,FXD,MICA DI:10PF,5%,500V	00853	D155C10000
C293	281-0209-00		CAP,VAR,AIR DI:1.3-5.4PF,250V	74970	187-0303-105
C319	281-0773-00		CAP,FXD,CER DI:0.01UF,10%,100V	04222	MA201C103KAA
C321	281-0812-00		CAP,FXD,CER DI:1000PF,10%,100V	04222	MA101C102KAA
C323	283-0637-00		CAP,FXD,MICA DI:20PF,2.5%,500V	00853	D155E200D0
C324	281-0226-00		CAP,VAR,PLASTIC:4-38PF,100V	52769	GXD38000
C329	281-0773-00		CAP,FXD,CER DI:0.01UF,10%,100V	04222	MA201C103KAA
C333	281-0812-00		CAP,FXD,CER DI:1000PF,10%,100V	04222	MA101C102KAA

Replaceable Electrical Parts - TS615

Component No.	Tektronix Part No.	Serial/Assembly No. Effective Dscont	Name & Description	Mfr. Code	Mfr. Part No.
C334	281-0812-00		CAP,FXD,CER DI:1000PF,10%,100V	04222	MA101C102KAA
C335	281-0812-00		CAP,FXD,CER DI:1000PF,10%,100V	04222	MA101C102KAA
C336	283-0779-00		CAP,FXD,MICA DI:27 PF,2%,500V	00853	D155E270G0
C337	281-0812-00		CAP,FXD,CER DI:1000PF,10%,100V	04222	MA101C102KAA
C338	283-0779-00		CAP,FXD,MICA DI:27 PF,2%,500V	00853	D155E270G0
C341	281-0773-00		CAP,FXD,CER DI:0.01UF,10%,100V	04222	MA201C103KAA
C342	281-0812-00		CAP,FXD,CER DI:1000PF,10%,100V	04222	MA101C102KAA
C347	281-0064-00		CAP,VAR,PLASTIC:0.25-1.5PF,600V	52769	ER-530-013
C354	281-0791-00		CAP,FXD,CER DI:270PF,10%,100V	04222	MA101C271KAA
C355	283-0730-00		CAP,FXD,MICA DI:274PF,1%,500V	00853	D155F2740F0
C356	281-0775-00		CAP,FXD,CER DI:0.1UF,20%,50V	04222	MA205E104MAA
C357	283-0636-00		CAP,FXD,MICA DI:36PF,1.4%,100V	00853	D155E360G0
C358	281-0773-00		CAP,FXD,CER DI:0.01UF,10%,100V	04222	MA201C103KAA
C361	281-0775-00		CAP,FXD,CER DI:0.1UF,20%,50V	04222	MA205E104MAA
C362	281-0773-00		CAP,FXD,CER DI:0.01UF,10%,100V	04222	MA201C103KAA
C363	290-0745-00		CAP,FXD,ELCTLT:22UF,+50-10%,25V	54473	ECE-A25V22L
C364	281-0775-00		CAP,FXD,CER DI:0.1UF,20%,50V	04222	MA205E104MAA
C368	283-0638-00		CAP,FXD,MICA DI:130PF,1%,100V	00853	D155F131F0
C371	283-0631-00		CAP,FXD,MICA DI:95PF,1%,500V	00853	D155F950F0
C376	281-0775-00		CAP,FXD,CER DI:0.1UF,20%,50V	04222	MA205E104MAA
C377	283-0779-00		CAP,FXD,MICA DI:27 PF,2%,500V	00853	D155E270G0
C385	283-0784-00		CAP,FXD,MICA DI:40PF,2%,500V	00853	D155E400G0
C387	283-0636-00		CAP,FXD,MICA DI:36PF,1.4%,100V	00853	D155E360G0
C391	283-0730-00		CAP,FXD,MICA DI:274PF,1%,500V	00853	D155F2740F0
C392	283-0631-00		CAP,FXD,MICA DI:95PF,1%,500V	00853	D155F950F0
C393	283-0644-00		CAP,FXD,MICA DI:150PF,1%,500V	00853	D155F151F0
C394	283-0766-00		CAP,FXD,MICA DI:47 PF,1%,500V	00853	D155E470D0
C397	283-0604-00		CAP,FXD,MICA DI:304PF,2%,500V	00853	D155F3040G0
C398	283-0784-00		CAP,FXD,MICA DI:40PF,2%,500V	00853	D155E400G0
C400	283-0672-00		CAP,FXD,MICA DI:200PF,1%,500V	00853	D155F2010F0
C401	283-0785-00		CAP,FXD,MICA DI:250PF,1%,500V	00853	D155F251F0
C403	283-0649-00		CAP,FXD,MICA DI:105PF,1%,300V	00853	D155F1050F0
C405	283-0728-00		CAP,FXD,MICA DI:120PF,1%,500V	00853	D155F121F0
C407	281-0775-00		CAP,FXD,CER DI:0.1UF,20%,50V	04222	MA205E104MAA
C408	283-0784-00		CAP,FXD,MICA DI:40PF,2%,500V	00853	D155E400G0
C410	283-0669-00		CAP,FXD,MICA DI:360PF,1%,500V	00853	D155F361F0
C412	283-0596-00		CAP,FXD,MICA DI:528PF,1%,300V	00853	D153F5280F0
C413	283-0641-00		CAP,FXD,MICA DI:180PF,1%,100V	00853	D155F181F0
C414	283-0625-00		CAP,FXD,MICA DI:220PF,1%,500V	00853	D105F221F0
C416	281-0775-00		CAP,FXD,CER DI:0.1UF,20%,50V	04222	MA205E104MAA
C419	283-0648-00		CAP,FXD,MICA DI:10PF,5%,500V	00853	D155C100D0
C420	290-0745-00		CAP,FXD,ELCTLT:22UF,+50-10%,25V	54473	ECE-A25V22L
C421	290-0745-00		CAP,FXD,ELCTLT:22UF,+50-10%,25V	54473	ECE-A25V22L
C423	283-0666-00		CAP,FXD,MICA DI:890PF,2%,100V	00853	D151F891G0
C424	283-0641-00		CAP,FXD,MICA DI:180PF,1%,100V	00853	D155F181F0
C425	283-0630-00		CAP,FXD,MICA DI:110PF,1%,100V	00853	D155F111F0
C427	283-0648-00		CAP,FXD,MICA DI:10PF,5%,500V	00853	D155C100D0
C431	290-0745-00		CAP,FXD,ELCTLT:22UF,+50-10%,25V	54473	ECE-A25V22L
C432	283-0730-00		CAP,FXD,MICA DI:274PF,1%,500V	00853	D155F2740F0
C433	283-0779-00		CAP,FXD,MICA DI:27 PF,2%,500V	00853	D155E270G0
C434	283-0728-00		CAP,FXD,MICA DI:120PF,1%,500V	00853	D155F121F0
C435	283-0635-00		CAP,FXD,MICA DI:51PF,1%,100V	00853	D151E510F0
C439	283-0648-00		CAP,FXD,MICA DI:10PF,5%,500V	00853	D155C100D0
C442	283-0641-00		CAP,FXD,MICA DI:180PF,1%,100V	00853	D155F181F0
C443	281-0775-00		CAP,FXD,CER DI:0.1UF,20%,50V	04222	MA205E104MAA
C451	283-0636-00		CAP,FXD,MICA DI:36PF,1.4%,100V	00853	D155E360G0
C452	283-0784-00		CAP,FXD,MICA DI:40PF,2%,500V	00853	D155E400G0
C453	283-0602-00		CAP,FXD,MICA DI:53PF,5%,300V	00853	D155E530J0

Component No.	Tektronix Part No.	Serial/Assembly No. Effective	Discont	Name & Description	Mfr. Code	Mfr. Part No.
C454	283-0779-00			CAP, FXD, MICA D1:27 PF, 2%, 500V	00853	D155E270G0
C455	283-0663-00			CAP, FXD, MICA D1:16.8PF, +/-0.5PF, 500V	00853	D155C16R800
C456	283-0766-00			CAP, FXD, MICA D1:47 PF, 1%, 500V	00853	D155E470D0
C457	281-0775-00			CAP, FXD, CER D1:0.1UF, 20%, 50V	04222	MA205E104MAA
C458	283-0663-00			CAP, FXD, MICA D1:16.8PF, +/-0.5PF, 500V	00853	D155C16R800
C461	281-0226-00			CAP, VAR, PLASTIC:4-38PF, 100V	52769	GXD38000
C462	283-0602-00			CAP, FXD, MICA D1:53PF, 5%, 300V	00853	D155E530J0
C463	283-0636-00			CAP, FXD, MICA D1:36PF, 1.4%, 100V	00853	D155E360G0
C464	283-0779-00			CAP, FXD, MICA D1:27 PF, 2%, 500V	00853	D155E270G0
C467	283-0766-00			CAP, FXD, MICA D1:47 PF, 1%, 500V	00853	D155E470D0
C471	283-0641-00			CAP, FXD, MICA D1:180PF, 1%, 100V	00853	D155F181F0
C472	283-0784-00			CAP, FXD, MICA D1:40PF, 2%, 500V	00853	D155E400G0
C474	283-0663-00			CAP, FXD, MICA D1:16.8PF, +/-0.5PF, 500V	00853	D155C16R800
C477	283-0663-00			CAP, FXD, MICA D1:16.8PF, +/-0.5PF, 500V	00853	D155C16R800
C481	281-0226-00			CAP, VAR, PLASTIC:4-38PF, 100V	52769	GXD38000
C482	283-0636-00			CAP, FXD, MICA D1:36PF, 1.4%, 100V	00853	D155E360G0
C483	283-0602-00			CAP, FXD, MICA D1:53PF, 5%, 300V	00853	D155E530J0
C484	283-0784-00			CAP, FXD, MICA D1:40PF, 2%, 500V	00853	D155E400G0
C485	283-0779-00			CAP, FXD, MICA D1:27 PF, 2%, 500V	00853	D155E270G0
C486	283-0663-00			CAP, FXD, MICA D1:16.8PF, +/-0.5PF, 500V	00853	D155C16R800
C487	283-0766-00			CAP, FXD, MICA D1:47 PF, 1%, 500V	00853	D155E470D0
C488	283-0663-00			CAP, FXD, MICA D1:16.8PF, +/-0.5PF, 500V	00853	D155C16R800
C490	283-0641-00			CAP, FXD, MICA D1:180PF, 1%, 100V	00853	D155F181F0
C491	281-0226-00			CAP, VAR, PLASTIC:4-38PF, 100V	52769	GXD38000
CR101	152-0141-02			SEMICON DVC, D1:SW, SI, 30V, 150MA, 30V, DO-35	03508	DA2527 (1N4152)
CR102	152-0141-02			SEMICON DVC, D1:SW, SI, 30V, 150MA, 30V, DO-35	03508	DA2527 (1N4152)
CR110	152-0141-02			SEMICON DVC, D1:SW, SI, 30V, 150MA, 30V, DO-35	03508	DA2527 (1N4152)
CR111	152-0141-02			SEMICON DVC, D1:SW, SI, 30V, 150MA, 30V, DO-35	03508	DA2527 (1N4152)
CR112	152-0141-02			SEMICON DVC, D1:SW, SI, 30V, 150MA, 30V, DO-35	03508	DA2527 (1N4152)
CR113	152-0141-02			SEMICON DVC, D1:SW, SI, 30V, 150MA, 30V, DO-35	03508	DA2527 (1N4152)
CR128	152-0141-02			SEMICON DVC, D1:SW, SI, 30V, 150MA, 30V, DO-35	03508	DA2527 (1N4152)
CR135	152-0322-00	8011716		SEMICON DVC, D1:SCHOTTKY BARRIER, SI, 15V	50434	5082-2672
CR181	152-0141-02			SEMICON DVC, D1:SW, SI, 30V, 150MA, 30V, DO-35	03508	DA2527 (1N4152)
CR182	152-0141-02			SEMICON DVC, D1:SW, SI, 30V, 150MA, 30V, DO-35	03508	DA2527 (1N4152)
CR202	152-0141-02			SEMICON DVC, D1:SW, SI, 30V, 150MA, 30V, DO-35	03508	DA2527 (1N4152)
CR203	152-0141-02			SEMICON DVC, D1:SW, SI, 30V, 150MA, 30V, DO-35	03508	DA2527 (1N4152)
CR211	152-0141-02			SEMICON DVC, D1:SW, SI, 30V, 150MA, 30V, DO-35	03508	DA2527 (1N4152)
CR213	152-0141-02			SEMICON DVC, D1:SW, SI, 30V, 150MA, 30V, DO-35	03508	DA2527 (1N4152)
CR216	152-0141-02			SEMICON DVC, D1:SW, SI, 30V, 150MA, 30V, DO-35	03508	DA2527 (1N4152)
CR217	152-0141-02			SEMICON DVC, D1:SW, SI, 30V, 150MA, 30V, DO-35	03508	DA2527 (1N4152)
CR218	152-0141-02			SEMICON DVC, D1:SW, SI, 30V, 150MA, 30V, DO-35	03508	DA2527 (1N4152)
CR232	152-0141-02			SEMICON DVC, D1:SW, SI, 30V, 150MA, 30V, DO-35	03508	DA2527 (1N4152)
CR251	152-0141-02			SEMICON DVC, D1:SW, SI, 30V, 150MA, 30V, DO-35	03508	DA2527 (1N4152)
CR252	152-0141-02			SEMICON DVC, D1:SW, SI, 30V, 150MA, 30V, DO-35	03508	DA2527 (1N4152)
CR254	152-0141-02			SEMICON DVC, D1:SW, SI, 30V, 150MA, 30V, DO-35	03508	DA2527 (1N4152)
CR263	152-0141-02			SEMICON DVC, D1:SW, SI, 30V, 150MA, 30V, DO-35	03508	DA2527 (1N4152)
CR271	152-0141-02			SEMICON DVC, D1:SW, SI, 30V, 150MA, 30V, DO-35	03508	DA2527 (1N4152)
CR272	152-0141-02			SEMICON DVC, D1:SW, SI, 30V, 150MA, 30V, DO-35	03508	DA2527 (1N4152)
CR282	152-0141-02			SEMICON DVC, D1:SW, SI, 30V, 150MA, 30V, DO-35	03508	DA2527 (1N4152)
CR283	152-0141-02			SEMICON DVC, D1:SW, SI, 30V, 150MA, 30V, DO-35	03508	DA2527 (1N4152)
CR284	152-0141-02			SEMICON DVC, D1:SW, SI, 30V, 150MA, 30V, DO-35	03508	DA2527 (1N4152)
CR291	152-0141-02			SEMICON DVC, D1:SW, SI, 30V, 150MA, 30V, DO-35	03508	DA2527 (1N4152)
CR293	152-0141-02			SEMICON DVC, D1:SW, SI, 30V, 150MA, 30V, DO-35	03508	DA2527 (1N4152)
CR338	152-0226-00			SEMICON DVC, D1:ZEN, SI, 5.1V, 5%, 0.4W, DO-7	04713	SZ12262RL
L108	114-0278-00			COIL, RF: VARIABLE, 4.6-16.7UH	80009	114-0278-00
L117	114-0254-00			COIL, RF: VARIABLE, 30-60UH	80009	114-0254-00
L331	114-0280-00			COIL, RF: VARIABLE, 12-43UH	80009	114-0280-00
L335	114-0311-00			COIL, RF: VARIABLE, 65-190UH	80009	114-0311-00

Replaceable Electrical Parts - TSG15

Component No.	Tektronix Part No.	Serial/Assembly No. Effective Discont	Name & Description	Mfr. Code	Mfr. Part No.
L379	114-0278-00		COIL, RF: VARIABLE, 4.6-16.7UH	80009	114-0278-00
L381	114-0259-00		COIL, RF: VARIABLE, 150-240UH	80009	114-0259-00
L389	114-0278-00		COIL, RF: VARIABLE, 4.6-16.7UH	80009	114-0278-00
L392	114-0259-00		COIL, RF: VARIABLE, 150-240UH	80009	114-0259-00
L395	114-0311-00		COIL, RF: VARIABLE, 65-190UH	80009	114-0311-00
L402	114-0259-00		COIL, RF: VARIABLE, 150-240UH	80009	114-0259-00
L405	114-0311-00		COIL, RF: VARIABLE, 65-190UH	80009	114-0311-00
L408	114-0310-00		COIL, RF: VARIABLE, 22-60UH	80009	114-0310-00
L412	114-0340-00		COIL, RF: VARIABLE, 350-750UH	80009	114-0340-00
L415	114-0311-00		COIL, RF: VARIABLE, 65-190UH	80009	114-0311-00
L425	114-0324-00		COIL, RF: VARIABLE, 200-300UH	80009	114-0324-00
L432	114-0254-00		COIL, RF: VARIABLE, 30-60UH	80009	114-0254-00
L445	114-0254-00		COIL, RF: VARIABLE, 30-60UH	80009	114-0254-00
L451	114-0280-00		COIL, RF: VARIABLE, 12-43UH	80009	114-0280-00
L455	114-0278-00		COIL, RF: VARIABLE, 4.6-16.7UH	80009	114-0278-00
L465	114-0278-00		COIL, RF: VARIABLE, 4.6-16.7UH	80009	114-0278-00
L468	114-0303-00		COIL, RF: VARIABLE, 6.5-23UH	80009	114-0303-00
L472	114-0257-00		COIL, RF: VARIABLE, 6-11UH	80009	114-0257-00
L474	114-0222-00		COIL, RF: VARIABLE, 2-6UH	80009	114-0222-00
L476	114-0222-00		COIL, RF: VARIABLE, 2-6UH	80009	114-0222-00
L479	114-0222-00		COIL, RF: VARIABLE, 2-6UH	80009	114-0222-00
L491	114-0220-00		COIL, RF: VARIABLE, 1-3UH	80009	114-0220-00
L494	114-0353-00		COIL, RF: VARIABLE, 0.6-1.0UH	24226	ORDER BY DESC
L496	114-0353-00		COIL, RF: VARIABLE, 0.6-1.0UH	24226	ORDER BY DESC
L499	114-0353-00		COIL, RF: VARIABLE, 0.6-1.0UH	24226	ORDER BY DESC
Q103	151-0341-00		TRANSISTOR: NPN, SI, TO-106	04713	SPS6919
Q110	151-0341-00		TRANSISTOR: NPN, SI, TO-106	04713	SPS6919
Q114	151-0341-00		TRANSISTOR: NPN, SI, TO-106	04713	SPS6919
Q115	151-0341-00		TRANSISTOR: NPN, SI, TO-106	04713	SPS6919
Q118	151-0341-00		TRANSISTOR: NPN, SI, TO-106	04713	SPS6919
Q135	151-0190-00		TRANSISTOR: NPN, SI, TO-92	80009	151-0190-00
Q137	151-0220-00		TRANSISTOR: PNP, SI, TO-92	80009	151-0220-00
Q138	151-0220-00		TRANSISTOR: PNP, SI, TO-92	80009	151-0220-00
Q156	151-0220-00		TRANSISTOR: PNP, SI, TO-92	80009	151-0220-00
Q157	151-0341-00		TRANSISTOR: NPN, SI, TO-106	04713	SPS6919
Q184	151-0190-00		TRANSISTOR: NPN, SI, TO-92	80009	151-0190-00
Q194	151-0273-00		TRANSISTOR: SELECTED	03508	X16E3616
Q195	151-0190-00		TRANSISTOR: NPN, SI, TO-92	80009	151-0190-00
Q203	151-0273-00		TRANSISTOR: SELECTED	03508	X16E3616
Q204	151-0273-00		TRANSISTOR: SELECTED	03508	X16E3616
Q205	151-0273-00		TRANSISTOR: SELECTED	03508	X16E3616
Q206	151-0273-00		TRANSISTOR: SELECTED	03508	X16E3616
Q214	151-0190-00		TRANSISTOR: NPN, SI, TO-92	80009	151-0190-00
Q222	151-0190-00		TRANSISTOR: NPN, SI, TO-92	80009	151-0190-00
Q224	151-0190-00		TRANSISTOR: NPN, SI, TO-92	80009	151-0190-00
Q232	151-0192-00		TRANSISTOR: SELECTED	04713	SPS8801
Q233	151-0273-00		TRANSISTOR: SELECTED	03508	X16E3616
Q236	151-0220-00		TRANSISTOR: PNP, SI, TO-92	80009	151-0220-00
Q237	151-0402-00		TRANSISTOR: SELECTED	01295	SKA6814
Q241	151-0216-00		TRANSISTOR: PNP, SI, TO-92	04713	SPS8803
Q253	151-0273-00		TRANSISTOR: SELECTED	03508	X16E3616
Q254	151-0190-00		TRANSISTOR: NPN, SI, TO-92	80009	151-0190-00
Q256	151-0402-00		TRANSISTOR: SELECTED	01295	SKA6814
Q261	151-0273-00		TRANSISTOR: SELECTED	03508	X16E3616
Q262	151-0192-00		TRANSISTOR: SELECTED	04713	SPS8801
Q272	151-0192-00		TRANSISTOR: SELECTED	04713	SPS8801
Q273	151-0273-00		TRANSISTOR: SELECTED	03508	X16E3616
Q274	151-0190-00		TRANSISTOR: NPN, SI, TO-92	80009	151-0190-00

Component No.	Tektronix Part No.	Serial/Assembly No. Effective Dscont	Name & Description	Mfr. Code	Mfr. Part No.
Q275	151-0402-00		TRANSISTOR:SELECTED	01295	5KA6814
Q284	151-0190-00		TRANSISTOR:NPN, SI, TO-92	80009	151-0190-00
Q291	151-0220-00		TRANSISTOR:PNP, SI, TO-92	80009	151-0220-00
Q292	151-0192-00		TRANSISTOR:SELECTED	04713	SPS8801
Q293	151-0273-00		TRANSISTOR:SELECTED	03508	X16E3616
Q294	151-0402-00		TRANSISTOR:SELECTED	01295	5KA6814
Q325	151-0341-00		TRANSISTOR:NPN, SI, TO-106	04713	SPS6919
Q325	151-0341-00		TRANSISTOR:NPN, SI, TO-106	04713	SPS6919
Q334	151-0341-00		TRANSISTOR:NPN, SI, TO-106	04713	SPS6919
Q335	151-0341-00		TRANSISTOR:NPN, SI, TO-106	04713	SPS6919
Q349	151-0341-00		TRANSISTOR:NPN, SI, TO-106	04713	SPS6919
Q354	151-0220-00		TRANSISTOR:PNP, SI, TO-92	80009	151-0220-00
Q371	151-0190-00		TRANSISTOR:NPN, SI, TO-92	80009	151-0190-00
Q381	151-0216-00		TRANSISTOR:PNP, SI, TO-92	04713	SPS8803
Q392	151-0216-00		TRANSISTOR:PNP, SI, TO-92	04713	SPS8803
Q418	151-0103-00		TRANSISTOR:NPN, SI, TO-5	04713	SM1307
Q438	151-0220-00		TRANSISTOR:PNP, SI, TO-92	80009	151-0220-00
Q439	151-0220-00		TRANSISTOR:PNP, SI, TO-92	80009	151-0220-00
Q448	151-0367-00		TRANSISTOR:NPN, SI, X-55	04713	SPS 8811
Q449	151-0216-00		TRANSISTOR:PNP, SI, TO-92	04713	SPS8803
R101	315-0202-00		RES,FXD,FILM:2K OHM,5%,0.25W	57668	NTR25J-E 2K
R102	315-0101-00		RES,FXD,FILM:100 OHM,5%,0.25W	57668	NTR25J-E 100E
R103	315-0272-00		RES,FXD,FILM:2.7K OHM,5%,0.25W	57668	NTR25J-E02K7
R105	321-0260-00		RES,FXD,FILM:4.99K OHM,1%,0.125W,TC=TO	19701	5033ED4K990F
R106	321-0222-00		RES,FXD,FILM:2.00K OHM,1%,0.125W,TC=TO	19701	5033ED2K00F
R107	321-0260-00		RES,FXD,FILM:4.99K OHM,1%,0.125W,TC=TO	19701	5033ED4K990F
R108	321-0193-00		RES,FXD,FILM:1K OHM,1%,0.125W,TC=TO	19701	5033ED1K00F
R116	321-0260-00		RES,FXD,FILM:4.99K OHM,1%,0.125W,TC=TO	19701	5033ED4K990F
R117	321-0193-00		RES,FXD,FILM:1K OHM,1%,0.125W,TC=TO	19701	5033ED1K00F
R118	315-0101-00		RES,FXD,FILM:100 OHM,5%,0.25W	57668	NTR25J-E 100E
R123	315-0471-00		RES,FXD,FILM:470 OHM,5%,0.25W	57668	NTR25J-E470E
R124	315-0911-00		RES,FXD,FILM:910 OHM,5%,0.25W	57668	NTR25J-E910E
R125	315-0911-00		RES,FXD,FILM:910 OHM,5%,0.25W	57668	NTR25J-E910E
R131	315-0202-00		RES,FXD,FILM:2K OHM,5%,0.25W	57668	NTR25J-E 2K
R132	315-0103-00		RES,FXD,FILM:10K OHM,5%,0.25W	19701	5043CX10K00J
R134	315-0103-00		RES,FXD,FILM:10K OHM,5%,0.25W	19701	5043CX10K00J
R137	315-0203-00		RES,FXD,FILM:20K OHM,5%,0.25W	57668	NTR25J-E 20K
R138	315-0221-00		RES,FXD,FILM:220 OHM,5%,0.25W	57668	NTR25J-E220E
R144	315-0101-00		RES,FXD,FILM:100 OHM,5%,0.25W	57668	NTR25J-E 100E
R145	315-0302-00		RES,FXD,FILM:3K OHM,5%,0.25W	57668	NTR25J-E03K0
R146	315-0432-00		RES,FXD,FILM:4.3K OHM,5%,0.25W	57668	NTR25J-E04K3
R147	315-0681-00		RES,FXD,FILM:680 OHM,5%,0.25W	57668	NTR25J-E680E
R148	315-0432-00		RES,FXD,FILM:4.3K OHM,5%,0.25W	57668	NTR25J-E04K3
R149	315-0123-00		RES,FXD,FILM:12K OHM,5%,0.25W	57668	NTR25J-E12K0
R154	307-0541-00		RES NTWK,FXD,FI:(7)1K OHM,10%,1W	01121	108A102
R155	315-0103-00		RES,FXD,FILM:10K OHM,5%,0.25W	19701	5043CX10K00J
R156	315-0272-00		RES,FXD,FILM:2.7K OHM,5%,0.25W	57668	NTR25J-E02K7
R157	315-0182-00		RES,FXD,FILM:1.8K OHM,5%,0.25W	57668	NTR25J-E1K8
R159	315-0432-00		RES,FXD,FILM:4.3K OHM,5%,0.25W	57668	NTR25J-E04K3
R169	311-1241-00		RES,VAR,NORMW:TRMR,100K OHM,0.5W	32997	3386X-T07-104
R175	315-0302-00		RES,FXD,FILM:3K OHM,5%,0.25W	57668	NTR25J-E03K0
R176	315-0202-00		RES,FXD,FILM:2K OHM,5%,0.25W	57668	NTR25J-E 2K
R178	307-0542-00		RES NTWK,FXD,FI:(5)10K OHM,5%,0.125W	01121	106A103OR706A103
R179	311-1246-00		RES,VAR,NORMW:TRMR,50K OHM,0.5W	32997	3386X-T07-503
R182	315-0101-00	B010100	RES,FXD,FILM:100 OHM,5%,0.25W	57668	NTR25J-E 100E
R182	315-0750-00	B010102	RES,FXD,FILM:75 OHM,5%,0.25W	57668	NTR25J-E75E0
R183	321-1619-07		RES,FXD,FILM:8.334K OHM,0.1%,0.125W,TC=T9	19701	5033RE8K334B
R184	315-0302-00		RES,FXD,FILM:3K OHM,5%,0.25W	57668	NTR25J-E03K0

Replaceable Electrical Parts - TSG15

Component No.	Tektronix Part No.	Serial/Assembly No. Effective	Discont	Name & Description	Mfr. Code	Mfr. Part No.
R185	315-0202-00			RES, FXD, FILM: 2K OHM, 5%, 0.25W	57668	NTR25J-E 2K
R186	311-1198-00			RES, VAR, NONMW: TRMR, 20K OHM, 0.5W	32997	3386X-T07-203
R191	321-1603-04			RES, FXD, FILM: 14.61K OHM, 0.1%, 0.125W, TC=T2	19701	5033RC14K61B
R192	321-0632-00			RES, FXD, FILM: 9.41K OHM, 0.5%, 0.125W, TC=T2	19701	5033RC9K410C
R193	321-1701-04			RES, FXD, FILM: 5.22K OHM, 0.1%, 0.125W, TC=T2	07716	CEAC52200B
R198	307-0540-00			RES, NTK, FXD, FI: (5)1K OHM, 10%, 0.7W	11236	750-61-R1KOHM
R201	315-0102-00			RES, FXD, FILM: 1K OHM, 5%, 0.25W	57668	NTR25JE01K0
R202	315-0202-00			RES, FXD, FILM: 2K OHM, 5%, 0.25W	57668	NTR25J-E 2K
R204	315-0101-00			RES, FXD, FILM: 100 OHM, 5%, 0.25W	57668	NTR25J-E 100E
R205	315-0302-00	8010100	8010101	RES, FXD, FILM: 3K OHM, 5%, 0.25W	57668	NTR25J-E03K0
R205	315-0123-00	8010102		RES, FXD, FILM: 12K OHM, 5%, 0.25W	57668	NTR25J-E12K0
R209	307-0542-00			RES, NTK, FXD, FI: (5)10K OHM, 5%, 0.125W	01121	106A1030R706A103
R210	315-0125-00	8010658		RES, FXD, FILM: 1.2M OHM, 5%, 0.25W	19701	5043CX1M200J
R211	321-1711-04			RES, FXD, FILM: 12.18K OHM, 0.1%, 0.125W, TC=T2	07716	CEAC12181B
R212	315-0202-00			RES, FXD, FILM: 2K OHM, 5%, 0.25W	57668	NTR25J-E 2K
R213	321-1711-04			RES, FXD, FILM: 12.18K OHM, 0.1%, 0.125W, TC=T2	07716	CEAC12181B
R216	315-0302-00			RES, FXD, FILM: 3K OHM, 5%, 0.25W	57668	NTR25J-E03K0
R217	315-0102-00			RES, FXD, FILM: 1K OHM, 5%, 0.25W	57668	NTR25JE01K0
R218	315-0100-00			RES, FXD, FILM: 10 OHM, 5%, 0.25W	19701	5043CX10R00J
R220	315-0302-00			RES, FXD, FILM: 3K OHM, 5%, 0.25W	57668	NTR25J-E03K0
R221	321-0289-07			RES, FXD, FILM: 10.0K OHM, 0.1%, 0.125W, TC=T9	19701	5033RE10K00B
R222	315-0101-00			RES, FXD, FILM: 100 OHM, 5%, 0.25W	57668	NTR25J-E 100E
R223	321-0816-07			RES, FXD, FILM: 5K OHM, 0.1%, 0.125W, TC=T9	19701	5033RE5K000B
R224	321-0344-00			RES, FXD, FILM: 37.4K OHM, 1%, 0.125W, TC=T0	19701	5033ED 37K40F
R225	321-0666-07			RES, FXD, FILM: 3.04K OHM, 0.1%, 0.125W, TC=T9	19701	5033RE3K040B
R226	321-0666-07			RES, FXD, FILM: 3.04K OHM, 0.1%, 0.125W, TC=T9	19701	5033RE3K040B
R227	321-0770-03			RES, FXD, FILM: 4.204K OHM, 0.25%, 0.125W, TC=T2	19701	5033RC4K204C
R228	321-0422-00			RES, FXD, FILM: 243K OHM, 1%, 0.125W, TC=T0	07716	CEAD24302F
R229	311-1248-00			RES, VAR, NONMW: TRMR, 500 OHM, 0.5W	32997	3386X-T07-501
R232	321-0126-00			RES, FXD, FILM: 200 OHM, 1%, 0.125W, TC=T0	19701	5033ED200ROF
R233	321-0296-00			RES, FXD, FILM: 11.8K OHM, 1%, 0.125W, TC=T0	07716	CEAD11801F
R237	315-0202-00			RES, FXD, FILM: 2K OHM, 5%, 0.25W	57668	NTR25J-E 2K
R238	315-0102-00			RES, FXD, FILM: 1K OHM, 5%, 0.25W	57668	NTR25JE01K0
R239	311-1238-00			RES, VAR, NONMW: TRMR, 5K OHM, 0.5W	32997	3386X-DY6-502
R241	321-0126-00			RES, FXD, FILM: 200 OHM, 1%, 0.125W, TC=T0	19701	5033ED200ROF
R243	321-0983-00			RES, FXD, FILM: 4.5 MEG OHM, 1%, 0.125W, TC=T0	91637	CMF55116-G45003F
R245	315-0124-00			RES, FXD, FILM: 120K OHM, 5%, 0.25W	19701	5043CX120K0J
R246	321-0239-00			RES, FXD, FILM: 3.01K OHM, 1%, 0.125W, TC=T0	19701	5043ED3K010F
R247	321-0466-00			RES, FXD, FILM: 698K OHM, 1%, 0.125W, TC=T0	19701	5043EP698K0F
R249	315-0102-00			RES, FXD, FILM: 1K OHM, 5%, 0.25W	57668	NTR25JE01K0
R251	315-0302-00			RES, FXD, FILM: 3K OHM, 5%, 0.25W	57668	NTR25J-E03K0
R252	315-0101-00			RES, FXD, FILM: 100 OHM, 5%, 0.25W	57668	NTR25J-E 100E
R253	321-0353-00			RES, FXD, FILM: 46.4K OHM, 1%, 0.125W, TC=T0	07716	CEAD46401F
R254	321-0335-00			RES, FXD, FILM: 30.1K OHM, 1%, 0.125W, TC=T0	57668	RB14FXE30K1
R255	321-1712-06			RES, FXD, FILM: 4.4K OHM, 0.25%, 0.125W, TC=T9	07716	CEAE440000C
R256	315-0102-00			RES, FXD, FILM: 1K OHM, 5%, 0.25W	57668	NTR25JE01K0
R257	315-0102-00			RES, FXD, FILM: 1K OHM, 5%, 0.25W	57668	NTR25JE01K0
R258	311-1246-00			RES, VAR, NONMW: TRMR, 50K OHM, 0.5W	32997	3386X-T07-503
R259	311-1236-00			RES, VAR, NONMW: TRMR, 250 OHM, 0.5W	32997	3386X-T07-251
R261	321-0126-00			RES, FXD, FILM: 200 OHM, 1%, 0.125W, TC=T0	19701	5033ED200ROF
R264	321-0481-00			RES, FXD, FILM: 1M OHM, 1%, 0.125W, TC=T0	19701	5043ED1M000F
R265	315-0202-00			RES, FXD, FILM: 2K OHM, 5%, 0.25W	57668	NTR25J-E 2K
R266	321-0403-00			RES, FXD, FILM: 154K OHM, 1%, 0.125W, TC=T0	07716	CEAD15402F
R267	321-0289-00			RES, FXD, FILM: 10.0K OHM, 1%, 0.125W, TC=T0	19701	5033ED10K0F
R268	321-0387-00			RES, FXD, FILM: 105K OHM, 1%, 0.125W, TC=T0	07716	CEAD10502F
R269	311-1198-00			RES, VAR, NONMW: TRMR, 20K OHM, 0.5W	32997	3386X-T07-203
R274	315-0202-00			RES, FXD, FILM: 2K OHM, 5%, 0.25W	57668	NTR25J-E 2K
R275	321-0403-00			RES, FXD, FILM: 154K OHM, 1%, 0.125W, TC=T0	07716	CEAD15402F

Component No.	Tektronix Part No.	Serial/Assembly No. Effective Discnt	Name & Description	Mfr. Code	Mfr. Part No.
R276	321-1712-06		RES, FXD, FILM: 4.4K OHM, 0.25%, 0.125W, TC=T9	07716	CEAE440000C
R277	315-0102-00		RES, FXD, FILM: 1K OHM, 5%, 0.25W	57668	NTR25JE01K0
R278	315-0102-00		RES, FXD, FILM: 1K OHM, 5%, 0.25W	57668	NTR25JE01K0
R279	311-1236-00		RES, VAR, NONW: TRMR, 250 OHM, 0.5W	32997	3386X-T07-251
R280	321-0126-00		RES, FXD, FILM: 200 OHM, 1%, 0.125W, TC=T0	19701	5033ED200ROF
R281	315-0101-00		RES, FXD, FILM: 100 OHM, 5%, 0.25W	57668	NTR25J-E 100E
R282	315-0302-00		RES, FXD, FILM: 3K OHM, 5%, 0.25W	57668	NTR25J-E03K0
R283	321-0481-00		RES, FXD, FILM: 1M OHM, 1%, 0.125W, TC=T0	19701	5043ED1M000F
R284	321-0297-00		RES, FXD, FILM: 12.1K OHM, 1%, 0.125W, TC=T0	07716	CEAD12101F
R285	321-0387-00		RES, FXD, FILM: 105K OHM, 1%, 0.125W, TC=T0	07716	CEAD10502F
R286	321-1712-06		RES, FXD, FILM: 4.4K OHM, 0.25%, 0.125W, TC=T9	07716	CEAE440000C
R287	315-0102-00		RES, FXD, FILM: 1K OHM, 5%, 0.25W	57668	NTR25JE01K0
R288	315-0102-00		RES, FXD, FILM: 1K OHM, 5%, 0.25W	57668	NTR25JE01K0
R289	311-1198-00		RES, VAR, NONW: TRMR, 20K OHM, 0.5W	32997	3386X-T07-203
R290	321-0272-00		RES, FXD, FILM: 6.65K OHM, 1%, 0.125W, TC=T0	19701	5043ED6K650F
R291	321-0126-00		RES, FXD, FILM: 200 OHM, 1%, 0.125W, TC=T0	19701	5033ED200ROF
R292	321-1611-07		RES, FXD, FILM: 550 OHM, 0.1%, 0.125W, TC=T9	19701	5033RE550R08
R293	321-0481-00		RES, FXD, FILM: 1M OHM, 1%, 0.125W, TC=T0	19701	5043ED1M000F
R294	315-0202-00		RES, FXD, FILM: 2K OHM, 5%, 0.25W	57668	NTR25J-E 2K
R295	321-0403-00		RES, FXD, FILM: 154K OHM, 1%, 0.125W, TC=T0	07716	CEAD15402F
R296	321-0303-00		RES, FXD, FILM: 14.0K OHM, 1%, 0.125W, TC=T0	07716	CEAD 14001F
R297	321-0387-00		RES, FXD, FILM: 105K OHM, 1%, 0.125W, TC=T0	07716	CEAD10502F
R298	311-1236-00		RES, VAR, NONW: TRMR, 250 OHM, 0.5W	32997	3386X-T07-251
R299	311-1198-00		RES, VAR, NONW: TRMR, 20K OHM, 0.5W	32997	3386X-T07-203
R309	321-0666-07		RES, FXD, FILM: 3.04K OHM, 0.1%, 0.125W, TC=T9	19701	5033RE3K0408
R310	315-0203-00		RES, FXD, FILM: 20K OHM, 5%, 0.25W	57668	NTR25J-E 20K
R311	315-0203-00		RES, FXD, FILM: 20K OHM, 5%, 0.25W	57668	NTR25J-E 20K
R312	315-0203-00		RES, FXD, FILM: 20K OHM, 5%, 0.25W	57668	NTR25J-E 20K
R314	315-0203-00		RES, FXD, FILM: 20K OHM, 5%, 0.25W	57668	NTR25J-E 20K
R316	131-0566-00		BUS, CONDUCTOR: DUMMY RES, 0.094 x 0.225	24546	OMA 07
R317	321-0367-00		RES, FXD, FILM: 64.9K OHM, 1%, 0.125W, TC=T0	07716	CEAD64901F
R318	321-0817-07		RES, FXD, FILM: 949.2 OHM, 0.1%, 0.125W, TC=T9	19701	5033RE949R28
R319	321-0666-07		RES, FXD, FILM: 3.04K OHM, 0.1%, 0.125W, TC=T9	19701	5033RE3K0408
R321	315-0203-00		RES, FXD, FILM: 20K OHM, 5%, 0.25W	57668	NTR25J-E 20K
R324	315-0391-00		RES, FXD, FILM: 390 OHM, 5%, 0.25W	57668	NTR25J-E390E
R325	315-0391-00		RES, FXD, FILM: 390 OHM, 5%, 0.25W	57668	NTR25J-E390E
R326	315-0202-00		RES, FXD, FILM: 2K OHM, 5%, 0.25W	57668	NTR25J-E 2K
R327	315-0562-00		RES, FXD, FILM: 5.6K OHM, 5%, 0.25W	57668	NTR25J-E05K6
R328	315-0101-00		RES, FXD, FILM: 100 OHM, 5%, 0.25W	57668	NTR25J-E 100E
R329	315-0332-00		RES, FXD, FILM: 3.3K OHM, 5%, 0.25W	57668	NTR25J-E03K3
R331	315-0101-00		RES, FXD, FILM: 100 OHM, 5%, 0.25W	57668	NTR25J-E 100E
R333	315-0182-00		RES, FXD, FILM: 1.8K OHM, 5%, 0.25W	57668	NTR25J-E1K8
R336	315-0202-00		RES, FXD, FILM: 2K OHM, 5%, 0.25W	57668	NTR25J-E 2K
R337	315-0201-00		RES, FXD, FILM: 200 OHM, 5%, 0.25W	57668	NTR25J-E200E
R338	321-0126-00		RES, FXD, FILM: 200 OHM, 1%, 0.125W, TC=T0	19701	5033ED200ROF
R339	315-0201-00		RES, FXD, FILM: 200 OHM, 5%, 0.25W	57668	NTR25J-E200E
R341	315-0101-00		RES, FXD, FILM: 100 OHM, 5%, 0.25W	57668	NTR25J-E 100E
R342	321-0230-00		RES, FXD, FILM: 2.43K OHM, 1%, 0.125W, TC=T0	19701	5043ED2K430F
R343	321-0282-00		RES, FXD, FILM: 8.45K OHM, 1%, 0.125W, TC=T0	07716	CFAD84500F
R344	315-0751-00		RES, FXD, FILM: 750 OHM, 5%, 0.25W	57668	NTR25J-E750E
R345	315-0751-00		RES, FXD, FILM: 750 OHM, 5%, 0.25W	57668	NTR25J-E750E
R346	315-0202-00		RES, FXD, FILM: 2K OHM, 5%, 0.25W	57668	NTR25J-E 2K
R347	315-0182-00		RES, FXD, FILM: 1.8K OHM, 5%, 0.25W	57668	NTR25J-E1K8
R348	315-0152-00		RES, FXD, FILM: 1.5K OHM, 5%, 0.25W	57668	NTR25J-E01K5
R349	315-0152-00		RES, FXD, FILM: 1.5K OHM, 5%, 0.25W	57668	NTR25J-E01K5
R351	315-0102-00		RES, FXD, FILM: 1K OHM, 5%, 0.25W	57668	NTR25JE01K0
R352	315-0152-00		RES, FXD, FILM: 1.5K OHM, 5%, 0.25W	57668	NTR25J-E01K5
R353	315-0100-00		RES, FXD, FILM: 10 OHM, 5%, 0.25W	19701	5043CX10R00J

Replaceable Electrical Parts - TS615

Component No.	Tektronix Part No.	Serial/Assembly No. Effective	Discont	Name & Description	Mfr. Code	Mfr. Part No.
R354	315-0202-00			RES, FXD, FILM: 2K OHM, 5%, 0.25W	57668	NTR25J-E 2K
R355	321-0812-07			RES, FXD, FILM: 455 OHM, 0.1%, 0.125W, TC=T9	19701	5033RE455ROB
R356	321-0812-07			RES, FXD, FILM: 455 OHM, 0.1%, 0.125W, TC=T9	19701	5033RE455ROB
R357	315-0101-00			RES, FXD, FILM: 100 OHM, 5%, 0.25W	57668	NTR25J-E 100E
R358	315-0201-00			RES, FXD, FILM: 200 OHM, 5%, 0.25W	57668	NTR25J-E200E
R359	321-0231-00			RES, FXD, FILM: 2 49K OHM, 1%, 0.125W, TC=T0	19701	5033ED2K49F
R361	315-0223-00			RES, FXD, FILM: 22K OHM, 5%, 0.25W	19701	5043CX22K00J32U
R363	321-0729-06			RES, FXD, FILM: 786 OHM, 0.25%, 0.125W	07716	CEAE786ROC
R364	315-0102-00			RES, FXD, FILM: 1K OHM, 5%, 0.25W	57668	NTR25JE01K0
R365	315-0201-00			RES, FXD, FILM: 200 OHM, 5%, 0.25W	57668	NTR25J-E200E
R366	321-0729-06			RES, FXD, FILM: 786 OHM, 0.25%, 0.125W	07716	CEAE786ROC
R367	321-0126-00			RES, FXD, FILM: 200 OHM, 1%, 0.125W, TC=T0	19701	5033ED200ROF
R368	-----			(SELECTED)		
R369	311-1238-00			RES, VAR, NONW: TRMR, 5K OHM, 0.5W	32997	3386X-DY6-502
R371	315-0472-00			RES, FXD, FILM: 4 7K OHM, 5%, 0.25W	57668	NTR25J-E04K7
R372	315-0222-00			RES, FXD, FILM: 2.2K OHM, 5%, 0.25W	57668	NTR25J-E02K2
R373	315-0222-00			RES, FXD, FILM: 2 2K OHM, 5%, 0.25W	57668	NTR25J-E02K2
R374	315-0101-00			RES, FXD, FILM: 100 OHM, 5%, 0.25W	57668	NTR25J-E 100E
R378	315-0154-00			RES, FXD, FILM: 150K OHM, 5%, 0.25W	57668	NTR25J-E150K
R379	311-1915-00			RES, VAR, NONW: TRMR, 20K OHM, 10%, 0.5 W	32997	3386C-T07-203
R385	321-0729-06			RES, FXD, FILM: 786 OHM, 0.25%, 0.125W	07716	CEAE786ROC
R399	308-0299-00			RES, FXD, W: 300 OHM, 1%, 3W	01686	T2A-300R-F-10
R408	315-0272-00			RES, FXD, FILM: 2.7K OHM, 5%, 0.25W	57668	NTR25J-E02K7
R416	315-0302-00			RES, FXD, FILM: 3K OHM, 5%, 0.25W	57668	NTR25J-E03K0
R417	321-0085-00	B010100	B010657	RES, FXD, FILM: 75 OHM, 1%, 0.125W, TC=T0	57668	CRB14FXE 75 OHM
R417	321-0085-03	B010658		RES, FXD, FILM: 75 OHM, 0.25%, 0.125W, TC=T2	57668	CRB14 CYE 75 OHM
R418	321-0085-00	B010100	B010657	RES, FXD, FILM: 75 OHM, 1%, 0.125W, TC=T0	57668	CRB14FXE 75 OHM
R418	321-0085-03	B010658		RES, FXD, FILM: 75 OHM, 0.25%, 0.125W, TC=T2	57668	CRB14 CYE 75 OHM
R425	321-0729-06			RES, FXD, FILM: 786 OHM, 0.25%, 0.125W	07716	CEAE786ROC
R426	321-0126-00			RES, FXD, FILM: 200 OHM, 1%, 0.125W, TC=T0	19701	5033ED200ROF
R427	321-0161-00			RES, FXD, FILM: 464 OHM, 1%, 0.125W, TC=T0	07716	CEAD464ROF
R428	321-1296-03			RES, FXD, FILM: 12.0K OHM, 0.25%, 0.125W, TC=T2	07716	CEAC12001C
R429	311-1936-00			RES, VAR, NONW: TRMR, 50 OHM, 20%, 0.5W	32997	3386X-T07-500
R438	321-0239-03			RES, FXD, FILM: 3.01K OHM, 0.25%, 0.125W, TC=T2	19701	5033RC3K01C
R439	321-1133-02			RES, FXD, FILM: 240 OHM, 0.5%, 0.125W, TC=T2	24546	NC6002400D
R444	321-0612-00			RES, FXD, FILM: 500 OHM, 1%, 0.125W, TC=T0	07716	CEAD500ROF
R445	321-0612-03			RES, FXD, FILM: 500 OHM, 0.25%, 0.125W, TC=T2	19701	5033RC500ROC
R446	321-0928-03			RES, FXD, FILM: 250 OHM, 0.25%, 0.125W, TC=T2	19701	5033RC250ROC
R447	321-0927-07			RES, FXD, FILM: 125 OHM, 0.1%, 0.125W, TC=T9	19701	5033RE125ROB
R448	321-0143-00			RES, FXD, FILM: 301 OHM, 1%, 0.125W, TC=T0	07716	CEAD301ROF
R449	315-0102-00			RES, FXD, FILM: 1K OHM, 5%, 0.25W	57668	NTR25JE01K0
R455	311-1245-00			RES, VAR, NONW: TRMR, 10K OHM, 0.5W	32997	3386X-DY6-103
R456	321-0269-00			RES, FXD, FILM: 6 19K OHM, 1%, 0.125W, TC=T0	07716	CEAD61900F
R457	321-0774-03			RES, FXD, FILM: 4 5K OHM, 0.25%, 0.125W, TC=T2	19701	5033RC4K500C
R458	315-0101-00			RES, FXD, FILM: 100 OHM, 5%, 0.25W	57668	NTR25J-E 100E
R459	321-0227-00			RES, FXD, FILM: 2.26K OHM, 1%, 0.125W, TC=T0	01121	RNK2261F
S100	263-0016-04			SWITCH PB ASSY: 4 LATCH, 7.5 MM, 8 CONTACTS	80009	263-0016-04
S101	263-0016-04			SWITCH PB ASSY: 4 LATCH, 7.5 MM, 8 CONTACTS	80009	263-0016-04
S102	263-0016-04			SWITCH PB ASSY: 4 LATCH, 7.5 MM, 8 CONTACTS	80009	263-0016-04
S103	263-0016-04			SWITCH PB ASSY: 4 LATCH, 7.5 MM, 8 CONTACTS	80009	263-0016-04
S105	263-0010-00			SWITCH PB ASSY: 1 PUSH, 7.5 MM, W/2 CONTACTS	80009	263-0010-00
S106	263-0015-02			SWITCH PB ASSY: 3 LCH, 7.5MM, 6 CONTACTS	80009	263-0015-02
S107	263-0015-02			SWITCH PB ASSY: 3 LCH, 7.5MM, 6 CONTACTS	80009	263-0015-02
S108	263-0015-02			SWITCH PB ASSY: 3 LCH, 7.5MM, 6 CONTACTS	80009	263-0015-02
S301	263-0010-00			SWITCH PB ASSY: 1 PUSH, 7.5 MM, W/2 CONTACTS	80009	263-0010-00
S302	263-0010-00			SWITCH PB ASSY: 1 PUSH, 7.5 MM, W/2 CONTACTS	80009	263-0010-00
S303	263-0010-00			SWITCH PB ASSY: 1 PUSH, 7.5 MM, W/2 CONTACTS	80009	263-0010-00
S306	263-0010-00			SWITCH PB ASSY: 1 PUSH, 7.5 MM, W/2 CONTACTS	80009	263-0010-00

Component No.	Tektronix Part No.	Serial/Assembly No.		Name & Description	Mfr. Code	Mfr. Part No.
		Effective	Discont.			
S308	263-0010-00			SWITCH PB ASSY:1 PUSH,7 5 MM,W/2 CONTACTS	80009	263-0010-00
S309	263-0010-00			SWITCH PB ASSY:1 PUSH,7 5 MM,W/2 CONTACTS	80009	263-0010-00
S437	260-1771-00			SWITCH,PUSH:1 BUTTON,2 POLE,SLOPE	31918	ORDER BY DESC
T339	120-1071-00			TRANSFORMER,RE-TOROID	80009	120-1071-00
T358	120-1070-00			TRANSFORMER,RE-TOROID	80009	120-1070-00
U112	156-0382-02			MICROCKT,DGTL:QUAD 2 INP NAND GATE BURN	18324	N74LS00NB
U122	156-0383-02			MICROCKT,DGTL:QUAD 2-INP NOR GATE	18324	N74LS02NB
U133	156-0905-09			MICROCKT,DGTL:256 X 4 PROM,PROGRAMMED	80009	156-0905-09
U137	156-0733-02			MICROCKT,DGTL:DUAL MONOSTABLE MV W/ST INP	01295	SN74LS221N3
U153	156-0905-01			MICROCKT,DGTL:PROM,PROGRAMMED	80009	156-0905-01
U163	156-0391-02			MICROCKT,DGTL:HEX LATCH W/CLEAR	01295	SN74LS174NP3
U178	156-0479-02			MICROCKT,DGTL:QUAD 2-INP OR GATE	01295	SN74LS32NP3
U188	156-0905-10			MICROCKT,DGTL:256 X 4 PROM,PROGRAMMED	80009	156-0905-10
U208	156-0383-02			MICROCKT,DGTL:QUAD 2-INP NOR GATE	18324	N74LS02NB
U219	156-1114-01			MICROCKT,LINER:OPERATIONAL AMP,TESTED	02735	CA3160E/5
U349	156-0534-00	8010100	8010724	MICROCKT,LINER:DUAL DIFF AMPL	02735	CA3102E-98
U349	156-0534-01	8010725		MICROCKT,LINER:DUAL DIFF AMPL,BURN-IN	02735	CA3102EX
U353	156-1114-01			MICROCKT,LINER:OPERATIONAL AMP,TESTED	02735	CA3160E/5
U375	156-0742-00			MICROCKT,LINER:OPNL AMPL	01295	LM318P

REPLACEABLE MECHANICAL PARTS

PARTS ORDERING INFORMATION

Replacement parts are available from or through your local Tektronix, Inc. Field Office or representative

Changes to Tektronix instruments are sometimes made to accommodate improved components as they become available, and to give you the benefit of the latest circuit improvements developed in our engineering department. It is therefore important, when ordering parts, to include the following information in your order: Part number, instrument type or number, serial number, and modification number if applicable.

If a part you have ordered has been replaced with a new or improved part, your local Tektronix, Inc. Field Office or representative will contact you concerning any change in part number.

Change information, if any, is located at the rear of this manual.

ITEM NAME

In the Parts List, an Item Name is separated from the description by a colon (:). Because of space limitations, an Item Name may sometimes appear as incomplete. For further Item Name identification, the U.S. Federal Cataloging Handbook H6-1 can be utilized where possible.

FIGURE AND INDEX NUMBERS

Items in this section are referenced by figure and index numbers to the illustrations.

INDENTATION SYSTEM

This mechanical parts list is indented to indicate item relationships. Following is an example of the indentation system used in the description column:

```

1 2 3 4 5      Name & Description
Assembly and or Component
Attaching parts for Assembly and or Component
      **** END ATTACHING PARTS ****
Detail Part of Assembly and or Component
Attaching parts for Detail Part
      **** END ATTACHING PARTS ****
Parts of Detail Part
Attaching parts for Parts of Detail Part
      **** END ATTACHING PARTS ****

```

Attaching Parts always appear in the same indentation as the item it mounts, while the detail parts are indented to the right. Indented items are part of, and included with, the next higher indentation.

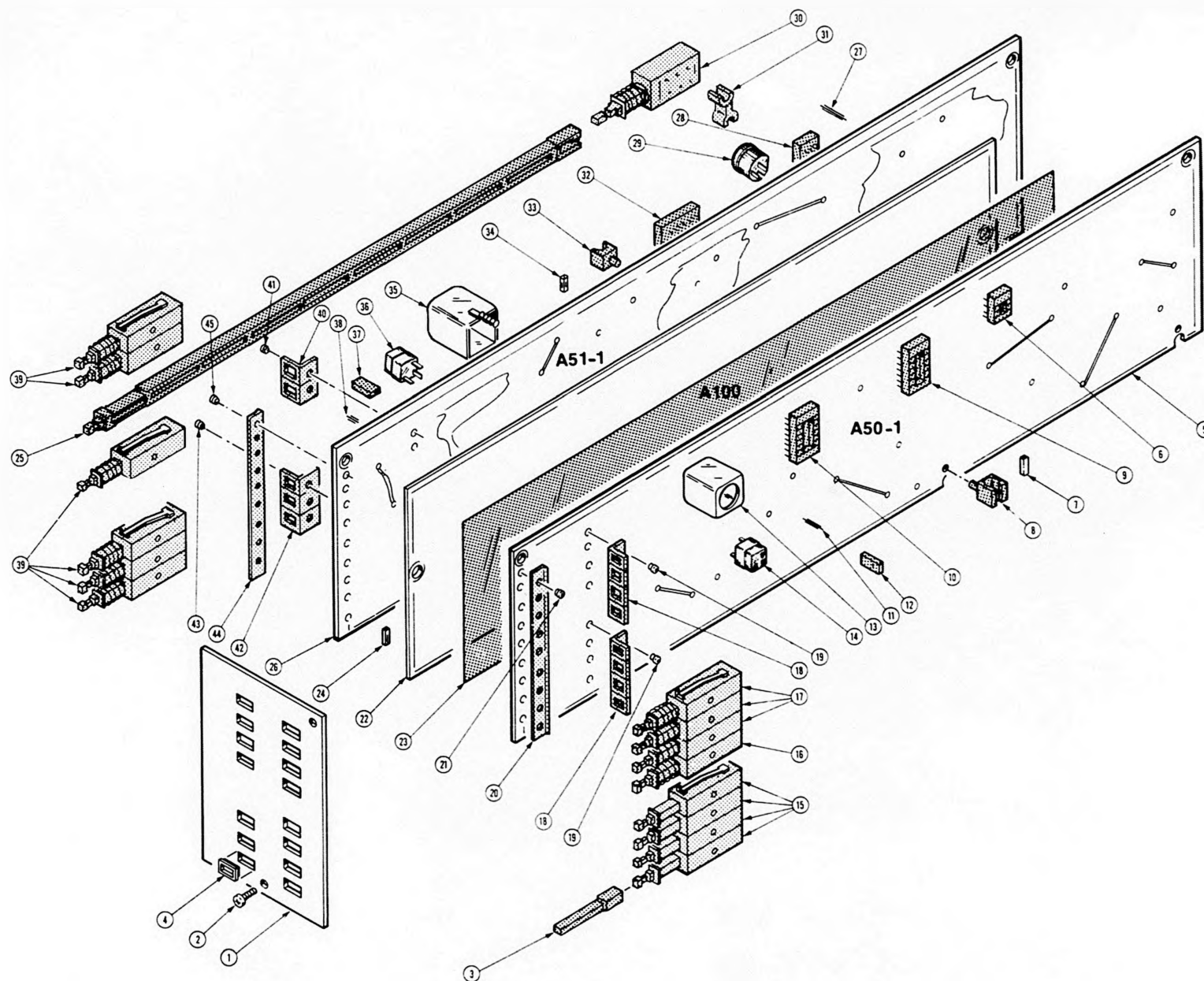
Attaching parts must be purchased separately, unless otherwise specified.

ABBREVIATIONS

ACTR	INCH	ELECTRN	ELECTRON	IN	INCH	SE	SINGLE END
ADPTR	NUMBER SIZE	ELEC	ELECTRICAL	INCAND	INCANDESCENT	SECT	SECTION
ALIGN	ACTUATOR	ELECTLT	ELECTROLYTIC	INSUL	INSULATOR	SEMICON	SEMICONDUCTOR
AL	ADAPTER	ELEM	ELEMENT	INTL	INTERNAL	SHLD	SHIELD
ASSEM	ALIGNMENT	EPL	ELECTRICAL PARTS LIST	LPHLDR	LAMPHOLDER	SHLDR	SHOULDERED
ASSY	ALUMINUM	EQPT	EQUIPMENT	MACH	MACHINE	SKT	SOCKET
ATTEN	ASSEMBLED	EXT	EXTERNAL	MECH	MECHANICAL	SL	SLIDE
AWG	ASSEMBLY	FIL	FILLISTER HEAD	MTG	MOUNTING	SLFLMG	SELF-LOCKING
BD	ATTENUATOR	FLEX	FLEXIBLE	NIP	NIPPLE	SLVG	SLEEVING
BRKT	AMERICAN WIRE GAGE	FLM	FLAT HEAD	NON WIRE	NOT WIRE WOUND	SPR	SPRING
BRS	BOARD	FLTR	FILTER	OBD	ORDER BY DESCRIPTION	SO	SQUARE
BRZ	BRACKET	FR	FRAME or FRONT	OD	OUTSIDE DIAMETER	SST	STAINLESS STEEL
BSMG	BRASS	FSTNR	FASTENER	OVH	OVAL HEAD	STL	STEEL
CAB	BRONZE	FT	FOOT	PH BRZ	PHOSPHOR BRONZE	SW	SWITCH
CAB	BUSHING	FXD	FIXED	PL	PLAIN or PLATE	T	TUBE
CAP	CABINET	GSKT	GASKET	PLSTC	PLASTIC	TERM	TERMINAL
CER	CAPACITOR	HDL	HANDLE	PN	PART NUMBER	THD	THREAD
CHAS	CERAMIC	HEX	HEXAGON	PNH	PAN HEAD	THK	THICK
CKT	CHASSIS	HEX HD	HEXAGONAL HEAD	PWR	POWER	TNSN	TENSION
COMP	CIRCUIT	HEX SOC	HEXAGONAL SOCKET	RCPT	RECEPTACLE	TPG	TAPPING
CONN	COMPOSITION	HLCPS	HELICAL COMPRESSION	RES	RESISTOR	TRM	TRUSS HEAD
COV	CONNECTOR	HLEXT	HELICAL EXTENSION	RGD	RIGID	V	VOLTAGE
CPLG	COVER	HV	HIGH VOLTAGE	RLF	RELIEF	VAR	VARIABLE
CRT	COUPLING	IC	INTEGRATED CIRCUIT	RTNR	RETAINER	W	WITH
DEG	CATHODE RAY TUBE	ID	INSIDE DIAMETER	SCM	SOCKET HEAD	WSMR	WASHER
DWR	DEGREE	IDNT	IDENTIFICATION	SCOPE	OSCILLOSCOPE	XFMR	TRANSFORMER
	DRAWER	IMPLR	IMPELLER	SCR	SCREW	XSTR	TRANSISTOR

CROSS INDEX - MFR. CODE NUMBER TO MANUFACTURER

Mfr. Code	Manufacturer	Address	City, State, Zip Code
07707	USM CORP SUB OF EMMART INDUSTRIES INC USM FASTENER DIV	510 RIVER RD	SHELTON CT 06484
09922	BURNDY CORP	RICHARDS AVE	NORWALK CT 06852
22526	DU PONT E I DE NEMOURS AND CO INC DU PONT CONNECTOR SYSTEMS	30 HUNTER LANE	CAMP HILL PA 17011
32436	SYSCON INTERNATIONAL, INC.	205 SYCAMORE ST.	SOUTH BEND, IN 46622
70485	ATLANTIC INDIA RUBBER WORKS INC	571 W POLK ST	CHICAGO IL 60607
71785	TRW INC TRW CINCH CONNECTORS	1501 MORSE AVE	ELK GROVE VILLAGE IL 60007
80009	TEKTRONIX INC	4900 S W GRIFFITH DR P O BOX 500	BEAVERTON OR 97077
93907	TEXTRON INC CAMCAR DIV	600 18TH AVE	ROCKFORD IL 61101



TSG15 PULSE & BAR TEST SIGNAL GENERATOR

Fig. & Index No.	Tektronix Part No.	Serial/Assembly No. Effective Dscont	Qty	12345 Name & Description	Mfr. Code	Mfr. Part No.
1-1	333-2360-00		1	PANEL, FRONT: (ATTACHING PARTS)	80009	333-2360-00
-2	213-0277-00		2	SCREW, TPG, TF: 2-56 X 0.312, SPCL TYPE, PMH, STL (END ATTACHING PARTS)	93907	234-01725-024
-3	366-1691-00		15	PUSH BUTTON: GY, 1.2 L	80009	366-1691-00
-4	426-1206-00		15	FRAME, PUSH BTN: MOMENTARY, GRAY PLASTIC	80009	426-1206-00
-5	-----		1	OKT BD ASSY: PULSE AND BAR LOGIC (SEE AS0-1 REPL)		
-6	136-0514-00		1	.SKT, PL-IN ELEK: MICROCIRCUIT, 8 DIP	09922	D1LB8P-108
-7	136-0328-03		63	.SOCKET, PIN TERM: U/W 0.025 SQ PINS	22526	47710
-8	214-2440-00		3	.RECEPTACLE, PIN: CIRCUIT BOARD	80009	214-2440-00
-9	136-0269-02		4	.SKT, PL-IN ELEK: MICROCKT, 14 DIP, PCB MT	09922	D1LB14P-108T
-10	136-0260-02		5	.SKT, PL-IN ELEK: MICROCKT, 16 DIP, PCB MT	09922	D1LB16P-108T
-11	131-0608-00		31	.TERMINAL, PIN: 0.365 L X 0.025 BRZ GLD PL	22526	48283-036
-12	131-0993-00		8	.BUS, CONDUCTOR: SHUNT ASSEMBLY, BLACK	22526	65474-005
-13	337-1417-00		1	.SHIELD, ELEC: 0.55 SQ X 0.685 INCH HIGH	32436	A-1020002-1
-14	136-0220-00	B010100	39	.SKT, PL-IN ELEK: TRANSISTOR 3 CONTACT	71785	133-23-11-034
	136-0220-00	B010276	10	.SKT, PL-IN ELEK: TRANSISTOR 3 CONTACT	71785	133-23-11-034
-15	-----		1	.SWITCH, PB ASSY: (SEE S100, S101, S102, S103 REPL)		
-16	-----		1	.SWITCH PB ASSY: (SEE S105 REPL)		
-17	-----		1	.SWITCH PB ASSY: (SEE S106, S107, S108 REPL)		
	348-0004-00	B011221	1	.GROMMET, RUBBER: BLACK, ROUND, 0.281 ID	70485	763
-18	343-0499-04	B010100	3	.CLIP, SWITCH: REAR, 7.5MM X 4 UNIT	80009	343-0499-04
	343-0499-13	B011456	3	.CLIP, SWITCH: 7.5MM X 4 UNIT (ATTACHING PARTS)	80009	343-0499-13
-19	210-3033-00		8	.EYELET, METALLIC: 0.059 OD X 0.156 L, BRS (END ATTACHING PARTS)	07707	SE-25
-20	343-0495-09		1	.CLIP, SWITCH: FRONT, 7.5MM X 9 UNIT (ATTACHING PARTS)	80009	343-0495-09
-21	210-3033-00		9	.EYELET, METALLIC: 0.059 OD X 0.156 L, BRS (END ATTACHING PARTS)	07707	SE-25
-22	-----		1	OKT BOARD ASSY: SHIELD (SEE A100 REPL)		
-23	255-0632-00		1	.PLASTIC SHEET: 2.5 X 11.0 X 0.002 ADM BACK	80009	255-0632-00
-24	136-0328-03		3	.SOCKET, PIN TERM: U/W 0.025 SQ PINS	22526	47710
-25	384-1058-00		1	EXTENSION SHAFT: 8.157 L	80009	384-1058-00
-26	-----		1	OKT BD ASSY: PULSE AND BAR OUTPUT (SEE AS1-1 REPL)		
-27	131-0589-00		4	.TERMINAL, PIN: 0.46 L X 0.025 SQ PH BRZ	22526	48283-029
-28	136-0514-00		2	.SKT, PL-IN ELEK: MICROCIRCUIT, 8 DIP	09922	D1LB8P-108
-29	136-0183-00	B010100	1	.SKT, PL-IN ELEK: TRANSISTOR, 3 CONTACT	80009	136-0183-00
-30	-----		1	.:	80009	-----
-31	361-0384-00		2	.SPACER, PB SW: 0.133 L, RED POLYCARBONATE	80009	361-0384-00
-32	136-0269-02		1	.SKT, PL-IN ELEK: MICROCKT, 14 DIP, PCB MT	09922	D1LB14P-108T
-33	214-2440-00		3	.RECEPTACLE, PIN: CIRCUIT BOARD	80009	214-2440-00
-34	136-0328-03		34	.SOCKET, PIN TERM: U/W 0.025 SQ PINS	22526	47710
-35	337-1417-00		26	.SHIELD, ELEC: 0.55 SQ X 0.685 INCH HIGH	32436	A-1020002-1
-36	136-0220-00	B010100	13	.SKT, PL-IN ELEK: TRANSISTOR 3 CONTACT	71785	133-23-11-034
-37	131-0993-00		3	.BUS, CONDUCTOR: SHUNT ASSEMBLY, BLACK	22526	65474-005
-38	131-0608-00		12	.TERMINAL, PIN: 0.365 L X 0.025 BRZ GLD PL	22526	48283-036
-39	-----		6	.SWITCH PB ASSY: (SEE S301, S302, S303, S306, S308, S309 REPL)		
-40	343-0499-02	B010100	1	.CLIP, SWITCH: REAR, 7.5MM X 2 UNIT	80009	343-0499-02
	343-0499-11	B010694	1	.CLIP, SWITCH: REAR, 7.5MM X 2 UNIT (ATTACHING PARTS)	80009	343-0499-11
-41	210-3033-00		5	.EYELET, METALLIC: 0.059 OD X 0.156 L, BRS (END ATTACHING PARTS)	07707	SE-25
-42	343-0499-03		3	.CLIP, SWITCH: REAR, 7.5MM X 3 UNIT (ATTACHING PARTS)	80009	343-0499-03
-43	210-3033-00		3	.EYELET, METALLIC: 0.059 OD X 0.156 L, BRS (END ATTACHING PARTS)	07707	SE-25
-44	343-0495-09		1	.CLIP, SWITCH: FRONT, 7.5MM X 9 UNIT (ATTACHING PARTS)	80009	343-0495-09
-45	210-3033-00		9	.EYELET, METALLIC: 0.059 OD X 0.156 L, BRS (END ATTACHING PARTS)	07707	SE-25

Replaceable Mechanical Parts - TS615

Fig. & Index No.	Tektronix Part No.	Serial/Assembly No. Effective Discnt	Qty	12345	Name & Description	Mfr. Code	Mfr. Part No.
1-					STANDARD ACCESSORIES		
	070-2479-00		1		MANUAL, TECH: INSTRUCTION	80009	070-2479-00